

Complete Physical Interface of the Neutrino Propagation Role Complex

Carrier-Neutral Invariants, Simultaneous Lawful Equivalence, and Constraint Closure

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Abstract

A preceding target-fixed analysis of neutrino oscillation established the nested physical role complex

$$(r_{\triangleright\Delta}, r_{\Delta}; r_{\Delta} \succ r_{\triangleright\Delta}), \quad r_{\Delta} \Rightarrow r_{\triangleright\Delta}, \quad r_{\triangleright\Delta} \not\Rightarrow r_{\Delta},$$

without identifying a mass term, mixing matrix, right-handed neutrino, Weinberg operator, sterile sector, defect, or any other microscopic occupant. The present paper determines the complete carrier-neutral physical interface that any lawful realization of that role complex must reproduce.

For same-target completions whose interface data admit one simultaneous physical realization, we define a canonical projection

$$\Pi_{\nu} : \mathfrak{C}_{\nu}^{\text{sim}} \longrightarrow \mathfrak{S}_{\nu}^{\text{sim}}.$$

The core interface consists of the charged-current preparation/detection boundary, the record-relevant effective propagation channel, cross-boundary support transfer, the finite boundary-visible response-mode packet inherited from the preceding analysis, mode-to-boundary incidence, environment and record transport, and the strict nesting of the two roles. The attachment data record the locality, causal-support, electroweak boundary, gauge-quotient, representation/charge, anomaly, vacuum, deformation, operator-availability, and RG/EFT conditions required for one realized physical interface. Global-form, higher-form, higher-group, defect, and non-invertible data enter only when an admitted target probe or attachment condition activates them.

The completeness proof is independent of the interface coordinate list. A target-facing trace $\tau_{\mathbb{T}_{\nu}^{(2)}} : \mathcal{R}_{\nu}^{\text{raw}} \rightarrow \mathcal{T}(\mathbb{T}_{\nu}^{(2)})^{\perp}$ is defined directly from the fixed target specification and the imported physical-role architecture. Semantic target uses and physical evaluators are characterized by same-target typing, lawful-redescription invariance, fail-closed physical standing, and extensionality with respect to this trace. Target-trace descent proves that every such use is a lawful functional of the five trace factors; saturation gives the exact target-use structure, and refinement into nine output sorts yields $\text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)}) = \text{Eval}(\mathbb{T}_{\nu}^{(2)})$ without assuming a finite evaluator list. The carrier-neutral profile $\mathcal{P}_{\nu}^{+} = (\mathcal{P}_{\nu}^0, \mathcal{J}_{\nu}^{\text{sim}})$ is then formed from the primitive evaluator signatures and is proved to recover the target-facing trace. Its joint-incidence component is the physical isomorphism class of the shared witness-overlap diagram, including common-cone existence. Raw coordinate products are restricted first to the shared-realization pullback and then to the fail-closed admissible subobject; the lawful quotient is formed only afterward.

For simultaneous same-target completions,

$$\Pi_{\nu}(X) = \Pi_{\nu}(Y) \iff \tilde{\mathfrak{S}}_{\nu}(X) \equiv_{\mathbb{T}_{\nu}^{(2)}} \tilde{\mathfrak{S}}_{\nu}(Y) \iff \mathcal{P}_{\nu}^{+}(\tilde{\mathfrak{S}}_{\nu}(X)) = \mathcal{P}_{\nu}^{+}(\tilde{\mathfrak{S}}_{\nu}(Y)).$$

Pullback membership and every active constraint are congruent on complete-profile classes, so

$$\mathfrak{S}_{\nu}^{\text{adm}} = \mathcal{M}_{\text{adm}} / \simeq_{\text{int}}$$

is well defined. For every already-specified completion,

$$\text{InterfaceAdeq}(X, T_{\nu}^{\text{osc}}) \iff \tilde{\mathfrak{S}}_{\nu}(X) \in \mathcal{M}_{\text{adm}},$$

and for simultaneous completions this is equivalent to $\Pi_{\nu}(X) \in \mathfrak{S}_{\nu}^{\text{adm}}$. The resulting interface is complete only relative to the declared target and comparison class: it is not full-theory identity, does not determine a microscopic carrier, and does not promote interface adequacy to physical realization.

Keywords: neutrino oscillation; physical interface; quantum channel; charged-current boundary; shared physical witnesses; lawful equivalence; interface completeness; generalized symmetry; anomaly compatibility; AASC; carrier neutrality.

Principal result

Fix the target, empirical response input, nested role complex, and comparison class stated below. Every lawful simultaneous realization admits one canonical carrier-neutral interface packet. Equality of complete packets is equivalent to indistinguishability under every semantic target-admitted evaluator, and hence to equality of every component value and physical joint-incidence value in $\mathcal{P}_v^+$. The strict relation $r_\Delta \succ r_{\triangleright\triangleleft}$ is preserved; no extended-operator or generalized-symmetry coordinate is universally forced in the point-local core fiber; and no microscopic occupant follows from the interface theorem alone.

Scope of the result

The argument has four logical layers.

1. **Prior role result.** The preceding target-fixed analysis supplies the empirical target, the definitions of $r_{\triangleright\triangleleft}$ and r_Δ , their strict nesting, and the associated lawful-redescription class [20]. These results are used as established inputs and are not rederived here.
2. **Operational interface.** The charged-current boundary and complete source–propagation–detection channel are formulated without specifying microscopic fields or a mass mechanism.
3. **Physical attachment and simultaneous incidence.** Locality, gauge, anomaly, vacuum, deformation, scale, and conditionally relevant generalized structures are typed, and their retained coordinates are required to belong to one shared physical witness diagram.
4. **Target-relative semantic completeness.** The target-facing trace and semantic evaluator class are defined independently of the comparison profile; the profile is then proved to recover that trace, simultaneous and admissible subobjects are formed, and the canonical lawful quotient is taken only after those reductions.

The strongest conclusion is target-relative interface completeness. It is not candidate-universe completeness, role-occupancy uniqueness, or particle prediction.

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1 Introduction and theorem strategy

1.1 From a necessary role to a complete physical interface

A theorem establishing a necessary physical role is logically distinct from a theorem identifying a microscopic carrier of that role. The preceding analysis showed that the realized neutrino-oscillation target cannot be preserved unless propagation violates loss-tolerant charged-current sector nonmixing and unless the common reactor boundary exhibits at least two lawfully distinct visible response modes. It further established that the second obligation is strictly stronger than the first and that the minimal renormalizable massless-neutrino Standard Model realizes neither role [20]. These results establish a nonempty residual physical obligation, but they do not identify what carries it.

The next mathematical problem is therefore to determine the exact target-relative comparison object that every possible realization must reproduce. Without such an object, a candidate analysis is vulnerable to three opposed errors. A coarse signature can merge physically distinct completions and create false uniqueness. An overfine signature can preserve basis, gauge, dilation, or categorical presentation differences and create false multiplicity. A componentwise signature can also combine individually admissible data that do not belong to any single physical realization.

The present paper addresses all three problems simultaneously. It first defines the complete target-facing trace as the restriction of a raw presentation to the fixed target specification and the imported physical witness sorts, independently of any proposed interface coordinates. Semantic physical uses and evaluators are then defined only by lawful same-target extensionality with respect to that trace. Target-trace descent yields exact saturated use and evaluator structures. The canonical interface is subsequently built so as to preserve every value and joint-incidence distinction visible to the semantic target trace, remove merely presentational differences, and require every retained coordinate to belong to one physical witness system.

The operational language of quantum theory supplies the natural core formalism: preparations, channels, effects, instruments, and outcome probabilities [10, 12, 26, 29]. The AASC framework supplies the fixed-domain constraints needed to control target identity, lawful redescription, and standing [17, 21]. The realized-physical-interior and Standard-Model symmetry results constrain attachment to locality, gauge structure, anomaly freedom, vacuum support, and scale [16, 22, 24]. The simultaneous-constraint result of [19] adds the crucial requirement that separately admissible projections must describe one realization through one shared witness system; pairwise compatibility alone is insufficient.

1.2 Four distinct notions of completeness

The term *complete* is used in a typed sense throughout.

1. **Interface completeness:** every physical value or joint-incidence distinction admitted by the target is represented by the canonical interface packet.
2. **Candidate completeness:** every admissible microscopic carrier class is generated. This is a separate downstream problem.
3. **Simultaneous-survivor completeness:** every generated candidate is tested under one coupled physical constraint bundle. This too is downstream of the present theorem.
4. **Role-realization completeness:** candidate, quotient, role, eligibility, compatibility, live-residue, and failure completeness jointly license realization through Role-Occupancy Closure [18].

Only the first notion is proved here. A complete interface can have many microscopic preimages, and even a unique interface-adequate candidate would not by itself constitute a realization theorem.

1.3 Proof architecture

The argument uses six recurring proof forms.

Operational formation. A physical source–propagation–detection realization induces an effective map

from admitted preparations to admitted detector records. Under the stated subsystem assumptions this is represented by a quantum instrument whose total map is completely positive and trace preserving.

Deletion and merge. A datum is retained only if removing it destroys a target operation, merges two target-distinct interfaces, makes downstream compatibility ill typed, or changes a target anchor.

Shared-realization pullback. Separately admissible coordinate values must factor through one typed physical witness diagram; pairwise compatibility is insufficient.

Target-use and evaluator generation. The fixed target determines every standing-bearing use of its record, continuation/support, substrate-attachment, activated-extension, and simultaneous-co-instantiation data. Every target-admitted evaluator is then generated from the corresponding typed primitive evaluations.

First-difference exhaustion. Any same-target semantic evaluator disagreement must appear in one of the value or joint-incidence loci through which the evaluator algebra factors, rather than in a enumerated candidate list.

Lawful quotient after closure. Simultaneous redescriptions preserving the complete value-and-incidence profile are quotiented only after the raw coordinate product has been restricted to the shared-realization and admissible subobjects.

1.4 Main results

The theorem chain may be summarized as follows.

- R1.** The target and nested role complex are imported without allowing completion-specific variables to define any interface coordinate.
- R2.** The fixed target and upstream physical-role architecture independently determine the complete target-facing trace and the semantic classes of target-admitted uses and evaluators. Target-trace descent then proves exact saturated use/evaluator generation before the comparison profile is introduced.
- R3.** Every lawful realization admits the same operational charged-current boundary up to simultaneous boundary redescription.
- R4.** Every lawful realization induces a record-complete effective propagation interface sufficient to evaluate every target record without microscopic reconstruction.
- R5.** The Boolean role obligations are refined into a full nonalignment interface and a full visible-response interface, including boundary incidence and environment/record transport.
- R6.** Target-complete visible-response multiplicity implies nonalignment, while a one-mode nonaligned channel witnesses failure of the converse.
- R7.** Every lawful interface carries the target-required locality, causal support, physical consistency, electroweak attachment, gauge quotient, anomaly, vacuum, deformation, operator, and scale data needed to remain the same physical problem.
- R8.** Global-form and generalized-symmetry data are activated only by target-admitted extended probes or demonstrated attachment effects; no such coordinate is universally active in the point-local core.
- R9.** The profile $\mathcal{P}_\nu^+$, built from primitive evaluator signatures and physical joint incidence, is a complete invariant for the independently generated evaluator algebra $\text{Eval}(\mathbb{T}_\nu^{(2)})$.
- R10.** Core, attachment, and active extended coordinates factor through one shared physical witness system. The overlap diagram and common-cone status form a target-visible joint-incidence profile.

- R11.** Raw tuples reduce to $\mathcal{M}_{\text{pull}}$ and $\mathcal{M}_{\text{adm}}$. Complete-profile equivalence is congruent with both reductions, so quotienting cannot transfer admissibility across representatives.
- R12.** Equality of canonical simultaneous packets is equivalent to target-relative indistinguishability under every evaluator in $\text{Eval}(\mathbb{T}_v^{(2)})$, and equivalently to equality of $\mathcal{P}_v^+$.
- R13.** The resulting packet is sufficient to define a subsequent source-first candidate comparison while licensing no microscopic occupant.

Theorem 1.1 (Main interface-completeness theorem, informal). *For simultaneous lawful completions $X, Y \in \mathfrak{C}_v^{\text{sim}}$,*

$$\Pi_v(X) = \Pi_v(Y) \iff \tilde{\mathfrak{S}}_v(X) \equiv_{\mathbb{T}_v^{(2)}} \tilde{\mathfrak{S}}_v(Y) \iff \mathcal{P}_v^+(\tilde{\mathfrak{S}}_v(X)) = \mathcal{P}_v^+(\tilde{\mathfrak{S}}_v(Y)),$$

where $\equiv_{\mathbb{T}_v^{(2)}}$ is indistinguishability under the independently defined semantic target-evaluator class (equivalently the proved saturated algebra) and $\mathcal{P}_v^+ = (\mathcal{P}_v^0, \mathcal{J}_v^{\text{sim}})$ is its complete invariant. Under the imported target/role theorems, resolved relevance gates, shared-ledger constraints, and congruence results, every already-specified completion also satisfies

$$\text{InterfaceAdeq}(X, T_v^{\text{osc}}) \iff \tilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{adm}}.$$

For $X \in \mathfrak{C}_v^{\text{sim}}$, this is equivalent to $\Pi_v(X) \in \mathfrak{S}_v^{\text{adm}}$. The projection preserves $r_\Delta \succ r_{\text{b}\Delta}$ and omits every microscopic carrier coordinate.

The precise statements are theorems 13.20 and 14.3.

1.5 Why interface completeness is not model selection

A complete interface can have many microscopic preimages. Stinespring dilation already shows that a given completely positive map can admit nonunique larger-space representations, with uniqueness only up to the appropriate isometry for minimal dilations [27, 29]. Distinct field-theoretic or open-system descriptions can likewise induce the same complete target-relative record and the same physical overlap diagram. They are interface-equivalent when no admitted target operation or attachment witness distinguishes them; this does not identify them as the same full theory.

Conversely, a microscopic difference that changes gauge attachment, anomaly status, operator availability, vacuum phase, scale transport, an activated defect response, or physical witness incidence is retained even if a restricted oscillation probability agrees. The canonical interface is therefore neither a quotient of probability tables alone nor a classifier of complete microscopic theories. It occupies the logically intermediate level needed to compare otherwise heterogeneous realizations of the same target.

1.6 Generalized symmetry as a conditional interface question

Higher-form symmetries act on extended operators, global forms alter line and monopole sectors, higher groups require intertwined background transformations, and non-invertible defects carry fusion and junction data [1, 5, 6, 8, 32]. None of these facts alone implies that the point-local neutrino-propagation target requires such a coordinate. The present analysis therefore proves an activation trichotomy rather than assuming generalized structure. The resulting negative statement for the core target is substantive: generalized structures may remain physically important in particular completions or enlarged target fibers, but they are not universal coordinates of the fixed point-local interface.

1.7 Organization

Sections 2–4 specify the upstream inputs, derive the target-use and target-evaluator structures, construct $\mathcal{P}_v^+$ from its primitive signatures, and define the raw interface presentation. Sections 5–7 establish the operational boundary, channel, and response structure. Sections 8–10 establish locality, environment,

gauge, anomaly, vacuum, deformation, and scale attachment. Section 11 determines when global-form and generalized-symmetry data are physically active. Sections 12–13 prove simultaneous compatibility, congruence, quotient ordering, and interface completeness. Sections 14–17 establish interface adequacy, the no-carrier theorem, representative stress tests, falsifiability, and the final result.

2 Upstream inputs and independence from microscopic models

2.1 The preceding role theorem

The present analysis begins from the target-fixed role theorem established in [20]. The imported object is not a conventional mass-mixing model but a set of target-relative results that remain fixed throughout the present argument.

Table 1: Established inputs from the preceding role analysis and their use here.

Established input	Role in the present analysis
Target fingerprint $\mathfrak{f}(T_\nu^{\text{osc}})$.	Same-target and target-fiber typing.
Operational charged-current boundary classes.	Input to boundary reconstruction, not a propagation eigenbasis.
Primary event-record packet.	Empirical target held fixed here; no new empirical anchor is introduced.
Effective channel comparison class.	General propagation coordinate admitting unitary and reduced nonunitary representatives.
Channel-native nonalignment role $r_{\triangleright\triangleleft}$.	Mandatory boundary/channel interface coordinate.
Boundary-visible mode-class set $\mathfrak{M}_{\text{vis}}(C_\nu)$ with $\rho_{\text{vis}} \geq 2$.	Mandatory response coordinate and incidence data.
Strict nesting $r_\Delta \Rightarrow r_{\triangleright\triangleleft}$, with a one-mode nonaligned counterexample to the converse.	Nested-interface relation and noncollapse witness.
Lawful-redescription class.	Lower bound on the representative equivalences used here.
Minimal massless-neutrino Standard-Model nonoccupancy.	Motivation for the residual role complex; not a premise for selecting an occupant.
Open-completion target family.	Excludes backflow from a mass term, PMNS matrix, ν_R , Weinberg operator, sterile sector, defect, or new-matter hypothesis.

The empirical lineage of the event-record input consists of the SNO flavor-transformation result, Super-Kamiokande atmospheric flavor change and oscillatory L/E structure, and the slow and fast reactor-response packets of KamLAND and Daya Bay [7, 11, 28, 30, 31]. Only their operational content is imported; phenomenological fit coordinates are not promoted to microscopic carrier data.

The established role complex is

$$\mathcal{R}_\nu^{\text{osc}} = (\{r_{\triangleright\triangleleft}, r_\Delta\}, r_\Delta \succ r_{\triangleright\triangleleft}), \quad r_\Delta \Rightarrow r_{\triangleright\triangleleft}, \quad r_{\triangleright\triangleleft} \not\Rightarrow r_\Delta. \quad (1)$$

The present paper may represent these obligations through a richer physical interface, but it does not redefine them. In particular, r_Δ remains the finite multiplicity of source-certified boundary-visible mode classes, not a microscopic spectral rank, and $r_\Delta \succ r_{\triangleright\triangleleft}$ remains a dependence relation between pieces of physical work rather than a demand for two distinct occupants.

2.2 General construction and physical-role inputs

A small number of general theorem families are used at restricted scope.

The AASC kernel theorem fixes admissibility, standing, reference, and irreversibility as the governance roles required for non-degenerate identity-preserving construction on a fixed domain. The associated fixed-domain results supply the binary admissibility-relevant quotient, the AMetric boundary, standing as reuse-stable admissibility, a unique admissible interior, standing conservation, and anti-weakening [17]. The AMetric export theorem permits reuse of the no-selector/no-parameter boundary only after the downstream domain has independently instantiated the relevant admissibility-bearing structure [21].

The Realized Physical Interior supplies the physical role architecture used here: non-null realization, bounded localization, local continuation domains, quotient-stable comparison, bounded transport, field mediation, localized particle classes, gauge-compatible presentation, deformation response, vacuum baseline, temporal order, causal dependence, and record fixation [22]. These roles constrain the physical interface but do not determine its neutrino-specific content by themselves.

The simultaneous-constraint result of [19] supplies the shared-realization architecture: projection-wise admissibility must descend to one common witness system, pairwise compatibility is insufficient, and standing-relevant freely factorized variation collapses unless it is representational or is promoted to a genuinely different coupled candidate. The Role-Occupancy Closure theorem [18] supplies the downstream boundary between interface adequacy and realized occupancy. Merely constructing an interface does not satisfy the hypotheses required for realization.

2.3 Standard-Model and symmetry inputs

The Standard-Model carrier and symmetry results are used only to fix the known source/detection boundary and to classify differences as target changes, lawful representatives, consistency obstructions, or conditionally relevant extension data.

SYM-5 supplies the distinction between physical flavor-limit actions, weak-basis/rephasing equivalence, spurion bookkeeping, and quotient-stable mixing invariants [15]. SYM-7 supplies target separation among neutrino fibers and prevents right-handed-neutrino, Majorana, Weinberg-operator, or higher-operator data from being treated as redescrptions of the minimal renormalizable target [13]. SYM-9 and SYM-11 supply global-form, line, higher-form, higher-group, non-invertible, anomaly, target-fiber, and first-difference classifications [16, 24]. The Standard-Model carrier-closure result supplies the local gauge/matter/Higgs/vacuum architecture at its declared scope [23].

These inputs provide typed boundary and consistency data against which the interface is tested; none establishes the interface-completeness theorem by itself.

2.4 Independence from downstream realizations

Theorem 2.1 (Upstream-independence theorem). *No downstream candidate representation, successful benchmark, interface factorization, or later realization result may be used as a premise for the preceding role theorem or for the independent comparison profile defined in this paper. In particular, the existence of a standard mass-mixing realization does not authorize a mass basis, PMNS matrix, microscopic spectral multiplicity, or finite-dimensional Hilbert-space structure in the interface definitions.*

Proof. The target and role packet are fixed prior to candidate generation. A downstream representation is admissible only after it is shown to reproduce that packet. If a feature of the representation were used to define the target or the comparison profile, the candidate would determine the test it is supposed to pass. This reverses the dependency order and turns a target-fixed necessity theorem into a model-dependent restatement. Under fixed-domain discipline, the move is either a target change, hidden load, or a circular reuse of downstream standing. The only admissible direction is therefore

$$\text{target and role packet} \longrightarrow \text{interface} \longrightarrow \text{candidate analysis},$$

not the reverse. □

Corollary 2.2 (Benchmark nonauthority). *The conventional three-flavor unitary oscillation model may demonstrate that a familiar completion maps to the interface, but no theorem in the present analysis may use its microscopic variables to establish interface necessity or completeness.*

2.5 Source roles and prohibited promotions

Source family	Status	Permitted use	Prohibited promotion
Preceding role theorem	Established target-relative result	Target, role complex, visible-mode input, lawful quotient floor.	New microscopic or numerical interpretation.
Kernel/AMetric	General construction theorem	Fixed-domain and no-selector discipline.	Direct generation of neutrino physics.
RPI	Physical role architecture	Locality, transport, gauge, vacuum, deformation, causal and record roles.	Specific dynamics, field content, or parameters.
Simultaneous constraint collapse	Shared-realization theorem architecture	Common witness system, pairwise insufficiency, variation collapse.	Interface completeness by metadata or componentwise checks alone.
Role-Occupancy Closure	Downstream realization criterion	Boundary between interface adequacy and realization.	Realization from interface membership.
SM/SYM results	Target and classification structure	Boundary typing, basis quotient, anomaly/global-form/defect relevance.	Candidate selection or unique matter representation.
Experiments	Empirical target input	Target records and visible response classes.	Candidate mechanism or microscopic ontology.
Mainstream QFT/channel theory	Mathematical support	CP maps, dilations, local support, anomaly and extended-operator vocabulary.	Automatic target relevance.

Table 2: Source status and the distinction between legitimate use and prohibited promotion.

Upstream reopening condition

If a faithful same-target completion defeats the imported nonalignment theorem, the visible-mode input, the nesting relation, or the minimal-target nonoccupancy claim, the preceding theorem must be reopened explicitly. An upstream failure cannot be repaired by redefining the interface downstream.

3 Target specification and independently derived evaluator structure

3.1 Target specification

The present analysis retains the target established in the preceding role theorem and adds only the data required to compare physical interfaces. The completion coordinate remains open. Write

$$\mathbb{T}_\nu^{(2)} = (\mathbf{f}(T_\nu^{\text{osc}}), \mathcal{B}_{\text{CC}}, \Lambda_{\text{prot}}, \mathcal{R}_{\text{emp}}, \mathfrak{M}_{\text{vis}}(C_\nu), \mathcal{P}_{\text{law}}, \mathcal{W}_{\text{known}}, \mathcal{O}_{\text{admitted}}, \mathcal{J}_{\text{one}}, \mathcal{X}_{\text{open}}). \quad (2)$$

The entries are the predecessor target fingerprint, the charged-current source/detection boundary, the protocol/environment domain, the empirical record packet, the finite visible-mode set, the lawful redescription rules, the known-substrate witness ledger, the admitted observable/operator category, the requirement that all mandatory interface witnesses belong to one physical completion, and the open completion coordinate.

The target does not contain a Hamiltonian, mass matrix, mixing matrix, state-space dimension, microscopic field inventory, or generalized-symmetry label. Those items may appear only in later candidate records.

3.2 Comparison class

Definition 3.1 (Target-relative comparison class). *The class $\mathfrak{C}_\nu$ consists of all putative completions X satisfying the following preconditions:*

1. *the same four-dimensional flat-background target, unless a controlled target fiber is declared;*
2. *the same operational charged-current source/detection semantics;*
3. *the same empirical record packet and visible-mode input;*
4. *a well-defined effective propagation family sufficient to compute every target record;*
5. *identity-preserving transport from candidate variables to event-level predictions;*
6. *explicit source, detector, medium, calibration, and protocol data;*
7. *no hidden target change through altered boundary interactions, operator category, record semantics, or background;*
8. *compatibility with the fixed electroweak/lepton substrate, or an explicitly declared richer target fiber;*
9. *sufficient regularity for lawful quotienting and shared-ledger comparison;*
10. *no appeal to fit quality, naturalness, typicality, or explanatory convenience as interface authority.*

The class is broad enough to contain unitary models, trace-preserving open-system channels, event-complete instruments with loss outcomes, reductions of larger state spaces, effective descriptions, and relevance-gated topological or defect extensions. It is not so broad that source/detection identity or the record comparison becomes undefined.

3.3 Same-target relation and target fibers

Definition 3.2 (Same-target relation). *For $X, Y \in \mathfrak{C}_\nu$, write $X \sim_{\text{tgt}}^{(2)} Y$ when the following data agree up to lawful equivalence fixed independently of the candidates:*

1. *the target fingerprint and active target-fiber identifier;*
2. *charged-current preparation and detection semantics;*
3. *empirical record and visible-mode packets;*
4. *protocol and environment domain;*
5. *admitted observable and operator category;*
6. *one-completion and simultaneous-realization semantics;*
7. *background and EFT order;*
8. *global-form identifier whenever the active operator category can detect it.*

A difference in any item is a target-fiber difference unless an explicit equivalence theorem preserves the entire target profile. Controlled fibers include source/detector interaction variants, altered global-form and line-sector targets, curved-background or explicitly nonlocal targets, EFT-order variants, and defect-completed operator categories. Edges between fibers are matching, restriction, extension, or projection maps; they do not transport standing automatically.

3.4 The raw evaluation domain

The completeness problem must be stated before the comparison profile is chosen. Let $\mathcal{R}_v^{\text{raw}}$ be the class of all well-typed raw source–propagation–detection interface presentations in one fixed target fiber. A member may contain redundant bases, gauges, Kraus lists, calibration coordinates, witness names, and candidate-internal variables. It may also be witness-split: its component ledgers need not yet admit one common physical realization. The only requirement at this stage is that every target-declared input and output type can be evaluated or can fail closed.

For every evaluation codomain V , write

$$V^\perp := V \sqcup \{\perp_{\text{eval}}\},$$

where $\perp_{\text{eval}}$ denotes failure of definedness or target typing, not a third physical value.

3.5 Target-facing trace and semantic uses before evaluators

A completeness theorem cannot begin by declaring a finite evaluator grammar complete. It must first specify, independently of that grammar, what it means for a physical use to be a use *of this target*. This is done by restricting a raw interface presentation to the complete target-facing data already present in the fixed target specification and imported role architecture.

Definition 3.3 (Target-facing trace). *For one fixed target fiber, let*

$$\mathcal{T}(\mathbb{T}_v^{(2)}) = \mathcal{T}_{\text{rec}} \times \mathcal{T}_{\text{cs}} \times \mathcal{T}_{\text{att}} \times \mathcal{T}_{\text{ext}} \times \mathcal{T}_{\text{sim}} \quad (3)$$

be the product of the following complete target-facing restrictions:

1. $\mathcal{T}_{\text{rec}}$: *the charged-current preparation/detection, protocol, empirical-record, calibration, and visible-mode data declared by $(\mathcal{B}_{\text{CC}}, \mathcal{R}_{\text{emp}}, \mathfrak{M}_{\text{vis}})$;*
2. $\mathcal{T}_{\text{cs}}$: *the complete target-declared continuation, support, localization, environment, causal-order, and lawful transport data over Λ_{prot} ;*
3. $\mathcal{T}_{\text{att}}$: *the complete known-substrate restriction of $\mathcal{W}_{\text{known}}$, including electroweak/lepton attachment, representation/charge, anomaly, vacuum/deformation, and EFT/matching witness data that the target is permitted to test;*
4. $\mathcal{T}_{\text{ext}}$: *every datum in the target-admitted operator category $\mathcal{O}_{\text{admitted}}$, with a distinguished inactive value in fibers in which no extended probe is activated;*
5. $\mathcal{T}_{\text{sim}}$: *the complete physical co-instantiation data supplied by $\mathcal{J}_{\text{one}}$: witness identity classes, target-authorized overlap maps, required commutative diagrams, and common-cone existence.*

Each factor is taken modulo the lawful presentation equivalences already fixed by $\mathcal{P}_{\text{law}}$; no interface coordinate introduced later in the paper is used in its definition. The target-facing trace map

$$\tau_{\mathbb{T}_v^{(2)}} : \mathcal{R}_v^{\text{raw}} \longrightarrow \mathcal{T}(\mathbb{T}_v^{(2)})^\perp \quad (4)$$

returns these five restrictions, or $\perp_{\text{eval}}$ if the raw presentation does not supply the target-declared types needed to form them.

The word “complete” in definition 3.3 is scope-relative rather than encyclopedic: $\tau_{\mathbb{T}_v^{(2)}}(I)$ is the full restriction of I to the data and physical witness sorts already contained in $\mathbb{T}_v^{(2)}$. Candidate-internal variables absent from those sorts are deliberately omitted. Conversely, admitting a new primitive target-facing input, physical witness sort, operator category, or measurement relation not determined by these restrictions changes the target fiber rather than silently enlarging the same target.

Definition 3.4 (Semantic target-admitted physical use). A semantic target-admitted physical use is a *totalized map*

$$U : \mathcal{R}_v^{\text{raw}} \longrightarrow W_U^\perp$$

to a target-lawful physical output quotient satisfying:

1. **same-target typing**: U is evaluated in one fixed target fiber;
2. **lawful-redescription invariance**: representative changes preserving the physical target input preserve the output up to its declared lawful equivalence;
3. **target extensionality**:

$$\tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J) \implies U(I) = U(J); \quad (5)$$

4. **standing-bearing effect**: the definedness or value of U can enter a target-relative physical judgment rather than merely a candidate-internal description;
5. **no selector authority**: candidate labels, fit ranking, familiarity, naturalness, or an expected mechanism do not define U .

The class of all such uses is denoted $\text{PhysUse}_{\text{sem}}(\mathbb{T}_v^{(2)})$. No derivation in a target-use grammar is required by this definition.

Lemma 3.5 (Target-trace descent). Every $U \in \text{PhysUse}_{\text{sem}}(\mathbb{T}_v^{(2)})$ descends uniquely to a target-lawful map

$$\bar{U} : \text{Im}(\tau_{\mathbb{T}_v^{(2)}}) \longrightarrow W_U^\perp$$

such that

$$U = \bar{U} \circ \tau_{\mathbb{T}_v^{(2)}}. \quad (6)$$

Proof. For $t \in \text{Im}(\tau_{\mathbb{T}_v^{(2)}})$, choose any I with $\tau_{\mathbb{T}_v^{(2)}}(I) = t$ and set $\bar{U}(t) := U(I)$. Target extensionality makes this independent of the chosen representative. Lawful-redescription invariance descends because the trace factors are themselves taken in the target-lawful quotients. Equation (6) is immediate, and uniqueness holds because $\tau_{\mathbb{T}_v^{(2)}}$ is surjective onto its image. $\square$

3.6 Saturated target-use grammar

The five primitive use families are now obtained as coordinate restrictions of the independently defined target trace, not as an assumed exhaustive list of physical functionals.

Definition 3.6 (Primitive target-use constructors). *Let*

$$\begin{aligned} u_{\text{rec}} &:= \pi_{\text{rec}} \circ \tau_{\mathbb{T}_v^{(2)}}, & u_{\text{cont}} &:= \pi_{\text{cont}} \circ \tau_{\mathbb{T}_v^{(2)}}, & u_{\text{att}} &:= \pi_{\text{att}} \circ \tau_{\mathbb{T}_v^{(2)}}, \\ u_{\text{ext}} &:= \pi_{\text{ext}} \circ \tau_{\mathbb{T}_v^{(2)}}, & u_{\text{sim}} &:= \pi_{\text{sim}} \circ \tau_{\mathbb{T}_v^{(2)}}. \end{aligned}$$

Their physical responsibilities are, respectively:

1. **record use** U_{rec} : source preparation, detector records, event/response comparison, calibration, and the inherited visible-mode record use;
2. **continuation/support use** U_{cont} : lawful restriction, composition, transport, localization, bounded support, environmental continuation, and causal record order;
3. **known-substrate compatibility use** U_{att} : electroweak/lepton, representation/charge, anomaly, vacuum/deformation, and EFT/matching attachment;
4. **activated extended-probe use** U_{ext} : target-activated line, surface, defect, junction, boundary, higher-group, non-invertible, or topological response;

5. **simultaneous-realization use** U_{sim} : same-witness identity, overlap maps, required commutativity, and existence of one common cone.

Their finite tuple is exactly the target-facing trace:

$$\tau_{\mathbb{T}_V^{(2)}} = \langle u_{\text{rec}}, u_{\text{cont}}, u_{\text{att}}, u_{\text{ext}}, u_{\text{sim}} \rangle \quad (7)$$

Definition 3.7 (Saturated target-use grammar). *Let $\text{Use}(\mathbb{T}_V^{(2)})$ be the class of all target-lawful maps that have a finite root normal form*

$$U = h_U \circ \langle u_{\text{rec}}, u_{\text{cont}}, u_{\text{att}}, u_{\text{ext}}, u_{\text{sim}} \rangle = h_U \circ \tau_{\mathbb{T}_V^{(2)}}, \quad (8)$$

where h_U is any lawful redescription-invariant physical map defined on $\text{Im}(\tau_{\mathbb{T}_V^{(2)}})$, valued in an existing target-lawful judgment codomain, whose value or definedness is standing-bearing at the target and whose definition uses no candidate selector. The class is therefore saturated under every lawful standing-bearing functional of the existing target trace, whether or not that functional has been separately enumerated. Familiar finite composition, tupling, restriction, transport, pullback/pushforward, calibration, matching, activation, incidence tests, and lawful postprocessing are constructive suboperations of this saturated class.

A new functional of existing trace data is not a target update: it is already included by (8). A genuine target update is required only when the proposed use needs a new primitive target-facing input, physical witness sort, operator category, or measurement relation not determined by $\mathcal{T}(\mathbb{T}_V^{(2)})$; a new numerical or logical function of existing trace data is already same-target.

Theorem 3.8 (Semantic target-use factorization and exhaustion). *Within every fixed target fiber,*

$$\boxed{\text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)}) = \text{Use}(\mathbb{T}_V^{(2)})}. \quad (9)$$

Equivalently, every same-target standing-bearing physical use has the normal form (8). In particular, an unenumerated lawful functional of already admitted target data cannot escape the target-use grammar merely because it is a new formula.

Proof. Let $U \in \text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)})$. By lemma 3.5, $U = \bar{U} \circ \tau_{\mathbb{T}_V^{(2)}}$ for a unique target-lawful $\bar{U}$. Since $\tau_{\mathbb{T}_V^{(2)}}$ is the finite tuple of the five primitive target restrictions by equation (7), this is exactly a normal form admitted by Definition 3.7. Hence $\text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)}) \subseteq \text{Use}(\mathbb{T}_V^{(2)})$.

Conversely, every map in $\text{Use}(\mathbb{T}_V^{(2)})$ is, by construction, a lawful, standing-bearing, selector-free map of the fixed target trace. It is therefore same-target typed, lawful-redescription invariant, and target-extensional, so $\text{Use}(\mathbb{T}_V^{(2)}) \subseteq \text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)})$. Thus equality holds.

The potentially problematic case of a new functional $U_\star$ acting only on existing target types is included automatically: target extensionality gives $U_\star = \bar{U}_\star \circ \tau_{\mathbb{T}_V^{(2)}}$, so $U_\star \in \text{Use}(\mathbb{T}_V^{(2)})$. If no such factorization exists, then two presentations with the same complete target trace receive different outputs; the proposed operation depends on data outside the fixed target and is not a same-target use at this scope. $\square$

Remark 3.9 (Independence of the completeness basis). *Definitions 3.3 and 3.4, lemma 3.5, and theorem 3.8 are stated before $\mathcal{P}_V^0$, $\mathcal{J}_V^{\text{sim}}$, the 23-coordinate interface packet, or any candidate universe is defined. The semantic class $\text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)})$ is defined without grammar membership; grammar completeness is a consequence of target-trace descent rather than a premise hidden in the definition of a use.*

3.7 Target-admitted physical evaluators

Definition 3.10 (Target-admitted physical evaluator). *A target-admitted physical evaluator is a totalized map*

$$E : \mathcal{R}_V^{\text{raw}} \longrightarrow V_E^\perp$$

satisfying all of the following:

1. **same-target typing:** E is evaluated within one fixed target fiber and is covariant under declared target-fiber maps;
2. **lawful-redescription invariance:** representative changes preserving the physical target input leave the output unchanged up to its declared lawful equivalence;

3. **target extensionality:**

$$\tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J) \implies E(I) = E(J);$$

4. **standing-bearing output:** the value or definedness of E is a target-relative physical judgment or enters one by a target-lawful output map;
5. **fail-closed typing:** absence of a required target input returns $\perp_{\text{eval}}$ rather than a repaired value;
6. **no selector authority:** candidate labels, fit ranking, naturalness, familiarity, and expected mechanism do not form evaluators.

The class of all such evaluators is denoted $\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$. No target-use derivation or evaluator-grammar derivation is part of this definition.

The evaluator class is therefore semantic and profile-independent. Every $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$ is a semantic target use with a physical judgment codomain, so lemma 3.5 and theorem 3.8 apply to it before any interface coordinate is introduced.

3.8 Primitive evaluator bearers

The five target-trace factors have a target-fixed refinement into nine physical output sorts:

$$\mathcal{T}(\mathbb{T}_v^{(2)}) \cong \mathcal{T}_{\text{rec}} \times \mathcal{T}_{\text{cont}} \times \mathcal{T}_{\text{loc}} \times \mathcal{T}_{\text{gauge}} \times \mathcal{T}_{\text{anom}} \times \mathcal{T}_{\text{vac}} \times \mathcal{T}_{\text{EFT}} \times \mathcal{T}_{\text{ext}} \times \mathcal{T}_{\text{sim}}. \quad (10)$$

Here $\mathcal{T}_{\text{cs}} \cong \mathcal{T}_{\text{cont}} \times \mathcal{T}_{\text{loc}}$ refines the continuation/support factor, while $\mathcal{T}_{\text{att}} \cong \mathcal{T}_{\text{gauge}} \times \mathcal{T}_{\text{anom}} \times \mathcal{T}_{\text{vac}} \times \mathcal{T}_{\text{EFT}}$ refines the known-substrate attachment factor. This is a typing decomposition of the target trace, not an interface hypothesis.

Define

$$\text{Gen}(\mathbb{T}_v^{(2)}) = \mathbf{G}_{\text{rec}} \sqcup \mathbf{G}_{\text{cont}} \sqcup \mathbf{G}_{\text{loc}} \sqcup \mathbf{G}_{\text{gauge}} \sqcup \mathbf{G}_{\text{anom}} \sqcup \mathbf{G}_{\text{vac}} \sqcup \mathbf{G}_{\text{EFT}} \sqcup \mathbf{G}_{\text{ext}} \sqcup \mathbf{G}_{\text{sim}}. \quad (11)$$

For every factor in (10), the corresponding family contains its target-trace coordinate projection and every target-lawful one-factor physical evaluation of that projection. Thus the nine coordinate projections themselves jointly reconstruct $\tau_{\mathbb{T}_v^{(2)}}$.

Record evaluators $\mathbf{G}_{\text{rec}}$. Source preparation, detector effects, event probabilities, event-response functionals, charged-current record tags, lawful calibration maps, and the inherited visible-mode record operations.

Continuation evaluators $\mathbf{G}_{\text{cont}}$. Baseline/energy/path dependence, channel composition, medium continuation, protocol restriction, source-to-detection transport, and matching of continuation histories.

Support evaluators $\mathbf{G}_{\text{loc}}$. Observable support, localization, bounded transport, causal record order, and tests excluding pathless target-visible influence.

Gauge-attachment evaluators $\mathbf{G}_{\text{gauge}}$. Attachment to the fixed charged-current electroweak boundary, representation/charge response, quotient-stable gauge behavior, and target-visible global-form attachment when activated.

Quantum-consistency evaluators $\mathbf{G}_{\text{anom}}$. Local, global, mixed, and background anomaly tests, including declared inflow or compensating-sector maps.

Vacuum/deformation evaluators G_{vac} . Compatibility with the non-null electroweak vacuum baseline and the response to admitted deformations.

Scale/matching evaluators G_{EFT} . EFT order, scale interval, threshold, scheme quotient, physical running, and UV/effective matching.

Activated extension evaluators G_{ext} . Line, surface, higher-group, defect, junction, non-invertible, boundary, and topological responses only in target fibers whose operator category or consistency witnesses activate them.

Simultaneous-incidence evaluators G_{sim} . Same-witness identity, restriction/projection/transport/matching incidence, commutativity of required overlap diagrams, and existence of a common cone over the mandatory witness packet.

Table 3: Primitive evaluator families obtained by refining the independently defined target-facing trace.

Family	Target source	Physical output type
G_{rec}	$\mathcal{B}_{\text{CC}}, \mathcal{R}_{\text{emp}}, \mathfrak{M}_{\text{vis}}(C_{\nu})$	Source/detection records, event responses, calibration classes, and visible-mode operations.
G_{cont}	$\Lambda_{\text{prot}}, \mathcal{P}_{\text{law}}$	Protocol, environment, path, composition, and matching continuation.
G_{loc}	RPI locality/transport roles	Support, localization, bounded transport, causal order, and no-pathless-influence tests.
G_{gauge}	$\mathcal{W}_{\text{known}}$: electroweak boundary	Gauge, representation, charge, quotient, and activated global-form attachment.
G_{anom}	$\mathcal{W}_{\text{known}}$: anomaly ledger	Local, global, mixed, background, inflow, and compensating-sector consistency.
G_{vac}	$\mathcal{W}_{\text{known}}$: vacuum/deformation ledger	Vacuum support and lawful deformation response.
G_{EFT}	$\mathcal{W}_{\text{known}}$: scale/matching ledger	EFT order, scale, threshold, scheme quotient, running, and matching.
G_{ext}	$\mathcal{O}_{\text{admitted}}$	Activated line, surface, defect, junction, higher-group, non-invertible, boundary, and topological probes.
G_{sim}	$\mathcal{I}_{\text{one}}, \mathcal{X}_{\text{open}}$	Same-witness incidence, overlap maps, commutativity, and common-cone existence.

Theorem 3.11 (Target-evaluator bearer factorization). *Every $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$ factors through the finite tuple of the nine target-trace projections in equation (10). Equivalently, there is a target-lawful map h_E such that*

$$E = h_E \circ (\tau_{\text{rec}}, \tau_{\text{cont}}, \tau_{\text{loc}}, \tau_{\text{gauge}}, \tau_{\text{anom}}, \tau_{\text{vac}}, \tau_{\text{EFT}}, \tau_{\text{ext}}, \tau_{\text{sim}}). \quad (12)$$

A genuinely one-factor primitive evaluator has a unique bearer family after lawful output recoding; a mixed evaluator may depend on several families through h_E .

Proof. By target extensionality and lemma 3.5, $E = \overline{E} \circ \tau_{\mathbb{T}_{\nu}^{(2)}}$. Substitute the target-fixed refinement (10) of $\tau_{\mathbb{T}_{\nu}^{(2)}}$ and take $h_E := \overline{E}$ on the corresponding product. Each factor projection belongs to the stated primitive family by construction. If an evaluator depends on exactly one refined target factor, the disjoint output typing fixes that bearer family; otherwise the dependence is explicitly mixed and is represented by the postprocessing map h_E . No finite grammar derivation is assumed. $\square$

3.9 Saturated evaluator algebra

Definition 3.12 (Target-evaluator grammar). *Let $\text{Eval}(\mathbb{T}_v^{(2)})$ be the saturated multi-sorted evaluator algebra consisting of all totalized maps*

$$E = h \circ (g_1, \dots, g_n), \quad n < \infty, \quad (13)$$

where each $g_i \in \text{Gen}(\mathbb{T}_v^{(2)})$ and h is any target-lawful redescription-invariant physical map on the corresponding existing output quotients. In particular, h need not have been enumerated in advance. The familiar operations of finite products/projections, target restriction, pullback/pushforward, calibration, transport, matching, finite logical incidence tests, and finite typed composition form an explicit constructive subalgebra. All operations propagate $\perp_{\text{eval}}$ fail closed.

Theorem 3.13 (Target-evaluator generation theorem). *Within the fixed target fiber,*

$$\boxed{\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) = \text{Eval}(\mathbb{T}_v^{(2)})} \quad (14)$$

up to lawful output equivalence. Hence every semantic target-admitted physical evaluator has the finite-root normal form (12); an unenumerated lawful functional of existing target data is already represented by the postprocessing map h_E .

Proof. Let $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$. By theorem 3.11, E has form (12). The nine factor projections are members of $\text{Gen}(\mathbb{T}_v^{(2)})$, and h_E is target-lawful, so $E \in \text{Eval}(\mathbb{T}_v^{(2)})$. Thus $\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) \subseteq \text{Eval}(\mathbb{T}_v^{(2)})$.

Conversely, every generator in $\text{Gen}(\mathbb{T}_v^{(2)})$ is a lawful evaluation of one target-trace factor and is therefore target-extensional. A target-lawful postprocessing of finitely many target-extensional evaluators remains target-extensional and lawful-redescription invariant, with fail-closed typing preserved. Restricting to standing-bearing physical outputs gives $\text{Eval}(\mathbb{T}_v^{(2)}) \subseteq \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$. Therefore the classes agree. $\square$

Definition 3.14 (Target-semantic equivalence). *For $I, J \in \mathcal{R}_v^{\text{raw}}$ in the same target fiber, define*

$$I \equiv_{\mathbb{T}_v^{(2)}} J \iff E(I) = E(J) \text{ for every } E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}), \quad (15)$$

where each output is compared in its target-lawful quotient and equality includes common definedness or a common fail-closed value. By theorem 3.13, the same relation is obtained by quantifying over $\text{Eval}(\mathbb{T}_v^{(2)})$.

Theorem 3.15 (Target-semantic equivalence relation). *The relation $\equiv_{\mathbb{T}_v^{(2)}}$ is an equivalence relation on $\mathcal{R}_v^{\text{raw}}$ within every fixed target fiber. It is fixed independently of the interface coordinate list and independently of evaluator-grammar membership.*

Proof. Pointwise equality of all semantic target-admitted evaluator outputs is reflexive, symmetric, and transitive. The evaluator class is defined by target typing, redescription invariance, target-trace extensionality, physical standing, and fail-closed behavior before any profile or candidate value is chosen. Hence the relation is candidate-independent and profile-independent. $\square$

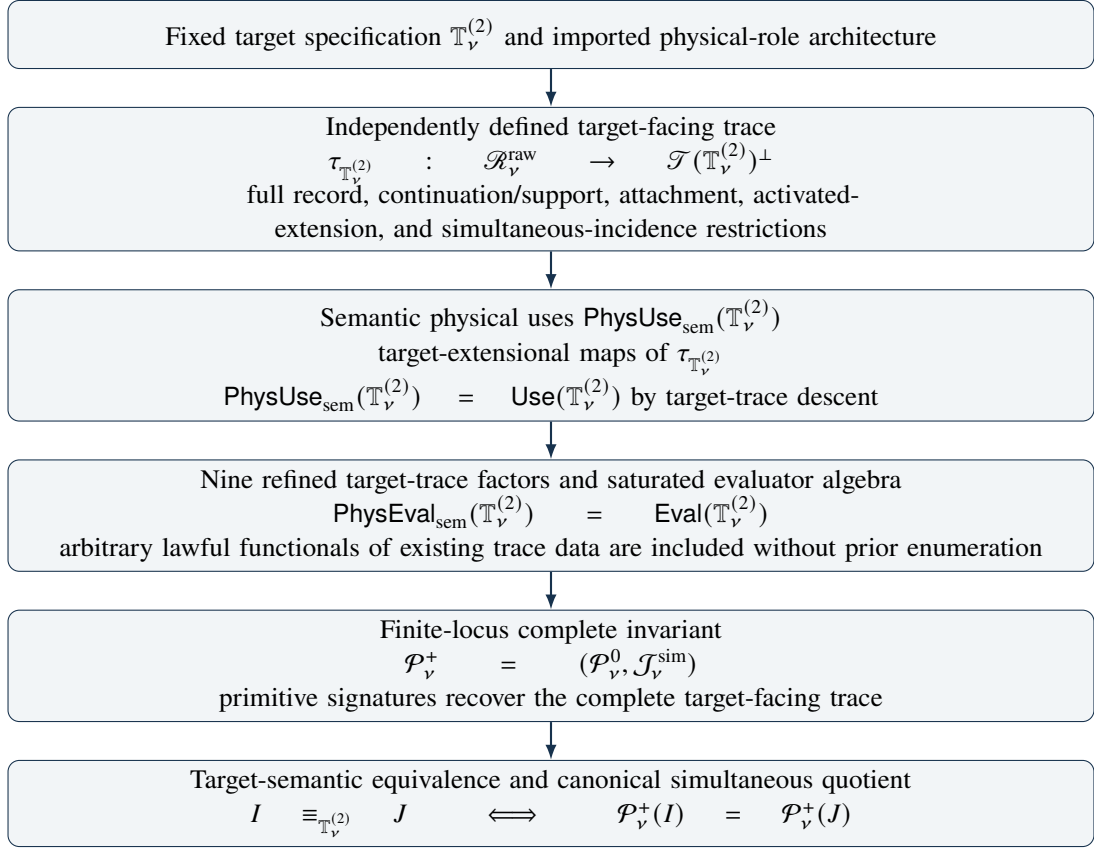


Figure 1: Semantic completeness is grounded upstream of the interface profile. The target-facing trace is defined directly from the fixed target before the use grammar, evaluator algebra, profile, or microscopic completion is introduced. Semantic uses and evaluators are lawful and extensional with respect to this trace; target-trace descent yields the saturated grammars. Because the primitive signature families contain the refined trace projections, $\mathcal{P}_\nu^+$ recovers the trace and is therefore a complete invariant for the semantic evaluator class.

3.10 Evaluator-signature component profile

The component profile is now constructed from the independently derived evaluator grammar rather than postulated as a complete list. For each primitive family $\mathbf{G}_\ell$, define its lawful output signature

$$\Sigma_\ell(I) := [(g(I))_{g \in \mathbf{G}_\ell}]_{\cong_\ell}, \quad (16)$$

where $\cong_\ell$ identifies only target-authorized output recodings and representative changes. The target fixes the index families and output equivalences; a candidate supplies only their values.

Define

$$\mathcal{P}_\nu^0(I) = (P_{\text{rec}}, C_{\text{cont}}, S_{\text{loc}}, G_{\text{attach}}, A_{\text{q}}, V_{\text{def}}, R_{\text{EFT}}, X_{\text{ext}})_I, \quad (17)$$

by the identifications

$$\begin{aligned} P_{\text{rec}} &:= \Sigma_{\text{rec}}, & C_{\text{cont}} &:= \Sigma_{\text{cont}}, & S_{\text{loc}} &:= \Sigma_{\text{loc}}, & G_{\text{attach}} &:= \Sigma_{\text{gauge}}, \\ A_{\text{q}} &:= \Sigma_{\text{anom}}, & V_{\text{def}} &:= \Sigma_{\text{vac}}, & R_{\text{EFT}} &:= \Sigma_{\text{EFT}}, & X_{\text{ext}} &:= \Sigma_{\text{ext}}. \end{aligned}$$

Thus each component is the complete lawful signature of one independently derived target-evaluator locus.

Record profile P_{rec} . The complete family of target-admitted source-to-detection probability and event-response evaluations, including protocol, environment, calibration, charged-current record use, and visible-mode operations.

Continuation profile C_{cont} . The lawful dependence of the interface on baseline, energy, path composition, medium state, source/detector localization, and allowed continuation of the protocol.

Support profile S_{loc} . Observable support, bounded transport, causal ordering, and no-pathless-influence evaluations inherited from the realized physical interior.

Gauge-attachment profile G_{attach} . The complete target-evaluator signature for attachment to the fixed electroweak charged-current boundary, including quotient-stable representation and charge witnesses.

Quantum-consistency profile A_q . The local gauge, global, mixed, and background anomaly evaluations relevant to the active target fiber, including declared inflow or compensating sectors.

Vacuum/deformation profile V_{def} . The target-evaluator signature of compatibility with the non-null electroweak vacuum baseline and lawful behavior under admitted perturbations.

Scale and matching profile R_{EFT} . The scale interval, operator order, matching maps, scheme quotient, physical running, and threshold evaluations required to compare UV and effective descriptions.

Extended-sector profile X_{ext} . Line, surface, higher-group, defect, junction, non-invertible, boundary, and topological evaluations only when activated by the target operator category or a mandatory consistency witness.

The component profile is not a complete theory profile: two UV completions may differ outside the target evaluator algebra and still map to the same target-relative interface.

3.11 Physical joint-incidence profile

Definition 3.16 (Physical witness-overlap diagram). *For a raw interface presentation I , let $\mathcal{D}_{\text{sim}}(I)$ be the typed diagram whose nodes are the lawful physical witness classes carried by*

$$W_{\text{tgt}}, W_{\text{CC}}, W_{\text{ch}}, W_{\text{rec}}, W_{\text{loc}}, W_{\text{env}}, W_{\text{gauge}}, W_{\text{anom}}, W_{\text{vac}}, W_{\text{def}}, W_{\text{EFT}}, W_{\text{ext}}, W_{\text{obs}},$$

and whose arrows record only target-admitted physical incidence: same-witness identity, restriction, projection, transport, matching, support, boundary attachment, anomaly evaluation, vacuum/deformation continuation, and operator-category inclusion. The diagram also records which required subdiagrams commute and whether a single globally compatible cone exists over all mandatory coordinates.

Two such diagrams are physically isomorphic when there is a type-preserving diagram isomorphism preserving the target fiber, lawful witness classes, arrow types, required commutative subdiagrams, and cone-existence status. Raw database IDs, hashes, file paths, provenance labels, and other certification metadata are forgotten.

Definition 3.17 (Joint-incidence profile). *The physical joint-incidence profile is both the simultaneous-evaluator signature and the physical diagram class:*

$$\mathcal{J}_v^{\text{sim}}(I) := \Sigma_{\text{sim}}(I) \cong [\mathcal{D}_{\text{sim}}(I)]_{\cong_{\text{phys}}}. \quad (18)$$

It records which interface coordinates are borne by the same physical witness classes, how their overlap maps compose, and whether the mandatory coordinate packet has one simultaneous physical realization. It does not record witness names or archival identities.

The isomorphism in equation (18) holds because G_{sim} contains exactly the same-witness, overlap-map, commutativity, and common-cone evaluations of the target incidence grammar. The node and arrow vocabulary is fixed by the target witness types before candidate inspection.

Definition 3.18 (Complete comparison profile). *The complete comparison object is*

$$\mathcal{P}_v^+(I) := (\mathcal{P}_v^0(I), \mathcal{J}_v^{\text{sim}}(I)). \quad (19)$$

It is the finite-locus typed normal form of the primitive target-evaluator signatures; each locus may contain an arbitrarily rich target-authorized evaluator family. Equality of the component values without equality of joint incidence is not physical interface equality.

Proposition 3.19 (Profile formation is candidate-independent). *The evaluator families, grammar operations, output equivalences, and component grouping defining $\mathcal{P}_v^+$ are fixed by $\mathbb{T}_v^{(2)}$ before candidate inspection. A candidate supplies evaluator values but cannot add, delete, merge, or retype an evaluator locus.*

Proof. The generator families arise from the target’s operational record, continuation/support, substrate-attachment, activated-operator, and simultaneous-incidence witness types by theorem 3.11. Their closure operations are those of definition 3.12. The profile is then the corresponding family of generator signatures. None depends on a candidate mechanism or survivor label. A newly required witness type therefore reopens the target grammar rather than modifying the profile inside one candidate evaluation. $\square$

Proposition 3.20 (Joint incidence is physical interface data). *Changing $\mathcal{J}_v^{\text{sim}}$ while holding $\mathcal{P}_v^0$ fixed can change whether the coordinate packet describes one physical interface. Hence $\mathcal{J}_v^{\text{sim}}$ is neither archival metadata nor representational skin; it is a carrier-neutral physical interface coordinate.*

Proof. Take three component packets whose numerical and algebraic values are fixed. In one presentation they are all evaluated on one environment, representation/charge, and global-form witness tuple. In another, the channel uses (e_0, q_0) , the anomaly ledger closes only on (q_1, Γ_0) , and the defect packet is defined only on (e_1, Γ_1) , with no lawful identifications $e_0 \simeq e_1$ or $q_0 \simeq q_1$. The component values can be copied unchanged, but only the first presentation admits a common physical cone. Relabeling witness IDs leaves the diagram isomorphism class fixed; changing the physical overlap changes it. Thus the retained difference is simultaneous physical incidence, not certification metadata. $\square$

3.12 Candidate-independent coordinate admission

Definition 3.21 (Coordinate-admission test). *A proposed mandatory interface coordinate c is admitted only if:*

1. **uniform typing:** $c(X)$ is defined for every X in the relevant comparison class or in a declared relevance fiber;
2. **physical witness:** changing c changes the value of some evaluator in $\text{Eval}(\mathbb{T}_v^{(2)})$ or destroys target adequacy;
3. **lawful quotient:** all presentation variants of c are identified by an explicit witness-preserving relation;
4. **carrier neutrality:** the definition requires no mass basis, field species, representation, defect label, or preferred mechanism;
5. **shared-ledger compatibility:** c can be evaluated over the same typed witness ledger as the other interface coordinates.

Theorem 3.22 (Coordinate-admission theorem). *A coordinate belongs to the mandatory core interface only if it passes all five clauses of definition 3.21. If uniform typing or physical witness fails, the coordinate is excluded from the core. If it is meaningful only in a controlled target fiber, it is relevance-gated. If lawful quotient or shared-ledger compatibility remains unproved, the strongest interface-completeness theorem is unavailable.*

Proof. A coordinate that is not uniformly typed cannot compare every object in the declared class and therefore cannot be mandatory. A coordinate whose variation changes no target evaluator performs no target-interface work and is skin or candidate-internal data. A coordinate with no lawful quotient either multiplies physical classes by representation or erases genuine differences without control. A candidate-dependent coordinate defines the target through one proposed occupant. Finally, a coordinate that cannot coexist on one witness ledger with the remaining packet cannot be an attribute of one physical interface. These cases exhaust the ways a proposed coordinate can fail the stated function. $\square$

3.13 Fixed anchors and open coordinates

Class	Fixed by the present target	Open in microscopic completion
Boundary	Interaction-defined charged-current source/detection semantics, record labels, and lawful calibration maps.	Microscopic source/detector completion in explicit target variants.
Propagation	Event-complete channel adequacy, cross-boundary transfer, visible-mode coverage, protocol/environment dependence.	Hamiltonian, Lindbladian, non-Hermitian, geometric, topological, or other mechanism.
Support	Observable localization, causal order, and bounded continuation.	Microscopic mediator, field content, or local dilation.
Gauge/consistency	Known electroweak boundary and required anomaly/vacuum compatibility.	New representations, counterterms, inflow, or UV sectors.
Joint incidence	Physical overlap diagram, common-witness requirement, and simultaneous-cone status.	Microscopic ontology of the common witnesses.
Generalized symmetry	Operator-category and target-fiber policy.	Optional global-form, line, higher-group, or non-invertible structure unless relevance is activated.
Numerical	Established empirical records and lawful visible-mode classes.	Masses, mixing coordinates, ordering, couplings, absolute scale, and new observables.

Table 4: Fixed anchors and open completion coordinates.

3.14 Target stability

Theorem 3.23 (Target stability under lawful comparison). *If $X \sim_{\text{tgt}}^{(2)} Y$, every lawful comparison between X and Y preserves the target specification $\mathbb{T}_v^{(2)}$. Conversely, a transformation that changes a target-coordinate entry without an explicit target-fiber map is not a lawful same-target redescription.*

Proof. The target coordinates are anchors of the comparison relation. A lawful arrow preserves their operational and record-bearing content. If an arrow changes a boundary tag, record family, operator category, background, or active global-form fiber, the source and target are different comparison objects. Calling the transformation same-target would erase an anchor difference and permit target repair after failure, contrary to the fixed-domain and irreversibility disciplines imported in section 2. $\square$

Breadth without target dissolution

The comparison class is intentionally completion-neutral, not unconstrained. A proposal that abolishes interaction-defined preparation/detection classes, causal record order, or a calculable source-to-detection response is not a more general occupant of the same target. It declines to instantiate the target interface and belongs outside $\mathfrak{C}_v$.

4 Interface presentations and lawful redescription

4.1 Raw interface object

For each $X \in \mathfrak{C}_\nu$, define the raw interface presentation

$$\tilde{\mathfrak{I}}_\nu(X) = (I_{\text{tgt}}, I_{\text{CC}}, I_{\text{ch}}, I_{\text{rec}}, I_{\text{loc}}, I_{\text{env}}, I_{\text{gauge}}, I_{\text{anom}}, I_{\text{vac}}, I_{\text{EFT}}, I_{\text{ext}}, I_{\text{sim}})_X. \quad (20)$$

The entries contain a target identifier; source/detection boundary data; an effective channel or instrument presentation; event-level and visible-mode records; localization and support data; medium/environment dependence; electroweak attachment; anomaly and quantum-consistency status; vacuum and deformation data; scale and matching data; any activated extended-operator packet; and the typed physical witness-incidence diagram joining those entries.

A raw presentation is deliberately redundant. It may include bases, phases, Kraus operators, dilations, gauges, calibration coordinates, and notation whose variation changes no physical witness. More importantly, a raw tuple may combine separately meaningful coordinates that do not belong to one common physical witness system. Such a tuple is not yet a physical interface.

4.2 Constructive lawful generators

Definition 4.1 (Constructive lawful-redescription groupoid). *The groupoid $\mathcal{G}_\nu^{\text{gen}}$ has raw interface presentations as objects. An arrow*

$$g : I \longrightarrow I'$$

is a finite composition of declared, explicitly realized transformations preserving the fixed target fiber and every target-admitted evaluator in $\text{Eval}(\mathbb{T}_\nu^{(2)})$. Constructive generator families include:

1. *simultaneous unitary or isometric changes of state and effect representation;*
2. *weak-basis and rephasing transformations preserving the charged-current boundary and every event-level kernel;*
3. *equivalent Kraus decompositions of the same completely positive map;*
4. *equivalent Stinespring dilations when no target-admitted evaluator distinguishes the dilation sector;*
5. *gauge transformations preserving the quotient-stable boundary and attachment ledgers;*
6. *source-authorized calibration and analysis-coordinate transformations preserving physical event records;*
7. *type-preserving witness recodings that induce a physical isomorphism of the witness-overlap diagram.*

The list is constructive, not asserted to be generator-complete for every mathematically possible presentation.

Theorem 4.2 (Constructive-generator soundness). *Every arrow of $\mathcal{G}_\nu^{\text{gen}}$ preserves the output of every evaluator in $\text{Eval}(\mathbb{T}_\nu^{(2)})$. Hence constructive orbit equivalence refines target-semantic equivalence $\equiv_{\mathbb{T}_\nu^{(2)}}$, and therefore also preserves $\mathcal{P}_\nu^+$.*

Proof. Each generator is admitted only after the target operations capable of detecting its action are shown to be unchanged: boundary and record evaluations, continuation and support evaluations, substrate-attachment and consistency evaluations, activated extension evaluations, and simultaneous-incidence evaluations. Those are the primitive families of equation (11). The evaluator grammar is closed under composition, restriction, matching, lawful postcomposition, and finite logical combination, so preservation of the generators implies preservation of every term in $\text{Eval}(\mathbb{T}_\nu^{(2)})$. The final profile-preservation claim follows because its entries are the primitive evaluator signatures defined in equations (16) to (18). $\square$

4.3 Semantic and profile prequotient relations

Definition 4.3 (Complete-profile equivalence). *For raw interface presentations I, J in the same fixed target fiber, write*

$$I \equiv_{\mathcal{P}^+} J \iff \mathcal{P}_v^+(I) = \mathcal{P}_v^+(J).$$

This relation is a finite-locus typed compression of the independently derived target-evaluator signature. Its equality with target-semantic equivalence is proved, not assumed, in theorem 13.3.

Definition 4.4 (Saturated lawful-redescription groupoid). *The relation groupoid $\mathcal{G}_v^{\text{law, sat, sim}}$ is the equivalence-relation groupoid of target-semantic equivalence $\equiv_{\mathbb{T}_v^{(2)}}$ inside a fixed target fiber. The constructive groupoid $\mathcal{G}_v^{\text{gen}}$ maps into $\mathcal{G}_v^{\text{law, sat, sim}}$ by theorem 4.2.*

The lawful groupoid is therefore fixed by the complete target-evaluator algebra, not by the enumerated generator library and not by the interface profile. The raw coordinate product is first restricted to the shared-ledger pullback and fail-closed survivor subobjects in section 12; only then is the semantic quotient formed in section 13. This ordering prevents a semantic class from conferring standing on a witness-split tuple.

Theorem 4.5 (Quotient-ordering principle). *No raw presentation outside the shared-ledger pullback acquires a canonical physical interface class merely because it is $\equiv_{\mathbb{T}_v^{(2)}}$ -equivalent to another raw presentation. The physical quotient is formed on the simultaneous subobject, and the admissible quotient on the survivor subobject.*

Proof. The raw product permits mutually incompatible coordinate ledgers. Quotienting that product before testing simultaneous incidence would allow a nonrealizable tuple to inherit the class of a realizable tuple if simultaneous-realizability status were not congruent. The target evaluator grammar includes the physical joint-incidence evaluators, and the complete profile records their signature, but the canonical physical domain is still the pullback subobject whose elements possess the required common witness realization. Restriction before quotient preserves the logical order: simultaneous physical object first, semantic presentation class second. $\square$

4.4 Core and extension packets

The provisional core representative associated with X is

$$\mathfrak{J}_v^{\text{core}}(X) = (\mathcal{B}_{\text{CC}}, [\mathcal{I}_\Lambda]_{\text{ch}}, \mathcal{K}_{\text{CC}}, \mathfrak{M}_{\text{vis}}(X), \mathcal{N}_{\Delta \bowtie}(X), \mathcal{L}_{\text{tr}}(X), \mathcal{E}_{\text{env}}(X), \mathcal{G}_{\text{attach}}(X), \mathcal{A}_{\text{q}}(X), \mathcal{V}_{\text{def}}(X), \mathcal{R}_{\text{EFT}}(X), \mathcal{J}_v^{\text{sim}}(X)). \quad (21)$$

Here $[\mathcal{I}_\Lambda]_{\text{ch}}$ is the class of an event-complete channel/instrument family, $\mathcal{K}_{\text{CC}}$ is the charged-current transition-kernel family, $\mathcal{N}_{\Delta \bowtie}$ records the strict nested relation between visible response multiplicity and nonalignment, and $\mathcal{J}_v^{\text{sim}}$ records the physical co-instantiation of all packet coordinates.

The extension packet is

$$\mathfrak{J}_v^{\text{ext}}(X) = (\Gamma_{\text{glob}}, \mathcal{L}_{\text{line}}, \mathcal{H}_{\text{higher}}, \mathcal{D}_{\text{NI}}, \mathcal{B}_{\text{bdy}}, \mathcal{T}_{\text{top}})_X, \quad (22)$$

with global-form, line/one-form, higher-group, non-invertible defect, boundary/edge, and topological-sector data. The entries are activated only by the relevance theorems of section 11.

4.5 Coordinate specifications

Definition 4.6 (Interface-coordinate specification). *Every retained coordinate c is accompanied by the specification*

$$\text{Cert}_c = (\text{type}_c, \text{source}_c, \text{witness}_c, \text{quotient}_c, \text{overlap}_c, \text{variation}_c, \text{scope}_c, \text{status}_c).$$

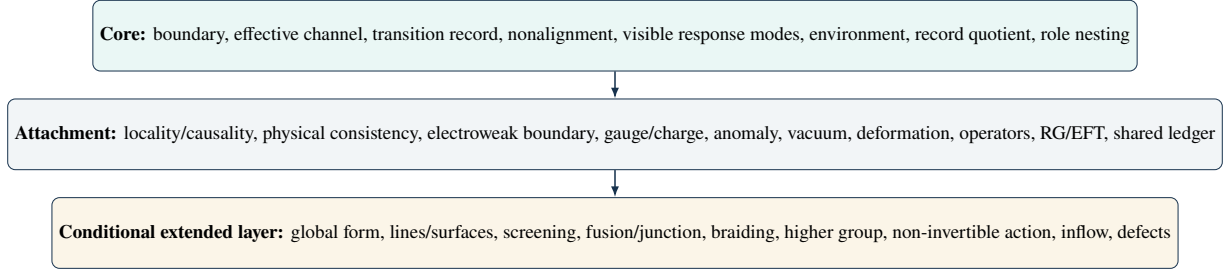


Figure 2: The carrier-neutral interface is layered by universal core data, universal attachment gates, and conditionally activated extended data.

The fields record its mathematical type, source theorem or target anchor, physical witness, lawful quotient, shared-ledger overlap keys, variation/deletion test, target scope, and proof status.

This specification makes the mathematical type, physical witness, quotient, overlap conditions, scope, and proof status explicit. It is expository bookkeeping and does not substitute for the proof that the coordinate is necessary or complete.

4.6 Lawful covariance before quotient

Theorem 4.7 (Lawful profile covariance). *Every arrow $g : I \rightarrow I'$ in $\mathcal{G}_v^{\text{law}, \text{sat}, \text{sim}}$ preserves every evaluator in $\text{Eval}(\mathbb{T}_v^{(2)})$ and therefore preserves $\mathcal{P}_v^+$; every arrow in $\mathcal{G}_v^{\text{gen}}$ is lawful in that saturated semantic sense. Each retained coordinate descends after its declared coordinate-specific quotient. The canonical physical quotient is nevertheless deferred until pullback and survivor congruence have been proved.*

Proof. An arrow of $\mathcal{G}_v^{\text{law}, \text{sat}, \text{sim}}$ relates presentations that are equal under every target evaluator by definitions 3.14 and 4.4. In particular, all primitive evaluator outputs agree, so their lawful signatures $\mathcal{P}_v^0$ and $\mathcal{J}_v^{\text{sim}}$ agree. Constructive-arrow soundness is theorem 4.2. Coordinate admission requires a target-admitted physical witness and an explicit lawful quotient. The final clause is theorem 4.5: covariance does not turn a witness-split raw tuple into one simultaneous physical interface. $\square$

Corollary 4.8 (No representative multiplication). *Equivalent bases, phases, gauges, Kraus lists, witness encodings, and source-authorized calibration coordinates do not produce distinct canonical interface classes once the presentations lie in the simultaneous domain.*

4.7 Examples of lawful and unlawful changes

Change	Lawful when	Unlawful or target-changing when
Kraus representation	The completely positive map and all admitted environment/dilation witnesses are unchanged.	A different dilation sector is physically observed.
Stinespring dilation	Reduced channel and every active witness and overlap map are intertwined.	Hidden-sector leakage, charge, topology, support, or joint incidence is admitted.
Weak basis/rephasing	Boundary tags, kernels, anomaly, record, and witness-incidence profiles are preserved.	Source/detection interactions or physical invariant phases change.
Gauge fixing	Gauge-invariant boundary and attachment data are unchanged.	A boundary charge or global sector changes.
Calibration coordinate	Source-authorized map preserves physical response records and their incidence.	It identifies distinct mode classes or changes event predictions without source authority.
EFT field redefinition	Full matched interface, common witness diagram, and target fiber are preserved.	Only a finite subset of records matches or operator order changes.

Table 5: Representative transformations and their scope.

4.8 Interface adequacy as a separate predicate

Definition 4.9 (Interface adequacy). *For $X \in \mathfrak{C}_v$, $\text{InterfaceAdeq}(X, T_v^{\text{osc}})$ holds when:*

1. *the target specification is preserved;*
2. *every target record is reproduced through the fixed source-analysis maps;*
3. *the nested role complex is occupied at interface level;*
4. *every mandatory transport, gauge, consistency, vacuum, deformation, and EFT constraint is satisfied;*
5. *every activated extension-fiber constraint is satisfied;*
6. *one shared physical witness diagram exists, all mandatory overlaps commute, and every active coordinate factors through it;*
7. *no selector, hidden target change, or after-the-fact repair is required.*

The predicate is defined before the canonical admissible quotient is characterized. The adequacy biconditional proved later is therefore not merely a definition repeated in theorem form; it is a factorization theorem connecting an independently stated physical predicate to the simultaneous survivor quotient.

No dilation overquotient

Operational equality of reduced channels is sufficient for interface equivalence only when the complete comparison profile contains no component or joint-incidence witness of the dilation sector. If a candidate carries a hidden-sector charge, line operator, decay product, causal support, or other admitted witness, the dilation data become an active extension coordinate and may distinguish the interfaces.

5 Charged-current boundary reconstruction

5.1 Operational rather than propagation-defined flavor

The source and detector labels used in neutrino experiments are fixed by interaction and record practice. They are not defined by a propagation eigenbasis. This distinction is essential: if the propagation model

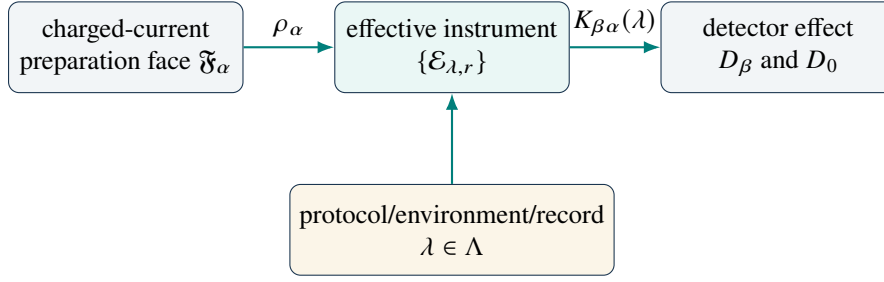


Figure 3: The core operational interface. Source/detection semantics are fixed; propagation is represented by the complete record-relevant instrument family.

were allowed to define the boundary classes, nonalignment could be removed by choosing a basis in which propagation is diagonal.

Let $\mathcal{A}_{\text{tag}}$ be the finite operational algebra generated by charged-current source and detector tags. For each admitted label α , let $\mathbf{P}_\alpha$ be the preparation class determined by the source interaction and record protocol, and let $\mathbf{F}_\alpha$ be the corresponding detector-effect class. At a concrete operator-system presentation one may use source states ρ_α and effects F_α , but the boundary object is their operational equivalence class.

Definition 5.1 (Charged-current boundary object). *The charged-current boundary is*

$$\mathcal{B}_{\text{CC}} = (\mathcal{A}_{\text{tag}}, \{\mathbf{P}_\alpha\}_{\alpha \in \mathcal{F}}, \{\mathbf{F}_\beta\}_{\beta \in \mathcal{F}}, \mathcal{R}_{\text{tag}}, \mathcal{C}_{\text{cal}}, \simeq_{\text{CC}}), \quad (23)$$

where $\mathcal{F}$ is the admitted flavor-tag set, $\mathcal{R}_{\text{tag}}$ is the record-use map, $\mathcal{C}_{\text{cal}}$ is the source-authorized calibration structure, and $\simeq_{\text{CC}}$ identifies presentations with the same preparation, effect, record, and calibration behavior.

The formulation is compatible with the ordinary charged-current description of weak interactions, but it does not assume a particular lepton mixing matrix. The same interaction-defined boundary was used in the preceding role theorem to establish nonalignment [20].

5.2 Boundary reconstruction

Theorem 5.2 (Boundary reconstruction). *The target-admitted source and detector records determine $\mathcal{B}_{\text{CC}}$ up to $\simeq_{\text{CC}}$. Any two concrete state/effect presentations that induce the same preparation classes, detector effects, record maps, and lawful calibrations define the same charged-current boundary interface.*

Proof. The target record packet fixes which source operations count as the same preparation and which detector operations count as the same effect for later evidential use. Two presentations that agree on every admitted preparation–effect pairing and on every record/calibration continuation make identical boundary predictions and support the same later use. Their distinction is therefore representational at the boundary. Conversely, if a presentation changes one preparation class, detector effect, or record-use relation, then some target-admitted source/detection statement changes. The difference is boundary-bearing and cannot be quotiented without changing the target. Thus the operational equivalence class is uniquely fixed by the target records. $\square$

Corollary 5.3 (No propagation-basis definition of the boundary). *No propagation eigenbasis, Hamiltonian diagonalization, mass label, or PMNS coordinate may define or modify $\mathcal{B}_{\text{CC}}$ at the present target-relative scope.*

Proof. Such objects belong to a candidate propagation representation. The boundary is reconstructed from source/detection interaction and record equivalence before a propagation map is specified. Allowing a later diagonalization to redefine the boundary would reverse the dependency and erase the interaction–propagation comparison that $r_{\triangleright\triangleleft}$ records. $\square$

5.3 Concrete operator-system presentations

A convenient finite-dimensional presentation uses an operator system $\mathfrak{A}_{\text{rec}}$ containing density operators ρ_α and effects $0 \leq F_\beta \leq 1$. Perfectly sharp flavor projectors are not required. The observable source-to-detection kernel is

$$K_{\beta\alpha}(\lambda) = \text{Tr}[F_\beta \mathcal{E}_\lambda(\rho_\alpha)].$$

Preparation and detector imperfections are carried by ρ_α and F_β , not hidden in a redefinition of the propagation map. If the same physical preparation or effect admits multiple operator representatives, the representatives are identified by $\approx_{\text{CC}}$ together with the channel equivalence defined in section 6.

Remark 5.4 (Why projectors remain useful). *The projector notation P_α used in the preceding analysis is the idealized sharp-boundary representative. The channel-native theorem extends to loss-tolerant nonmixing relations among operational faces generated by general effects and preparation cones. Projectors are retained in illustrative formulas, but the present analysis treats the operational preparation/effect classes as primary.*

5.4 Boundary variants and target change

Definition 5.5 (Boundary variant). *A source or detector modification is a boundary variant when it changes the preparation map, detector effect algebra, charged-current tag semantics, or record-use equivalence while leaving the propagation representation otherwise unspecified.*

Examples include nonstandard charged-current operators at production or detection, additional detector-selective final states, and flavor tags whose operational meaning changes. Such variants may be legitimate physical models. They are not same-target occupants of the fixed boundary unless an explicit equivalence theorem shows that every boundary and record witness is preserved.

Theorem 5.6 (Boundary target-fiber theorem). *If a completion changes $\mathcal{B}_{\text{CC}}$ on any target-admitted preparation, effect, record, or calibration witness, it belongs to a different target fiber. Equality of propagation kernels after a compensating redefinition does not restore same-target identity.*

Proof. A compensating redefinition can make numerical kernels look similar while changing which physical source and detector acts the indices denote. The target is individuated by those operational acts and their record use, not by an unlabeled matrix of numbers. Preserving only the matrix while changing its boundary interpretation fails reference fixation. A lawful same-target comparison therefore requires preservation of $\mathcal{B}_{\text{CC}}$, not merely numerical equality after retagging. $\square$

5.5 Boundary covariance

Proposition 5.7 (Boundary covariance under lawful redescription). *Let u be a simultaneous representation change acting on states, effects, and channel presentations. If*

$$\rho_\alpha \mapsto u\rho_\alpha u^\dagger, \quad F_\beta \mapsto uF_\beta u^\dagger, \quad \mathcal{E}_\lambda \mapsto u \circ \mathcal{E}_\lambda \circ u^{-1}$$

with the appropriate state/effect dual action and with every boundary record preserved, then u is boundary skin and leaves $\mathcal{B}_{\text{CC}}$ unchanged.

Proof. All preparation–effect probabilities and record maps are invariant under the simultaneous intertwining. The transformation changes only the concrete presentation of the operational classes. If the transformation acts on only one side or changes a record witness, it is not a lawful simultaneous redescription. $\square$

5.6 Boundary witness ledger

The boundary part of the shared witness ledger contains:

1. stable preparation IDs and source-provenance records;

2. detector effect IDs and detector-provenance records;
3. the charged-current tag algebra and allowed relabelings;
4. calibration maps and uncertainty domains;
5. event-record maps and retained-record equivalence;
6. overlap keys linking source/detector classes to the channel and response packets.

A later coordinate is not allowed to introduce a new boundary tag or use a different preparation/effect representative without updating these overlap keys and triggering full reevaluation.

Boundary/propagation separation

A model that attributes part of the observed flavor change to source or detector interactions is not rejected. It is classified as a controlled boundary target variant. The core theorem concerns completions of the fixed charged-current boundary, and the distinction prevents source/detector new physics from being smuggled into the propagation interface as if it were the same target.

6 Channel-native propagation interface

6.1 Event-complete channel representation

A propagation model need not be closed or unitary on the record-relevant sector. Decay, absorption, decoherence, leakage into unobserved states, and reduction from a larger Hilbert space can all produce effective nonunitarity. To compare these possibilities without selecting one, the present analysis uses an event-complete quantum instrument.

Definition 6.1 (Event-complete propagation instrument). *For each protocol/environment coordinate $\lambda \in \Lambda_{\text{prot}}$, an event-complete propagation instrument is a finite or countable family*

$$\mathcal{I}_{\lambda}^X = \{\mathcal{E}_{\lambda,o}^X : \mathfrak{A}_{\text{in}} \rightarrow \mathfrak{A}_{\text{out}}\}_{o \in \mathcal{O}_{\text{full}}}$$

of normal completely positive trace-nonincreasing maps such that

$$\sum_{o \in \mathcal{O}_{\text{full}}} \mathcal{E}_{\lambda,o}^X$$

is trace preserving. The outcome set includes all target-visible detection classes and an explicit null/loss class when necessary.

The event-complete sum is a CPTP channel. Ordinary unitary propagation is the special case $\mathcal{E}_{\lambda}(\rho) = U_{\lambda}\rho U_{\lambda}^{\dagger}$. A reduced nonunitary description is admissible only when its missing probability is assigned to explicit completion outcomes or when the target conditions justify a normalized conditional channel. This prevents unexplained loss from being used as a hidden response mechanism.

For source class α , detector class β , and any target-visible outcome selection, define

$$K_{\beta\alpha}^X(\lambda) = \text{Tr}[F_{\beta} \mathcal{E}_{\lambda}^X(\rho_{\alpha})], \quad (24)$$

where $\mathcal{E}_{\lambda}^X$ denotes the appropriate event-complete or conditionally normalized map declared in the target specification.

6.2 Channel equivalence

Definition 6.2 (Lawful channel intertwining). *Two channel families $\mathcal{I}^X$ and $\mathcal{I}^Y$ are lawfully intertwined when there are isomorphisms or isometries on the admitted input/output operator systems and outcome labels*

that:

1. preserve $\mathcal{B}_{\text{CC}}$ and every target record;
2. carry each completely positive outcome map to the corresponding map;
3. preserve protocol/environment incidence and the null/loss ledger;
4. preserve every active support, anomaly, vacuum, scale, and extension witness.

Theorem 6.3 (Channel presentation covariance). *Lawfully intertwined channel families define the same channel coordinate and the same transition-kernel family. Equivalent Kraus decompositions of one completely positive map and equivalent Stinespring dilations with no admitted dilation-sensitive witness are therefore interface-identical.*

Proof. Kraus decompositions and Stinespring dilations are presentations of the same completely positive map [12, 29]. Under a lawful intertwiner, source states and detector effects are transported simultaneously, so equation (24) is unchanged. The remaining interface witnesses are preserved by definition of the intertwining. The channel coordinate is therefore constant on the lawful presentation class. If the target admits a witness of the dilation sector, the final preservation clause fails and the dilations are not identified. $\square$

6.3 Boundary sectors and loss-tolerant nonmixing

In the sharp-boundary representative, let $\{P_\alpha\}$ be mutually orthogonal charged-current boundary projectors and

$$\mathfrak{F}_\alpha = \{\rho \geq 0 : \text{Tr } \rho = 1, \rho = P_\alpha \rho P_\alpha\}$$

the associated preparation face. Event-complete propagation may contain an explicit null/loss sector, so preservation of a visible flavor sector must not be defined by forbidding loss. The primitive condition is instead absence of transfer between distinct target-visible charged-current sectors.

Definition 6.4 (Charged-current sector nonmixing). *A complete effective channel $\mathcal{E}_\lambda$ is charged-current nonmixing when, for every $\alpha \neq \beta$,*

$$P_\alpha \mathcal{E}_\lambda^\dagger(P_\beta) P_\alpha = 0. \quad (25)$$

Equivalently, every state prepared in $\mathfrak{F}_\alpha$ has zero target-visible detection probability in sector $\beta \neq \alpha$. Probability may still flow to an explicitly recorded null/loss outcome. In the event-complete, no-loss, sharply resolved special case, simultaneous nonmixing reduces to preservation of each boundary face and to

$$\mathcal{E}_\lambda^\dagger(P_\alpha) = P_\alpha \quad \text{for every } \alpha.$$

Lemma 6.5 (Channel no-transfer lemma). *If $\mathcal{E}_\lambda$ is charged-current nonmixing, then for $\rho_\alpha \in \mathfrak{F}_\alpha$ and $\beta \neq \alpha$,*

$$\text{Tr}[P_\beta \mathcal{E}_\lambda(\rho_\alpha)] = 0.$$

Conversely, if this trace vanishes for every $\rho_\alpha \in \mathfrak{F}_\alpha$, then equation (25) holds.

Proof. For $\rho_\alpha = P_\alpha \rho_\alpha P_\alpha$,

$$\text{Tr}[P_\beta \mathcal{E}_\lambda(\rho_\alpha)] = \text{Tr}[P_\alpha \mathcal{E}_\lambda^\dagger(P_\beta) P_\alpha \rho_\alpha].$$

The right-hand side vanishes under equation (25). Conversely, $P_\alpha \mathcal{E}_\lambda^\dagger(P_\beta) P_\alpha$ is positive. If its expectation vanishes on every density operator supported in P_α , the positive operator itself is zero. $\square$

Definition 6.6 (Channel-native nonalignment coordinate). *The nonalignment interface coordinate $I_{\text{ns}}(X)$ is positive precisely when charged-current nonmixing fails somewhere in the complete effective channel family above the declared boundary leakage floor: there exist $\alpha \neq \beta$, a protocol/environment point λ , and a*

source-supported state $\rho_\alpha \in \mathfrak{F}_\alpha$ for which the target-visible cross-boundary detection probability is certified nonzero after calibrated boundary leakage is removed. Loss into a null sector by itself does not occupy $I_{\beta\alpha}$.

Theorem 6.7 (Boundary-sector nonalignment necessity). *Every completion $X \in \mathfrak{C}_v$ that realizes the established oscillation target has $I_{\beta\alpha}(X) = 1$.*

Proof. The established empirical target contains active cross-boundary flavor transformation. If the complete channel family were charged-current nonmixing, lemma 6.5 would force every target-visible cross-boundary kernel to vanish, irrespective of any separately recorded loss outcome. Therefore nonmixing fails. This is exactly the channel-native role $r_{\beta\alpha}$ established in the preceding analysis [20]. $\square$

6.4 Unitary representative

For $\mathcal{E}_\lambda(\rho) = U_\lambda \rho U_\lambda^\dagger$, charged-current nonmixing is equivalent to

$$[P_\alpha, U_\lambda] = 0 \quad \text{for all } \alpha.$$

Thus the familiar commutator criterion is a representative of the channel theorem, not its definition.

Corollary 6.8 (Unitary nonalignment criterion). *In a unitary representative, any nonzero cross-boundary transition requires*

$$[P_\alpha, U_\lambda] \neq 0$$

for at least one α and λ .

6.5 Realistic boundary leakage

Perfectly sharp source and detector classes are idealizations. Let $\epsilon_{\beta\alpha}(\lambda)$ be a calibrated upper bound on cross-tag preparation/detection leakage in the absence of propagation transfer.

Proposition 6.9 (Robust nonalignment witness). *If an observed or target-certified kernel satisfies*

$$K_{\beta\alpha}(\lambda) > \epsilon_{\beta\alpha}(\lambda) \quad (\beta \neq \alpha),$$

then the complete interface cannot remain charged-current nonmixing up to the declared boundary uncertainty.

Proof. Under charged-current nonmixing, all cross-tag probability is bounded by the calibrated leakage floor. A value exceeding that floor requires an additional transfer contribution in the complete channel or a change of the boundary target. The latter is excluded in the core comparison class, so nonalignment follows. $\square$

6.6 Dilation-sensitive distinctions

Proposition 6.10 (Dilation relevance rule). *Two enlarged realizations inducing the same reduced event-complete channel are interface-equivalent exactly when the active augmented comparison profile contains no component or joint-incidence witness that distinguishes their dilation sectors. A hidden-sector charge, decay product, support relation, anomaly, line operator, or causal record activates a distinct extension coordinate.*

Proof. If no admitted witness distinguishes the dilations, their difference changes neither $\mathcal{P}_v^+$ nor $\mathcal{J}_v^{\text{sim}}$ and is interface-inert. If a witness is admitted, the two completions differ on the support, consistency, extension, or joint-incidence profile and cannot be lawfully identified. This is the no-overquotient principle applied to dilation data. $\square$

6.7 Channel coordinate in the shared physical witness system

The channel packet exports:

1. a stable channel-family ID and target fiber;

2. source/effect overlap keys with $\mathcal{B}_{CC}$;
3. complete outcome and loss normalization;
4. protocol/environment incidence;
5. transition-kernel functions, not merely fitted values;
6. charged-current nonmixing and nonalignment status;
7. lawful intertwining and dilation policy;
8. source provenance for every event-level response map.

No unitary overrestriction

A decohering, decaying, absorptive, stochastic, or enlarged-state-space completion is not excluded because it lacks a unitary reduced propagator. It must still supply an event-complete channel, preserve the fixed boundary and record semantics, satisfy support and consistency constraints, and occupy the same target role complex.

7 Boundary-visible response multiplicity and the nested interface

7.1 Operational mode classes

The preceding role analysis did not define visible multiplicity by counting datasets, microscopic eigenvalues, or every Fourier character inside an uncertainty band. Each experiment contributed a finite phenomenological response record together with the collaboration's response model, nuisance treatment, and a source-certified phase-frequency interval. Those records were quotiented only by source-authorized calibration, environment, and analysis-coordinate transports. KamLAND and Daya Bay then supplied two inequivalent mode classes on the same operational reactor $\bar{\nu}_e$ boundary [7, 11, 20].

The present analysis uses the resulting finite set $\mathfrak{M}_{\text{vis}}(C_\nu)$ and retains more structure than its cardinality: each class is incident with a boundary channel, protocol range, environment profile, response-analysis map, and published analysis record.

Definition 7.1 (Candidate visible-mode projection). *For each target protocol j , let*

$$\Phi_j : (\mathcal{T}_j, \mathcal{L}_j, \Theta_j) \longrightarrow \mathfrak{R}_j$$

be the source-analysis map fixed by the preceding empirical specification, carrying a candidate event-level prediction, likelihood or test statistic, and nuisance domain to a finite response record. Applying the same Φ_j to a candidate X and quotienting the resulting records by the imported lawful mode equivalence gives the finite set $\mathfrak{M}_{\text{vis}}(X)$.

The multiplicity

$$\rho_{\text{vis}}(X) := |\mathfrak{M}_{\text{vis}}(X)|$$

is therefore a boundary-visible mode-class number. It is not an independently discovered linear-algebraic rank and not a microscopic spectrum.

7.2 Response covariance

Theorem 7.2 (Visible-response covariance). *If X and Y are related by a lawful interface redescription, then there is a canonical bijection*

$$\mathfrak{M}_{\text{vis}}(X) \cong \mathfrak{M}_{\text{vis}}(Y)$$

preserving boundary, protocol, environment, and source-analysis incidence. In particular, $\rho_{\text{vis}}(X) = \rho_{\text{vis}}(Y)$.

Proof. A lawful redescription preserves the event-level prediction family, the source-analysis maps Φ_j , the nuisance domains, and all source-authorized transports. The response record produced in each protocol is therefore carried to a lawfully equivalent record, and inequivalent records cannot be identified because the transformation preserves the imported mode equivalence. The map on records descends to a bijection of mode classes with all incidence data intact. $\square$

Corollary 7.3 (No hidden microscopic multiplication). *Additional candidate eigenvalues, phases, states, gaps, poles, or defect labels that do not change the event-level response through the fixed analysis maps do not create new target-visible mode classes.*

7.3 Target coverage

Definition 7.4 (Mode-incidence preserving injection). *A map*

$$\iota_X : \mathfrak{M}_{\text{vis}}(C_\nu) \hookrightarrow \mathfrak{M}_{\text{vis}}(X)$$

is mode-incidence preserving when it preserves the charged-current boundary, source-analysis map, protocol and environment fiber, lawful-equivalence provenance, and the established target incidence that attaches the visible-mode packet to active charged-current flavor transformation. The last clause is part of the fixed target specification: visible-mode cardinality without that active boundary incidence is not, by itself, the role r_Δ .

Theorem 7.5 (Visible response coverage). *Every completion $X \in \mathfrak{C}_\nu$ realizing T_ν^{osc} admits a mode-incidence preserving injection*

$$\mathfrak{M}_{\text{vis}}(C_\nu) \hookrightarrow \mathfrak{M}_{\text{vis}}(X).$$

Consequently,

$$\rho_{\text{vis}}(X) \geq |\mathfrak{M}_{\text{vis}}(C_\nu)| \geq 2.$$

Proof. Realizing the target requires reproducing every source-certified event-response packet through the same analysis and nuisance maps, together with the target-certified active flavor-transformation incidence established in the preceding analysis. Each target mode class must therefore have a candidate response record in the same boundary/protocol/environment incidence class. If two distinct target classes mapped to one candidate class, the candidate would induce a lawful identification forbidden by the target's mode equivalence. Hence the map is injective. The cardinality bound follows. Preservation of the active incidence prevents a multi-scale loss-only or sterile-disappearance response from being mistaken for the established role packet merely because it has several fitted modes. $\square$

Remark 7.6 (Why a candidate may have more visible modes). *The interface theorem imposes a lower bound, not exact equality. A candidate can contain additional visible structure only if it remains compatible with the complete target record and every shared constraint. Whether such structure survives later candidate exhaustion is not decided here.*

7.4 Response multiplicity coordinate

Definition 7.7 (Response-multiplicity coordinate). *The coordinate $I_\Delta(X)$ is positive when X carries a mode-incidence preserving image of the complete established response packet. Thus it must reproduce at least two lawfully inequivalent visible response classes in one completion, and preserve their target-certified incidence with active charged-current flavor transformation. The numerical condition $\rho_{\text{vis}}(X) \geq 2$ is necessary but is not sufficient by itself.*

This coordinate represents the role r_Δ at interface level. It is stronger than the assertion that some transition probability is nonconstant and stronger than bare mode counting. It preserves the full nested response object already established in the preceding analysis without assigning a microscopic interpretation to it.

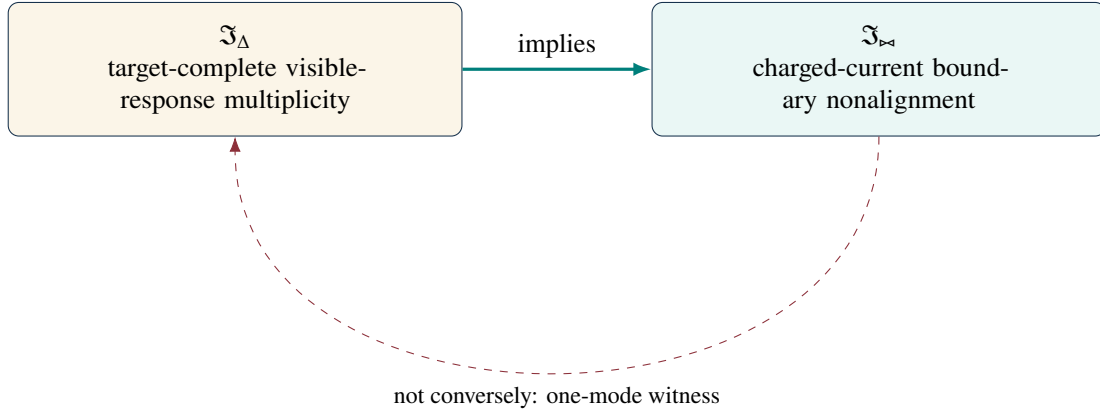


Figure 4: Strict role nesting is an interface dependency, not a demand for two microscopic occupants.

7.5 Nested-role incidence

Definition 7.8 (Nested-role incidence object). *The object $\mathcal{N}_{\Delta \triangleright \triangleleft}(X)$ consists of:*

1. *the truth values $I_{\Delta}(X)$ and $I_{\triangleright \triangleleft}(X)$;*
2. *the target-certified implication witness $I_{\Delta} \Rightarrow I_{\triangleright \triangleleft}$;*
3. *the lawful one-mode nonaligned counterexample class witnessing failure of the converse;*
4. *the declaration that one or several physical occupants may discharge the ordered role complex.*

Theorem 7.9 (Nested-interface theorem). *Within the fixed comparison class,*

$$I_{\Delta}(X) = 1 \quad \Rightarrow \quad I_{\triangleright \triangleleft}(X) = 1,$$

while

$$I_{\triangleright \triangleleft}(X) = 1 \not\Rightarrow I_{\Delta}(X) = 1.$$

Thus the interface preserves the strict order $r_{\Delta} \succ r_{\triangleright \triangleleft}$.

Proof. The forward implication is the interface preservation of the established nested-role theorem, not an inference from mode cardinality alone. By [definition 7.7](#), $I_{\Delta} = 1$ includes preservation of the target-certified active flavor-transformation incidence. By [theorem 6.7](#), that incidence requires failure of charged-current nonmixing, hence $I_{\triangleright \triangleleft} = 1$. A channel with two visible disappearance scales caused solely by loss, decay, or leakage would not satisfy [definition 7.7](#) unless it also reproduced the active cross-boundary target incidence.

For the converse, take a two-state unitary representative with boundary projectors P_0, P_1 and

$$U_t = e^{-it\omega\sigma_x}.$$

Then

$$K_{10}(t) = \sin^2(\omega t)$$

is nonzero and the boundary is nonaligned, but the response has only one lawfully visible frequency class. Hence $I_{\triangleright \triangleleft} = 1$ and $I_{\Delta} = 0$ relative to the two-class target. The model is a comparison witness, not a candidate claimed to realize the target. $\square$

Corollary 7.10 (No two-occupant inference). *The strict order $r_{\Delta} \succ r_{\triangleright \triangleleft}$ does not imply two particles, two fields, two operators, two mechanisms, or two independent occupants. One physical structure may discharge both coordinates, or several coupled structures may jointly do so.*

7.6 Same-target deletion at interface level

Theorem 7.11 (Response-coordinate deletion). *Let X be an interface-adequate completion with $I_\Delta(X) = 1$. Replace its event-level response by a same-boundary, same-environment, same-nonalignment response whose lawful visible-mode image has cardinality one, while leaving the target specification and all other interface anchors fixed. The altered object fails I_Δ and is not interface-adequate for T_v^{osc} .*

Proof. The target contains two inequivalent visible mode classes. A one-class candidate image cannot admit the injection of theorem 7.5. Since the alteration acts on the candidate response rather than on the target record or equivalence relation, it is a genuine same-target deletion of the response-multiplicity coordinate. Nonalignment may remain, as the counterexample in theorem 7.9 shows. $\square$

7.7 Three layers of multiplicity

Multiplicity	Meaning	Status in the present interface
Record multiplicity	Several datasets, bins, or experimental contexts.	Not itself a physical role.
Visible mode-class multiplicity	Several lawfully inequivalent response classes after source-authorized quotient.	Mandatory target interface, $\rho_{\text{vis}} \geq 2$.
Microscopic multiplicity	Masses, eigenvalues, fields, states, poles, defects, couplings, or carrier components.	Open candidate data; no inference from ρ_{vis} .

Table 6: Distinct multiplicity notions.

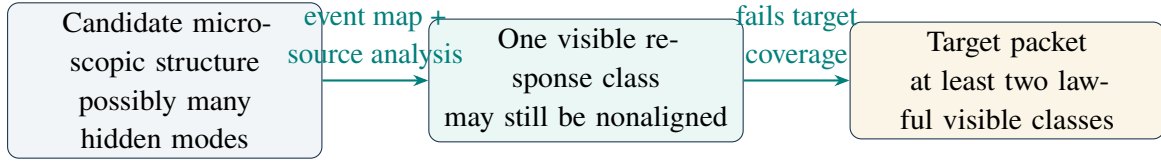


Figure 5: Microscopic richness, nonalignment, and target-visible multiplicity are distinct.

Empirical reopening condition

The response-multiplicity coordinate remains conditional on the phenomenological response input inherited from the preceding analysis. A carrier-neutral common-boundary reconstruction that lawfully collapses the KamLAND and Daya Bay response packets to one mode class would reopen r_Δ and therefore the present interface theorem. The present paper does not insulate that empirical assumption from future analysis.

8 Locality, transport, environment, and causal continuation

8.1 Observable transport rather than microscopic ontology

The present analysis does not assume that every admissible completion is described by the same microscopic spacetime variables or by a local Hamiltonian density. It does require that the realized source–medium–detector relation be a standing-bearing physical continuation rather than a pathless correlation or an undeclared global coordination rule. The relevant import from the Realized Physical Interior is therefore operational: bounded localization, lawful continuation, bounded perturbation transport, temporal ordering, and witness-dependent causal dependence [22].

The ordinary wave-packet and quantum-field-theoretic treatments of neutrino propagation illustrate how source localization, coherent propagation, separation, and detection enter a standard realization [2, 3, 9].

These treatments are used as physical illustrations and consistency checks, not as a unique microscopic premise.

Definition 8.1 (Observable transport ledger). *The ledger $\mathcal{L}_t(X)$ records:*

1. *source preparation support and event localization;*
2. *the protocol-ordered continuation from source through admitted medium to detector;*
3. *detector effect support and record localization;*
4. *the dependence of the effective channel on path, baseline, and environment;*
5. *the no-undeclared-signalling and no-pathless-influence conditions relevant to the target;*
6. *any local or causal dilation that explains an effective nonlocal kernel;*
7. *overlap keys linking support data to the channel, environment, and record packets.*

8.2 Transport necessity

Theorem 8.2 (Observable transport necessity). *Every completion realizing the fixed target must supply a witnessable source-to-detection continuation structure that preserves source and detector localization, protocol order, and the admitted causal/support relations. A proposal whose target-visible effect appears without any declared continuation, medium, boundary, or causal witness either changes the target, imports hidden load, or fails physical interface adequacy.*

Proof. The target records are source-indexed and detector-indexed, and their use presupposes that a preparation event is lawfully connected to a later detection event. If the target-visible dependence is not carried through any declared continuation, the same detector record could be attributed to an unrelated source without a witness-preserving relation. Reference and causal dependence then fail. If the missing relation is supplied by an external global coordinator, the coordinator is hidden physical load; if the source/detector semantics are changed to avoid the need for a continuation, the target changes. Therefore a target realization must carry the declared observable transport relation. $\square$

Remark 8.3 (No claim of a universal microscopic light cone). *The theorem is not a derivation of a particular propagator support theorem or a fundamental finite speed for every proposed completion. It requires that the target's local source and detector records be connected by a causally and operationally controlled interface. A richer theory may realize that control through a local QFT, an emergent causal structure, or a lawful dilation.*

8.3 Effective nonlocality

Definition 8.4 (Lawfully dilatable effective nonlocality). *An effective kernel is lawfully dilatable when there exists a declared richer realization whose local/causal witness ledger projects to the effective kernel and whose projection preserves every target record and active consistency coordinate.*

Proposition 8.5 (Effective nonlocality classification). *An effective nonlocal channel can remain in the declared comparison class if it is lawfully dilatable in the sense of definition 8.4. Fundamental pathless influence with no declared causal/support witness is not same-target interface adequacy.*

Proof. A lawfully dilatable kernel preserves the augmented comparison profile because its apparent nonlocality is a reduced presentation of a richer admissible continuation. The dilation may be candidate-distinct but interface-equivalent if no target witness detects the extra sector. Without such a dilation or an alternative causal witness, the target-visible dependence has no standing-bearing continuation and fails theorem 8.2. $\square$

8.4 Environment and matter response

Neutrino propagation can depend strongly on matter density and path composition. In the conventional framework this is represented by the Wolfenstein and Mikheyev–Smirnov matter potentials [25, 34]. The interface does not assume that form. It requires that environment dependence be explicit and covariant.

Definition 8.6 (Environment packet). *The packet $\mathcal{E}_{\text{env}}(X)$ contains:*

1. *the physical medium profile or environment state used by the channel;*
2. *its coordinate and calibration representations;*
3. *path-composition and ordering data;*
4. *the response map $\lambda \mapsto \mathcal{I}_\lambda^X$;*
5. *uncertainty and admissible-deformation domains;*
6. *source and detector interaction data kept distinct from propagation-medium data.*

Theorem 8.7 (Environment covariance). *Two environment descriptions represent the same physical interface when they are related by a lawful coordinate or calibration transformation that preserves the physical medium profile, path ordering, channel family, event-level predictions, and all shared-ledger witnesses.*

Proof. The physical witness is the environment-dependent continuation and its event-level effect, not a particular density coordinate, discretization, or profile parametrization. A lawful transport of the environment data that preserves those witnesses changes only representation. If the channel or record response changes, the transformation is tensor-level and defines a new interface candidate. $\square$

Theorem 8.8 (No hidden medium). *A candidate may not use an undeclared medium interaction, background field, density profile, or environment-dependent source/detector correction to discharge $r_{\bowtie}$ or r_Δ while retaining the original target ID.*

Proof. The target records are indexed by an explicit protocol/environment domain. An undeclared interaction changes the map from that domain to the channel or changes the boundary preparation/effect classes. In the first case the environment packet is incomplete; in the second the target boundary changes. Treating the new interaction as if it were already present would import hidden load and violate shared-ledger agreement. $\square$

8.5 Source/detector versus propagation effects

A source or detector nonstandard interaction can alter apparent flavor probabilities. The analysis therefore maintains the trichotomy:

1. propagation-medium dependence inside the fixed environment packet;
2. lawful calibration or representation change preserving the boundary;
3. altered source/detector interaction, which is a boundary target variant.

No numerical fit can collapse these categories by itself.

Proposition 8.9 (Boundary/environment separation). *If a modification changes preparation or detector effect classes, it is a boundary target variant even when its net event-level response can be written as an effective propagation kernel. If it changes only the medium-dependent continuation while preserving $\mathcal{B}_{\text{CC}}$, it is a propagation/environment variation requiring full interface reevaluation.*

8.6 Causal record order

Let $s \prec m \prec d$ denote source, admitted medium continuation, and detection in the target record order. The interface requires that the effective prediction respect this order as a record-use relation. Retrocausal, acausal, or indefinite-order models are not dismissed globally, but they lie outside the core target unless an explicit target fiber supplies a coherent record semantics and comparison profile.

Theorem 8.10 (Causal-order target-fiber rule). *A completion that changes the source–medium–detector record order or makes the target records order-indefinite belongs to a different target fiber unless it proves an operational equivalence preserving every record, continuation, and causal witness.*

Proof. The temporal and causal ordering of the target records is part of their standing-bearing use. A different order changes which event is evidence for which preparation and therefore changes the augmented target comparison profile. Only an equivalence preserving the complete record-use relation can avoid a target change. $\square$

8.7 Transport and environment interface theorem

Theorem 8.11 (Transport/environment interface closure). *For every interface-adequate X , the pair*

$$(\mathcal{L}_{\text{tr}}(X), \mathcal{E}_{\text{env}}(X))$$

is jointly defined on the same protocol/environment domain as the channel and visible-mode packets, is invariant under lawful coordinate descriptions, and contains no undeclared source/detector or causal load.

Proof. Interface adequacy requires the channel to compute the target records over the fixed protocol/environment domain. The transport ledger supplies the support and causal continuation of that channel, and the environment packet supplies the physical medium dependence. If either uses a different domain or boundary, shared-ledger agreement fails. Covariance follows from theorem 8.7 and the lawful groupoid. Hidden boundary or causal load is excluded by theorems 8.8 and 8.10 and proposition 8.9. $\square$

Interpretive boundary. The transport condition is operational rather than ontological. Any microscopic completion must state how its source-to-detection response is supported and causally continued, or else belong to a controlled richer target fiber. This allows local quantum-field-theoretic, open-system, enlarged-state-space, and topological descriptions to be compared without prejudging their microscopic mechanism.

9 Gauge attachment and anomaly interface

9.1 Attachment to the fixed electroweak boundary

The target does not float free of the known electroweak theory. Its preparation and detection tags are realized by the charged-current boundary of the fixed Standard-Model substrate. A completion may add new propagating, effective, or topological structure, but the manner in which that structure attaches to the boundary is physical interface data.

Definition 9.1 (Gauge-attachment ledger). *The ledger $\mathcal{G}_{\text{attach}}(X)$ records:*

1. *the boundary representation-response classes used by source and detector acts;*
2. *the charge and residual electromagnetic witnesses relevant to those acts;*
3. *the map from candidate propagation variables to the boundary preparation/effect classes;*
4. *invariance under gauge, weak-basis, and phase redescrptions;*
5. *any new boundary current, representation, or charge introduced by a controlled target variant;*

6. *overlap keys with the channel, anomaly, vacuum, and EFT packets.*

The ledger uses the architecture/operation distinction established in the symmetry analysis. Gauge representatives are skin on the physical quotient, while the product, transport, representation, charge, and anomaly-free architecture remains load-bearing [15, 23, 24].

Theorem 9.2 (Boundary attachment theorem). *Every same-target completion must attach to $\mathcal{B}_{CC}$ through a gauge-compatible map preserving the operational flavor tags, the target-relevant representation/charge witness ledger, and the residual electromagnetic boundary. A proposal that changes these data defines a boundary target variant unless an explicit equivalence theorem preserves the complete comparison profile.*

Proof. The source and detector acts are physically implemented within the fixed electroweak boundary. If a candidate uses different representation or charge data for the same record labels, the labels no longer denote the same preparation/effect classes. If the difference is a gauge or weak-basis representative preserving every boundary and record witness, it is removed by the lawful groupoid. Otherwise the boundary attachment changes, and the candidate lies in another target fiber. Thus same-target realization requires a gauge-compatible attachment map. $\square$

Proposition 9.3 (Gauge representative nonauthority). *Gauge fixing, weak-basis choice, field rephasing, and representative connection coordinates cannot create, remove, or select a canonical interface class. Only quotient-stable boundary, response, charge, holonomy, anomaly, or extended-sector witnesses may distinguish interfaces.*

Proof. Representative transformations are lawful only when the full comparison profile is preserved. If they preserve it, they are arrows of $\mathcal{G}_v^{\text{law, sat, sim}}$ and disappear in Π_v . If they change a witness, they are not skin and must be evaluated as a new coupled interface candidate. $\square$

9.2 Quantum-consistency ledger

An anomaly can have several logically distinct meanings. A local gauge anomaly is an inconsistency of the quantum gauge architecture. A global current anomaly obstructs current conservation. A 't Hooft anomaly constrains gauging and infrared realization without making the global symmetry meaningless. A global anomaly can depend on the faithful group rather than only its Lie algebra. Inflow and matching can relocate, but not hide, the obstruction [4, 14, 16, 24, 33].

Definition 9.4 (Anomaly and quantum-consistency ledger). *The ledger $\mathcal{A}_q(X)$ is the target-fiber-indexed record of:*

1. *local perturbative gauge anomalies;*
2. *global gauge anomalies and faithful-group dependence;*
3. *mixed gauge, gravitational, and background anomalies when admitted;*
4. *global-current and 't Hooft anomaly data relevant to the interface;*
5. *explicit inflow, counterterm, spectator, or topological sectors used to cancel or match an obstruction;*
6. *the common representation, charge, global-form, and operator-category witnesses used in the calculation.*

Theorem 9.5 (Anomaly-interface necessity). *A lawful interface occupant must preserve the anomaly-free quantum gauge architecture of the fixed charged-current boundary. It may not create an uncanceled local or global gauge anomaly, inconsistent mixed obstruction, or ill-defined background coupling in the active target fiber. Any inflow or compensating sector must be explicit in the same shared witness ledger.*

Proof. The target boundary is a quantum gauge interface. An uncanceled gauge anomaly prevents gauge-redescription equivalence, destroys the physical quotient, or makes the boundary current inconsistent; the target then fails rather than becoming a differently realized version of the same interface. If a compensating sector cancels or transports the obstruction, that sector changes the candidate’s witness ledger and must be evaluated simultaneously with its representation, charge, global-form, vacuum, and transport data. A hidden compensator is therefore inadmissible. $\square$

Proposition 9.6 (Anomaly matching and target scope). *A global or ’t Hooft anomaly enters the mandatory interface only when the active target fiber admits the corresponding background or extended-operator witness. Otherwise it is candidate or target-fiber data rather than a universal core coordinate.*

Proof. A global anomaly is interface-relevant only if it changes a target-admitted consistency or response witness. If the target does not admit the background, defect, or global action required to formulate the anomaly, its value cannot distinguish core point-local interfaces. Activating the relevant operator/background category creates an extension fiber in which the anomaly becomes mandatory. $\square$

9.3 Anomaly types and their interface roles

Anomaly type	Physical meaning	Interface role	Disposition
Local gauge anomaly	Inconsistent quantum gauge architecture.	Mandatory obstruction ledger.	Candidate fails unless explicitly cancelled in the same ledger.
Global gauge anomaly	Faithful-group/topological inconsistency.	Mandatory in active global-form fiber.	Candidate fails or target changes.
Anomalous global current	Quantum nonconservation of a global current.	Relevant only if that global action is target-admitted.	Records selection rules or target-fiber structure.
’t Hooft anomaly	Obstruction to gauging; UV/IR matching constraint.	Relevance-gated consistency witness.	May constrain candidate classes without choosing one.
Inflow/matching	Compensating bulk/defect or IR realization.	Explicit shared-ledger structure.	Cannot be hidden as a repair.

Table 7: Anomaly taxonomy for the target-relative interface.

9.4 No anomaly-to-matter inference

Theorem 9.7 (No anomaly-to-matter inference). *The anomaly ledger can exclude or retype a specified candidate, but the interface theorem does not infer a unique new local matter representation, multiplicity, or particle from anomaly preservation alone.*

Proof. Anomaly equations constrain the combined representation and charge content of a complete candidate. Distinct local matter spectra, vectorlike pairs, topological/inflow sectors, or target-fiber changes can in principle alter the same obstruction ledger. The present paper neither generates nor exhausts those possibilities. A uniqueness claim would therefore require independent candidate completeness, simultaneous survivor analysis, and role-occupancy closure. Promoting the consistency constraint to a carrier identity here would assume results not established by the interface theorem. $\square$

9.5 Shared representation and charge data

The representation and charge data used in $\mathcal{G}_{\text{attach}}$ must be the same data used in $\mathcal{A}_q$. This is a principal overlap key in the shared witness ledger. A candidate may not use one charge normalization for boundary coupling and another for anomaly cancellation unless the lawful relation between them is explicit and preserves all witnesses.

Lemma 9.8 (Gauge–anomaly overlap). *If the gauge-attachment and anomaly ledgers of a raw interface cannot be evaluated on one common representation/charge/global-form record, the interface does not belong to $\mathfrak{S}_v^{\text{adm}}$, even when each ledger is separately satisfiable.*

Proof. Separate satisfiability does not establish that one physical candidate carries both structures. A common candidate must have one representation and charge assignment at the boundary. If the ledgers require incompatible assignments, no shared witness object exists; the pullback condition of section 12 fails. $\square$

9.6 Gauge/anomaly interface closure

Theorem 9.9 (Gauge and anomaly interface closure). *For every interface-adequate completion, $\mathcal{G}_{\text{attach}}$ and $\mathcal{A}_q$ descend to the lawful interface quotient, preserve the fixed charged-current boundary, and factor through one common representation/charge/global-form witness ledger.*

Proof. Descent follows from gauge representative nonauthority and the explicit preservation of quotient-stable data. Boundary preservation is theorem 9.2. Quantum consistency is theorem 9.5. Common factorization follows from lemma 9.8; otherwise the interface describes incompatible projections rather than one completion. $\square$

No hard-coded anomaly row

A candidate description may not treat “anomaly free” as an independent Boolean while changing the representation, charge, or global-form witness on which the anomaly is computed. The anomaly status must be derived or certified from the same shared entries used by the boundary attachment.

10 Vacuum, deformation, and EFT/matching interface

10.1 Vacuum compatibility as a shared target condition

The target is situated in the realized electroweak vacuum and uses its charged-current boundary. A candidate completion can add new vacuum-sensitive structure, but it cannot silently replace the vacuum, change the residual electromagnetic boundary, or use an unrecorded scalar/background value to select an otherwise failing interface. The Realized Physical Interior treats vacuum as the non-null no-particle baseline of the same field, gauge, transport, and deformation architecture [22]; the Standard Model and electroweak symmetry analyses type the Higgs/vacuum realization as a downstream structural specialization rather than as a free repair reservoir [23, 24].

Definition 10.1 (Vacuum and deformation packet). *The packet $\mathcal{V}_{\text{def}}(X)$ contains:*

1. *the vacuum or background state on which the interface is defined;*
2. *compatibility with the fixed electroweak charged-current boundary and residual electromagnetic structure;*
3. *every candidate-specific order parameter, condensate, scalar background, or topological vacuum datum that changes a target witness;*
4. *an admitted small-deformation family $\mathcal{U}_\epsilon(X)$;*
5. *the continuation of boundary, channel, visible-mode, transport, anomaly, and support data under those deformations;*
6. *explicit target-fiber IDs for vacuum-changing proposals.*

Theorem 10.2 (Vacuum-support necessity). *Every same-target interface-adequate completion must coexist with the fixed electroweak vacuum realization and preserve the target charged-current boundary. A vacuum*

expectation value, condensate, scalar offset, or background representative cannot function as a hidden selector, anomaly repair, or source of interface standing unless its physical role appears explicitly in $\mathcal{V}_{\text{def}}$ and the shared witness ledger.

Proof. Changing the vacuum can change boundary representations, residual charges, anomaly status, propagation media, and the response channel. If the change affects none of those witnesses, it is interface-inert presentation or an unobservable candidate-internal distinction. If it affects one, it is tensor or anchor data and must be declared. Using an undeclared vacuum datum to make a failed candidate pass would add physical work after the target and candidate were fixed, violating the no-repair and no-hidden-load disciplines. $\square$

10.2 Deformation continuity

The interface theorem requires the interface itself to remain well typed under every deformation that the target declares admissible. This is not a naturalness or robustness demand: a numerical fit may hold only at one measured parameter value, and phase transitions, resonances, protected zeros, or singular fibers may be physically legitimate. The issue is only whether the relevant point or singular locus has been typed explicitly as a target-fiber anchor rather than used to change the interface definition after the fact.

Definition 10.3 (Admitted deformation family). *An admitted deformation family $\mathcal{U}_\epsilon(X)$ is a target-declared neighborhood of candidate/interface data in which source/detection semantics, active target fiber, operator category, and record comparison remain well typed. It may include changes of medium profile, effective coefficients, matching parameters, or other lawful physical tensors. Pure representative changes are already quotiented and are not deformations.*

Theorem 10.4 (Deformation-continuation theorem). *For every admitted deformation of an interface-adequate completion, the target identity, boundary object, channel quotient, visible-mode incidence, transport ledger, gauge/anomaly status, and active extension-fiber data must remain well typed. The theorem does not require the deformed candidate to continue fitting the fixed empirical target. If the interface is defined only on an isolated or singular locus, that locus must be declared as a target-fiber anchor; outside it, target adequacy may simply fail or become inapplicable.*

Proof. The interface coordinates are intended to be comparable under lawful continuation wherever the target declares a deformation admissible. If an admitted deformation makes the boundary, quotient, or consistency data ill typed, then the deformation leaves the declared comparison fiber or the interface specification is incomplete. Declaring an isolated point, threshold, or singular locus as an anchor yields the correct narrower fiber. No requirement of parameter insensitivity, genericity, radiative naturalness, or probabilistic typicality is imposed. $\square$

Remark 10.5 (What deformation continuity does not say). *A tuned parameter value, resonance, exact zero, phase boundary, or isolated phenomenological fit is not excluded. Quantitative predictions may change sharply or cease to match the target under deformation. The present analysis requires only explicit typing of the interface and of the fiber on which it is defined.*

10.3 Scale, operator order, and matching

A high-scale completion and a low-energy effective theory can have different fields and operators while producing the same target interface. Conversely, agreement at a few energies does not establish full interface identity. The scale/matching packet makes that distinction explicit.

Definition 10.6 (EFT and matching packet). *The packet $\mathcal{R}_{\text{EFT}}(X)$ records:*

1. *the scale interval and operator order of the description;*
2. *the active degrees of freedom and target-fiber ID;*

3. *matching maps between descriptions;*
4. *scheme and field-redefinition equivalences;*
5. *physical running, threshold, and nondecoupling effects;*
6. *the preservation of every target-relative comparison-profile component under matching.*

Theorem 10.7 (EFT-fiber discipline). *A UV completion and its low-energy effective description define the same canonical interface class only if a lawful matching map preserves the complete target specification, event-response family, continuation/support profile, gauge/anomaly/vacuum ledgers, visible-mode incidence, and every activated extension witness. Agreement on finitely many records or fitted coefficients is insufficient.*

Proof. Interface identity means equality of the complete augmented comparison profile, not numerical coincidence on a subset. A lawful matching map must therefore carry every active component and physical joint-incidence relation. If it does, the two descriptions differ by scale-dependent presentation and candidate content outside the visibility of the present target; they share one interface image. If it does not, the descriptions are distinct interface classes or different target fibers. $\square$

Proposition 10.8 (Scheme and field-redefinition covariance). *Renormalization scheme, basis, and local field redefinitions are representational equivalences when the complete matched interface is preserved. Physical running, thresholds, anomalous dimensions, and nondecoupling effects are interface tensor data when they alter $\mathcal{P}_v^+$.*

Proof. A scheme or field coordinate that changes no target witness is a lawful redescription. A change in scale dependence, threshold structure, or event response changes the matching profile and cannot be quotiented as skin. This is the general scheme/physical split applied to the present interface. $\square$

Theorem 10.9 (No RG or matching selector). *A renormalization scale, matching scale, scheme, truncation convention, or fitted Wilson coefficient cannot choose among otherwise distinct interface classes. Such data are either lawful representative coordinates, physical tensor data requiring coupled reevaluation, or target-fiber anchors.*

Proof. If the datum changes no profile witness, it is representative. If it changes a witness, it is physical data and cannot serve as a meta-level selector without being part of the candidate. If it changes the operator order or active scope, it changes the target fiber. These cases exhaust its roles in the present interface. $\square$

10.4 Vacuum/EFT overlap

A vacuum-sensitive operator can run, match, or disappear across thresholds. The vacuum and EFT packets must therefore share the same background and operator witnesses.

Lemma 10.10 (Vacuum–matching overlap). *If a matching map changes the vacuum/background data used by the boundary or channel, it is not interface-preserving unless the change is explicitly represented in both $\mathcal{V}_{\text{def}}$ and $\mathcal{R}_{\text{EFT}}$ and preserves the complete augmented comparison profile.*

Proof. A vacuum change can alter masses, charges, residual sectors, matter response, and anomaly conditions. A matching claim that ignores the change uses different physical ledgers on its two sides. Shared-ledger identity therefore fails unless the vacuum transformation is explicit and complete-profile-preserving. $\square$

10.5 Vacuum/deformation/EFT closure

Theorem 10.11 (Vacuum, deformation, and scale interface closure). *For every interface-adequate completion, the packets $\mathcal{V}_{\text{def}}$ and $\mathcal{R}_{\text{EFT}}$ descend to the lawful interface quotient, share common background and target-fiber witnesses, and preserve the nested role complex under every admitted deformation and lawful matching map.*

Proof. Vacuum support and deformation continuity are theorems 10.2 and 10.4. Scale and matching covariance are theorem 10.7 and proposition 10.8. The common background condition is lemma 10.10. If the nested role packet were lost under an admitted deformation or lawful matching, the candidate would cease to reproduce the target and the map would not be interface-preserving. $\square$

No naturalness premise

The interface theorem does not rank candidates by parameter size, radiative stability, tuning, typicality, or Bayesian weight. Deformation continuity is a typing and continuation condition, not a naturalness criterion. Candidate ranking, if ever admitted, would require a separate target and standing theorem.

11 Global form and generalized-symmetry relevance

11.1 Why a relevance theorem is required

Different faithful global forms of a gauge theory can share the same local Lie algebra and perturbative particle spectrum while differing in line operators, monopole sectors, theta periodicities, and generalized symmetries [1, 16, 32]. Higher-form symmetries act on extended operators rather than ordinary point particles [8]. Higher-group structure can intertwine ordinary flavor/background actions with one-form sectors [6]. Non-invertible defects can carry exact physical actions with non-group fusion and may be supported by defect-localized topological degrees of freedom rather than new bulk asymptotic states [5].

These facts make blanket exclusion untenable. They do not make every such structure part of the neutrino propagation target. The relevant question is target-relative: does the fixed interface require the distinction?

11.2 Core operator category

The mandatory core interface admits:

1. interaction-defined local source preparation classes;
2. local detector effect and event-record classes;
3. effective source-to-detection channel/instrument data;
4. local and causal continuation/support witnesses;
5. gauge, anomaly, vacuum, deformation, and EFT data required to make those local records consistent.

It does not, by default, admit Wilson, 't Hooft, or dyonic line operators; monopole sectors; background higher-form fields; topological junctions; or non-invertible defect insertions. Those objects define controlled extension fibers when introduced.

Definition 11.1 (Core-forgetful projection). *For an extended interface I^{ext} , let*

$$F_{\text{core}} : I^{\text{ext}} \longrightarrow I^{\text{core}}$$

forget line, surface, defect, junction, and topological data while retaining every point-local target record and mandatory consistency witness.

11.3 Global-form relevance

Let

$$G_{\Gamma} = \frac{SU(3)_c \times SU(2)_L \times U(1)_Y}{\Gamma}, \quad \Gamma \leq \mathbb{Z}_6.$$

The local charged-current boundary can be written in every faithful global-form fiber compatible with the known local representations, but extended sectors can differ.

Theorem 11.2 (Global-form relevance trichotomy). *For the fixed target, global-form data have exactly one of three statuses:*

1. **mandatory core:** *if changing Γ changes $\mathcal{B}_{CC}$, the core channel/record profile, or a mandatory anomaly/vacuum witness for every completion;*
2. **relevance-gated fiber:** *if Γ is detected only after line, monopole, theta, or other extended-sector data are admitted;*
3. **core-inert:** *if changing Γ changes no component of the core comparison profile.*

No fourth same-target status exists.

Proof. A global-form distinction either changes a core target witness or it does not. If it changes a witness uniformly, it is mandatory. If it changes no core witness but changes an extended-sector witness in a target that admits that sector, it is a fiber coordinate. If it changes no active witness, it is interface-inert at that scope. Any alleged fourth status would have to be physically relevant while changing neither core nor extension witnesses, which is impossible by the coordinate-admission test. $\square$

Proposition 11.3 (Core global-form status). *For the point-local core target, the faithful global-form coordinate is not forced as a discriminating interface coordinate by the oscillation records alone. It remains target metadata and becomes active only in extension fibers where line/monopole sectors or global-form-sensitive anomaly data enter the comparison profile.*

Proof. The core record and channel packets contain only local preparation/effect classes and their source-to-detection responses. These data factor through the local electroweak representation and are insensitive to the choice among globally distinct fibers that share the same local action. SYM-9 and SYM-11 show that the distinction appears in line, monopole, theta, or background sectors rather than in the sharp-local record alone [16, 24]. Therefore the coordinate is not a core discriminator. It remains an anchor for any extension fiber in which those sectors are admitted. $\square$

11.4 Higher-form and line sectors

Theorem 11.4 (Higher-form admission rule). *A one-form or line-sector coordinate belongs to the interface only when:*

1. *the active target fiber admits the relevant extended operators;*
2. *their endability, screening, linking, or charge action changes a target-admitted observable, consistency witness, or occupancy capability;*
3. *the coordinate is defined and quotiented at physical, not merely categorical, level;*
4. *it factors through the same witness ledger as the point-local boundary and channel data.*

Heavy probe lines, mathematical completions, and optional monopole policies do not automatically become core propagation-interface data.

Proof. Without an admitted extended operator, a one-form action has no target witness and fails the coordinate-admission test. If an extended operator is admitted but its variation changes no target profile component, the coordinate is inert. If it changes a witness, it is physically relevant in that fiber and must be included with its screening/endability and consistency data. The shared-ledger condition prevents an extended sector from being appended independently of the candidate that carries it. $\square$

11.5 Higher-group structure

Proposition 11.5 (Higher-group relevance test). *A higher-group coordinate is mandatory in a target fiber only if removing the mixed zero-form/one-form background transformation law while preserving the ordinary boundary and channel data destroys a target-admitted response or anomaly witness. If the point-local interface closes without that law, the higher-group data are optional or fiber-specific.*

Proof. The mixed transformation law is physically relevant only through its action on admitted backgrounds, extended operators, or anomaly data. If no such witness changes, it is not a mandatory coordinate. If a witness changes, deletion violates target adequacy in that fiber and the coordinate is required. This is the variation test applied to higher-group structure. $\square$

11.6 Non-invertible defects

Definition 11.6 (Non-invertible extension packet). *When activated, the non-invertible packet may include defect objects, fusion multiplicities N_{ab}^c , junction spaces V_{ab}^c , associators or F -data, braiding/linking, module action on local and extended operators, endability/screening, anomaly/inflow, gauging/condensation, and RG transport. Only entries that change the active comparison profile are retained.*

Theorem 11.7 (Non-invertible relevance theorem). *For the fixed target, non-invertible defect data are exactly:*

1. **required**, if every lawful realization of the nested role complex must carry the same target-visible defect action, fusion/junction structure, or selection rule;
2. **optional**, if some candidate completions carry such a defect while the core interface image is unchanged;
3. **irrelevant**, if the defect changes no active comparison-profile coordinate;
4. **target-changing**, if its definition requires a different massless limit, gauging choice, operator category, boundary, or UV target.

For the point-local core target, the present source packet does not establish the first branch.

Proof. The four branches are exhaustive by the coordinate-admission theorem and target-fiber discipline. A required defect must have a universal target witness. An optional defect has a candidate-specific witness in an activated extension fiber but is not forced by the core role complex. An irrelevant defect has no target witness. A defect requiring different target data is target-changing. The preceding role theorem and the present core record packet contain no defect, junction, or fusion observable shared by every realization, so they do not establish universal necessity. $\square$

Corollary 11.8 (No non-invertible-to-matter inference). *Even in an extension fiber where non-invertible data are active, their existence does not by itself imply new bulk propagating matter. Defect-localized topological support remains a distinct candidate class until later exhaustion.*

11.7 Resolved relevance result

Theorem 11.9 (Generalized-symmetry resolution). *The mandatory point-local core interface contains no independent global-form, one-form, higher-group, or non-invertible coordinate beyond the target metadata and consistency data already present. These structures are relevance-gated extension-fiber coordinates. They become mandatory only when an activated target fiber admits a distinguishing extended-operator or anomaly witness.*

Proof. The independently fixed point-local comparison profile admits boundary, channel, response, transport, gauge, anomaly, vacuum, deformation, and EFT witnesses, but no target-admitted extended-operator, background higher-form, or defect probe. By proposition 11.3, global form is not separated by those records

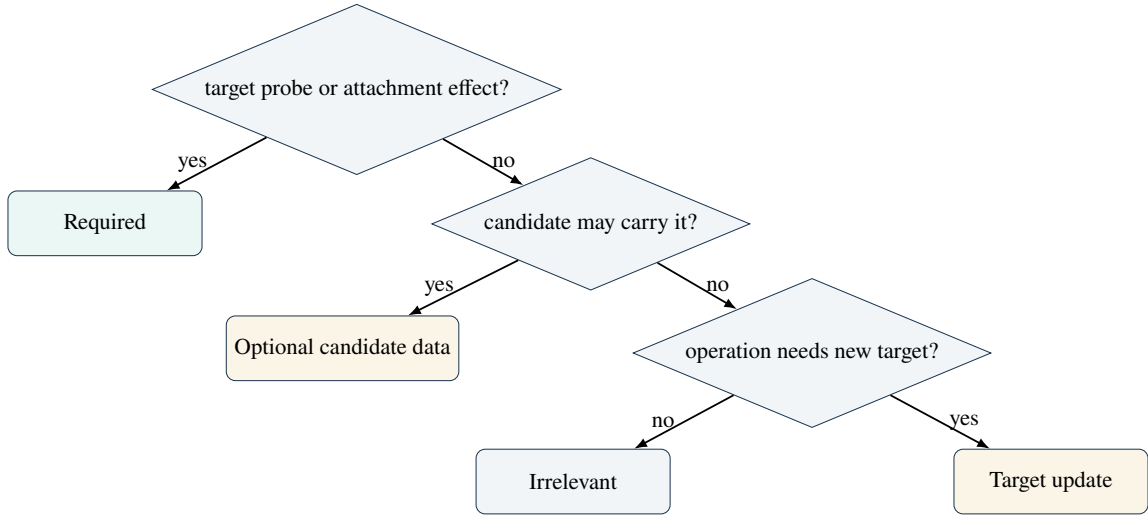


Figure 6: Pre-candidate activation discipline for global-form and generalized-symmetry data.

alone. By theorems 11.4 and 11.7 and proposition 11.5, the remaining generalized structures require extended operators, backgrounds, or defect observables not present in the core target. Their nonactivation in the core is therefore a comparison-profile relevance result, not an antecedent completeness assumption and not a denial of their physical existence. Activating the appropriate target fiber adds the corresponding packet and constraints. $\square$

11.8 Extension-fiber data requirements

When a microscopic completion activates an extension fiber, it must supply:

1. the faithful global-form and operator-category ID;
2. genuine line/surface/defect objects and their action;
3. screening, endability, linking, fusion, and junction data as applicable;
4. anomaly and inflow data on the same representation/global-form ledger;
5. locality/support and vacuum/deformation compatibility;
6. the projection to the core interface;
7. a proof that the additional data are candidate-level, target-mandatory, or interface-inert.

Structure	Core status	Activation condition	Blocked shortcut
Global form	Target metadata; not a core discriminator.	Line/monopole or global anomaly witness.	Same Lie algebra means same extended interface.
One-form/line sector	Relevance-gated.	Admitted extended operators with changed screening/endability or consistency.	Probe line equals dynamical particle.
Higher-group	Relevance-gated.	Mixed background law required by target response/anomaly.	Background notation becomes mandatory automatically.
Non-invertible defect	Optional/target-changing at core scope.	Universal or candidate-specific defect witness in active fiber.	Non-invertibility predicts new matter.

Table 8: Resolved generalized-symmetry relevance.

Resolved optionality is closure, not omission

The result does not imply that generalized symmetry is irrelevant to all neutrino completions. It proves only that the point-local target does not force these coordinates. Completions that activate an extension fiber must still be compared with the additional consistency and shared-realization conditions carried by that fiber.

12 Simultaneous compatibility and shared physical witnesses

12.1 The gluing problem

A candidate interface is not a list of separately acceptable statements. A channel can reproduce the target records using one medium profile; a gauge ledger can be consistent using one representation; an anomaly calculation can close using another; and a defect sector can be well defined in a different global-form fiber. Every pair can be presented as compatible under a suitable local identification while no single physical completion carries all of them simultaneously. This is the interface analogue of the freely factored gauge-arena problem isolated by SR1 [19].

The interface therefore requires one typed witness ledger through which every standing-bearing coordinate factors.

12.2 Shared physical witness structure

Definition 12.1 (Shared physical witness ledger). *For a raw interface $I = \tilde{\mathfrak{F}}_\nu(X)$, define*

$$\text{Wit}_\nu(I) = (W_{\text{tgt}}, W_{\text{CC}}, W_{\text{ch}}, W_{\text{rec}}, W_{\text{loc}}, W_{\text{env}}, W_{\text{gauge}}, W_{\text{anom}}, W_{\text{vac}}, W_{\text{def}}, W_{\text{EFT}}, W_{\text{ext}}, W_{\text{obs}}). \quad (26)$$

Each entry denotes a typed physical witness class. Archival identifiers and provenance annotations may accompany these witnesses, but such metadata are not part of the physical witness class.

Definition 12.2 (Coordinate factorization). *An interface coordinate I_k factors through $\text{Wit}_\nu(I)$ when every target-relevant assertion made by I_k is supported by a typed witness in the ledger, uses the same physical overlap entries as every coordinate with which it interacts, and is invariant under every lawful redescription preserving the physical witness diagram.*

Examples of required overlap include: the boundary labels used by the channel are the boundary labels used by the response analysis; the medium profile used in the channel is the profile used in the transport and deformation tests; the representation and charge data used in boundary attachment are the data used in

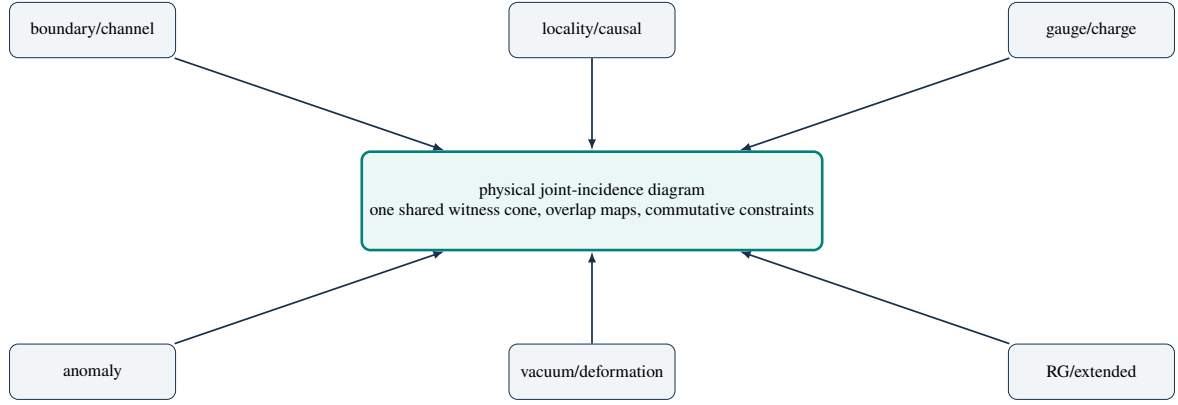


Figure 7: Separate projection checks define one physical interface only when they factor through a common witness cone. The canonical joint-incidence profile retains this gluing structure while forgetting witness names and archival identifiers.

anomaly evaluation; and any extended operator is defined in the same global-form and operator-category fiber as its anomaly and support ledgers.

12.3 Constraint family

Let

$$\mathcal{K}_v^{(2),\text{core}} = \{K_{\text{tgt}}, K_{\text{bdy}}, K_{\text{ch}}, K_{\text{resp}}, K_{\text{nest}}, K_{\text{loc}}, K_{\text{env}}, K_{\text{gauge}}, K_{\text{anom}}, K_{\text{vac}}, K_{\text{def}}, K_{\text{EFT}}, K_{\text{law}}\}. \quad (27)$$

Let $\mathcal{K}_v^{(2),\text{rel}}$ contain only those global-form or generalized-symmetry predicates activated by [theorem 11.9](#). The full family is

$$\mathcal{K}_v^{(2)} = \mathcal{K}_v^{(2),\text{core}} \cup \mathcal{K}_v^{(2),\text{rel}}.$$

Define the interface survivor predicate

$$\text{Surv}_v^{\text{int}}(I) \iff \bigwedge_{K \in \mathcal{K}_v^{(2)}} K(I). \quad (28)$$

Each predicate is fail-closed. A failed witness is not repaired by a later explanation or by changing only its display.

12.4 Raw, pullback, and admissible presentation spaces

For every coordinate k , let $\mathcal{M}_k$ be its presentation-quotiented moduli object after removing only independently certified skin. Define

$$\mathcal{M}_{\text{raw}} = \prod_k \mathcal{M}_k. \quad (29)$$

This is not the physical interface space. It is the formal product obtained by choosing each coordinate independently.

Let $\mathcal{M}_{\text{pull}} \subseteq \mathcal{M}_{\text{raw}}$ be the subobject of tuples admitting one shared physical witness ledger with agreement on every mandatory overlap. Let

$$\mathcal{M}_{\text{adm}} = \left\{ m \in \mathcal{M}_{\text{pull}} : K(m) = 1 \text{ for every } K \in \mathcal{K}_v^{(2)} \right\}. \quad (30)$$

The simultaneous-collapse chain is

$$\mathcal{M}_{\text{adm}} \hookrightarrow \mathcal{M}_{\text{pull}} \hookrightarrow \mathcal{M}_{\text{raw}}. \quad (31)$$

Define the candidate subclasses

$$\mathfrak{C}_v^{\text{sim}} := \{X \in \mathfrak{C}_v : \widetilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{pull}}\}, \quad (32)$$

$$\mathfrak{C}_v^{\text{adm}} := \{X \in \mathfrak{C}_v : \widetilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{adm}}\}. \quad (33)$$

A candidate outside $\mathfrak{C}_v^{\text{sim}}$ is not a different presentation of one physical interface. It is a witness-split coordinate product. A candidate in $\mathfrak{C}_v^{\text{sim}} \setminus \mathfrak{C}_v^{\text{adm}}$ is simultaneous but fails at least one active physical constraint.

12.5 Pullback necessity

Theorem 12.3 (Shared-ledger pullback necessity). *If $\text{InterfaceAdeq}(X, T_v^{\text{osc}})$ holds, then $X \in \mathfrak{C}_v^{\text{adm}} \subseteq \mathfrak{C}_v^{\text{sim}}$. Equivalently, every interface-adequate completion has one shared physical witness ledger and satisfies all overlap and active constraint clauses.*

Proof. Suppose an interface-adequate completion contains a coordinate that does not factor through a common ledger. Then at least one of four situations occurs: the coordinate uses an undeclared witness; it changes a shared witness without updating the other coordinates; it is evaluated in a different target fiber; or it claims physical force from presentation data not preserved by the lawful quotient. The first is hidden load, the second is an inconsistent coupled candidate, the third is target drift, and the fourth is representative privilege. Each contradicts a clause of definition 4.9. Hence $\widetilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{pull}}$. The remaining adequacy clauses are exactly the active constraints, so $\widetilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{adm}}$. $\square$

12.6 Pairwise compatibility is insufficient

Theorem 12.4 (Pairwise compatibility insufficiency). *Pairwise compatibility among interface coordinates does not imply the existence of a globally compatible shared witness ledger.*

Proof. Consider three coordinate packets C , A , and D representing channel/environment, anomaly/gauge, and defect/global-form data. Let the channel packet use environment/representation witness w_0 , the anomaly packet be neutral only on w_1 , and the defect packet be defined only in target fiber w_2 . Suppose there are local pairwise identifications

$$\varphi_{CA} : w_0 \sim w_1, \quad \varphi_{AD} : w_1 \sim w_2, \quad \varphi_{CD} : w_0 \sim w_2,$$

each valid only after forgetting a different witness component, but the composite around the triple is not identity on the full target ledger. Then every pair admits a reduced common subledger, while no single ledger preserves environment, representation, anomaly, global-form, and operator-category data simultaneously. Thus pairwise compatibility does not imply global compatibility. A theorem converting the pairwise data into a global ledger would be an additional completion principle and is not available from the target or the AASC kernel. $\square$

Example 12.5 (Finite witness-split obstruction). *Let $W_{\text{env}} = \{e_0, e_1\}$, $W_{\text{rep}} = \{q_0, q_1\}$, and $W_{\text{glob}} = \{\Gamma_0, \Gamma_1\}$. The channel closes only on (e_0, q_0) , the anomaly only on (q_1, Γ_0) , and a defect only on (e_1, Γ_1) . Pairwise projections can be made to agree by forgetting respectively e , Γ , or q , but no tuple (e, q, Γ) satisfies all three. The raw product contains the three component records; the pullback does not.*

12.7 Agreement clauses

Every simultaneous interface satisfies at least the following agreement clauses:

1. boundary labels in W_{CC} are the labels used in W_{ch} and W_{rec} ;
2. the event predictions from which $\mathfrak{M}_{\text{vis}}$ is extracted are the predictions of the same channel packet;
3. the environment profile used in the channel is the profile used in support and deformation tests;

4. representation, charge, and global-form entries used in gauge attachment are those used in anomaly evaluation;
5. the vacuum packet does not silently alter the boundary, environment, or anomaly ledger;
6. EFT matching preserves the same interface and active target fiber, not only a finite record subset;
7. an activated line or defect sector uses the same operator category, global form, support, anomaly, and vacuum witnesses;
8. every coordinate and normalization change is either lawful skin or a declared tensor-level candidate variation.

12.8 The joint-incidence profile determines pullback membership

For a physical overlap diagram D , write $\text{Cone}_{\text{req}}(D) \neq \emptyset$ when there exists one typed cone whose legs realize every mandatory coordinate-to-witness factorization and whose required overlap subdiagrams commute.

Theorem 12.6 (Pullback determination). *For every raw interface presentation I ,*

$$I \in \mathcal{M}_{\text{pull}} \iff \text{Cone}_{\text{req}}(\mathcal{D}_{\text{sim}}(I)) \neq \emptyset. \quad (34)$$

Consequently pullback membership is a property of $\mathcal{J}_v^{\text{sim}}(I)$, not of witness identifiers or archival encodings.

Proof. An element of $\mathcal{M}_{\text{pull}}$ is, by definition, a coordinate tuple admitting one shared typed witness ledger with all mandatory overlaps. Such a ledger is precisely a cone over $\mathcal{D}_{\text{sim}}(I)$ satisfying the required commutative diagrams. Conversely, a required cone supplies the common ledger and all factorization maps. Type-preserving physical diagram isomorphisms preserve existence of such a cone, so the property descends to $\mathcal{J}_v^{\text{sim}}(I)$. $\square$

Theorem 12.7 (Pullback congruence). *If $I \equiv_{\mathcal{P}^+} J$, then*

$$I \in \mathcal{M}_{\text{pull}} \iff J \in \mathcal{M}_{\text{pull}}.$$

Thus complete-profile saturation cannot identify a simultaneous interface presentation with a witness-split raw tuple.

Proof. Complete-profile equality includes $\mathcal{J}_v^{\text{sim}}(I) = \mathcal{J}_v^{\text{sim}}(J)$. By theorem 12.6, pullback membership is determined by the existence of a required cone in that diagram class. Hence the membership values agree. $\square$

12.9 Constraint factorization and admissibility congruence

Theorem 12.8 (Constraint factorization through the complete profile). *For every active $K \in \mathcal{K}_v^{(2)}$, there exists a target-fixed predicate $\widehat{K}$ on complete profiles such that for all $I \in \mathcal{M}_{\text{pull}}$,*

$$K(I) = \widehat{K}(\mathcal{P}_v^+(I)). \quad (35)$$

Proof. The target, boundary, channel, response, nesting, locality, environment, gauge, anomaly, vacuum, deformation, EFT, and activated extension predicates are evaluated from the corresponding components of $\mathcal{P}_v^0$. Whether those components are evaluated on the same physical target, boundary, environment, representation/charge, vacuum, scale, and operator witnesses is determined by $\mathcal{J}_v^{\text{sim}}$. The lawful-quotient predicate is fixed by the coordinate-specific quotient rules and their preservation in the same diagram class. No active predicate uses a candidate name, an archival ID, or data outside these two profile components. Therefore each predicate factors through $\mathcal{P}_v^+ = (\mathcal{P}_v^0, \mathcal{J}_v^{\text{sim}})$. $\square$

Theorem 12.9 (Admissibility congruence). *If $I, J \in \mathcal{M}_{\text{pull}}$ and $I \equiv_{\mathcal{P}^+} J$, then*

$$I \in \mathcal{M}_{\text{adm}} \iff J \in \mathcal{M}_{\text{adm}}.$$

Hence the survivor predicate is constant on complete-profile classes restricted to the simultaneous domain.

Proof. By theorem 12.8, every active constraint has the same value on I and J . Membership in $\mathcal{M}_{\text{adm}}$ is the conjunction of those values over $\mathcal{M}_{\text{pull}}$. The conjunction therefore agrees. $\square$

Corollary 12.10 (Survivor congruence). *The survivor predicate and every active constraint status descend to the complete-profile quotient on the simultaneous domain. In particular, no $\equiv_{\mathcal{P}^+}$ -class intersects both $\mathcal{M}_{\text{adm}}$ and $\mathcal{M}_{\text{pull}} \setminus \mathcal{M}_{\text{adm}}$.*

Proof. This is the class-level form of theorem 12.9. If one representative of a simultaneous complete-profile class satisfies every active constraint, then every simultaneous representative does; if one fails, every representative in that class fails the same factored constraint profile. $\square$

12.10 Variation collapse

Theorem 12.11 (Interface variation collapse). *Let $I \in \mathcal{M}_{\text{adm}}$. A standing-relevant variation of one interface coordinate cannot remain admissible while every overlapping coordinate and shared witness is held fixed, unless the variation is ledger-neutral skin. Every non-skin variation defines a new coupled interface candidate and requires reevaluation of the complete constraint family.*

Proof. Because I lies in the pullback subobject, each coordinate is a function of shared witness entries. A standing-relevant variation changes at least one such entry or the coordinate's relation to it. Holding the overlapping coordinates fixed makes their witness maps disagree, so the varied tuple leaves $\mathcal{M}_{\text{pull}}$. If the variation changes no witness, it is skin. The only admissible non-skin route is to update the shared ledger and reevaluate every dependent coordinate, producing a new candidate. $\square$

Corollary 12.12 (No free factorization). *Boundary, channel, response, environment, gauge, anomaly, vacuum, scale, and activated generalized-symmetry data are not freely factored degrees of freedom on $\mathcal{M}_{\text{adm}}$.*

12.11 Fail-closed behavior

Theorem 12.13 (No interface repair reservoir). *There is no hidden reservoir of admissibility from which a failed interface can draw repair. Any attempted repair is exactly one of:*

1. *already present in the shared witness ledger;*
2. *ledger-neutral skin and therefore unable to change the failure;*
3. *a tensor-level change defining a new coupled candidate;*
4. *an anchor change defining a new target fiber;*
5. *an inadmissible post-hoc selector or narrative.*

Proof. The anchor/tensor/skin trichotomy classifies every proposed change relative to the fixed ledger. If the change is already present, no repair occurred. If it is skin, it changes no failed witness. If it changes a physical witness, it is tensor-level and produces a new candidate. If it changes target identity, it is an anchor move. A residual move that changes standing without changing any of these data is an untyped selector and violates the AMetric/no-repair discipline. The cases are exhaustive. $\square$

12.12 Simultaneous-compatibility closure theorem

Theorem 12.14 (Simultaneous interface closure). *The physical interface quotient must be formed after the fail-closed reductions*

$$\mathcal{M}_{\text{raw}} \supseteq \mathcal{M}_{\text{pull}} \supseteq \mathcal{M}_{\text{adm}}.$$

The relation $\equiv_{\mathcal{P}^+}$ is congruent with both subobjects by theorems 12.7 and 12.9. Therefore quotienting the simultaneous and survivor subobjects cannot confer standing across a witness-split or inadmissible representative.

Proof. Pullback necessity and pairwise insufficiency show that the raw product is not the physical domain. Pullback congruence prevents a complete-profile class from mixing pullback and nonpullback presentations. Constraint factorization and admissibility congruence prevent it from mixing admissible and inadmissible simultaneous presentations. The canonical quotients can therefore be formed on $\mathcal{M}_{\text{pull}}$ and $\mathcal{M}_{\text{adm}}$ without ambiguity. Variation behavior is theorem 12.11. $\square$

Representational boundary. The physical content of $\mathcal{J}_v^{\text{sim}}$ is the isomorphism class of the witness-overlap diagram: witness identity classes, overlap maps, commutativity requirements, and common-cone status. Archival identifiers and implementation-specific encodings are discarded. The shared-realization theorem therefore depends on physical incidence, not on any particular record-keeping convention.

13 Target-semantic invariant, canonical quotient, and interface completeness

13.1 The complete profile as an invariant of the target-evaluator algebra

The evaluator grammar of section 3 was derived from the fixed target before the comparison profile was formed. The first task is therefore not to define semantic equivalence by the profile, but to prove that the finite-locus profile is a complete invariant for the independently defined evaluator algebra.

Lemma 13.1 (Primitive-signature correspondence). *For same-target raw evaluation presentations $I, J \in \mathcal{R}_v^{\text{raw}}$,*

$$\mathcal{P}_v^+(I) = \mathcal{P}_v^+(J) \iff g(I) \cong_g g(J) \text{ for every } g \in \text{Gen}(\mathbb{T}_v^{(2)}), \quad (36)$$

where $\cong_g$ is the target-authorized output equivalence for the generator family containing g .

Proof. By equations (16) and (17), equality of $\mathcal{P}_v^0$ is exactly equality of the lawful output signatures of $\mathbf{G}_{\text{rec}}, \mathbf{G}_{\text{cont}}, \mathbf{G}_{\text{loc}}, \mathbf{G}_{\text{gauge}}, \mathbf{G}_{\text{anom}}, \mathbf{G}_{\text{vac}}, \mathbf{G}_{\text{EFT}}$, and the activated $\mathbf{G}_{\text{ext}}$ family. By equation (18), equality of $\mathcal{J}_v^{\text{sim}}$ is exactly equality of the $\mathbf{G}_{\text{sim}}$ signature. These nine families form $\text{Gen}(\mathbb{T}_v^{(2)})$. The reverse implication reads the same identifications from generator outputs to the profile tuple. $\square$

Lemma 13.2 (Target-trace recovery from primitive signatures). *For same-target raw evaluation presentations $I, J \in \mathcal{R}_v^{\text{raw}}$,*

$$\boxed{\mathcal{P}_v^+(I) = \mathcal{P}_v^+(J) \iff \tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J).} \quad (37)$$

Proof. If the complete profiles agree, then by lemma 13.1 every primitive evaluator agrees. By construction of $\text{Gen}(\mathbb{T}_v^{(2)})$ in equation (11), the nine coordinate projections of the refined target trace are themselves primitive evaluators. Their joint equality reconstructs equality of the five-factor target trace by equation (10).

Conversely, if the target traces agree, every primitive evaluator has equal output because each is a lawful one-factor evaluation of a target-trace projection. The primitive-signature correspondence then gives equality of $\mathcal{P}_v^0$ and $\mathcal{J}_v^{\text{sim}}$, hence of $\mathcal{P}_v^+$. $\square$

Theorem 13.3 (Complete-invariant theorem for target evaluators). *For same-target raw evaluation presentations $I, J \in \mathcal{R}_v^{\text{raw}}$,*

$$\boxed{I \equiv_{\mathbb{T}_v^{(2)}} J \iff \mathcal{P}_v^+(I) = \mathcal{P}_v^+(J).} \quad (38)$$

Thus $\mathcal{P}_v^+$ is a complete invariant for the target-evaluator algebra $\text{Eval}(\mathbb{T}_v^{(2)})$, not merely a declared list of comparison coordinates.

Proof. Suppose first that $I \equiv_{\mathbb{T}_v^{(2)}} J$. Every primitive generator is a semantic target-admitted evaluator by Definition 3.10 and equation (11); therefore its output agrees on I and J . The primitive-signature correspondence gives $\mathcal{P}_v^+(I) = \mathcal{P}_v^+(J)$.

Conversely, suppose the complete profiles agree. By lemma 13.2, $\tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J)$. Every $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$ is target-extensional by definition, so $E(I) = E(J)$ with common definedness. Hence $I \equiv_{\mathbb{T}_v^{(2)}} J$ by Definition 3.14. No finite evaluator derivation is used in this direction. The independent target-evaluator generation result (theorem 3.13) additionally shows that the saturated evaluator algebra $\text{Eval}(\mathbb{T}_v^{(2)})$ is exactly the same semantic class. $\square$

Corollary 13.4 (Profile relation equals target-semantic equivalence). *Within every fixed target fiber,*

$$I \equiv_{\mathcal{P}^+} J \iff I \equiv_{\mathbb{T}_v^{(2)}} J.$$

Consequently the saturated lawful groupoid of definition 4.4 may be represented either semantically by all target evaluators or finitely by the complete profile.

13.2 Lawful equivalence on the simultaneous domain

The complete profile is evaluable on every raw presentation, but the canonical physical quotient is formed only after the shared-ledger pullback has established one simultaneous physical interface.

Definition 13.5 (Lawful simultaneous-interface equivalence). *For $I, J \in \mathcal{M}_{\text{pull}}$ in the same fixed target fiber, write*

$$I \simeq_{\text{int}} J \iff I \equiv_{\mathbb{T}_v^{(2)}} J. \quad (39)$$

By theorem 13.3, equivalently

$$I \simeq_{\text{int}} J \iff I \equiv_{\mathcal{P}^+} J \iff \mathcal{P}_v^+(I) = \mathcal{P}_v^+(J).$$

A constructive zig-zag in $\mathcal{G}_v^{\text{gen}}$ is a sufficient witness, but absence of a known constructive zig-zag is not evidence of physical inequivalence.

Theorem 13.6 (Equivalence relation). *The relation $\simeq_{\text{int}}$ is an equivalence relation on $\mathcal{M}_{\text{pull}}$ in every fixed target fiber, and its restriction to $\mathcal{M}_{\text{adm}}$ is an equivalence relation there.*

Proof. Target-semantic equivalence is an equivalence relation by theorem 3.15. Restriction of an equivalence relation to a subset remains an equivalence relation. Pullback and admissibility congruence from theorems 12.7 and 12.9 ensure that $\mathcal{M}_{\text{pull}}$ and $\mathcal{M}_{\text{adm}}$ are unions of semantic-equivalence classes. $\square$

13.3 Canonical simultaneous and admissible interface spaces

Definition 13.7 (Canonical interface spaces). *Define*

$$\mathfrak{S}_v^{\text{sim}} := \mathcal{M}_{\text{pull}} / \simeq_{\text{int}}, \quad (40)$$

$$\mathfrak{S}_v^{\text{adm}} := \mathcal{M}_{\text{adm}} / \simeq_{\text{int}}. \quad (41)$$

By theorem 12.9, $\mathfrak{S}_v^{\text{adm}}$ is canonically identified with a subset of $\mathfrak{S}_v^{\text{sim}}$.

For $X \in \mathfrak{C}_v^{\text{sim}}$, define

$$\Pi_v(X) := [\tilde{\mathfrak{S}}_v(X)]_{\simeq_{\text{int}}} \in \mathfrak{S}_v^{\text{sim}}. \quad (42)$$

A candidate outside $\mathfrak{C}_v^{\text{sim}}$ has no canonical simultaneous physical interface image; it remains a witness-split raw coordinate product.

Theorem 13.8 (Quotient well-definedness). *Every mandatory core coordinate, every activated extension coordinate, the shared physical incidence profile, every target evaluator, and every active constraint status have well-defined values on $\simeq_{\text{int}}$ -classes. The projection $\Pi_{\mathbf{v}} : \mathfrak{C}_{\mathbf{v}}^{\text{sim}} \rightarrow \mathfrak{I}_{\mathbf{v}}^{\text{sim}}$ and the inclusion $\mathfrak{I}_{\mathbf{v}}^{\text{adm}} \hookrightarrow \mathfrak{I}_{\mathbf{v}}^{\text{sim}}$ are therefore well defined.*

Proof. Semantic equivalence makes every evaluator in $\text{Eval}(\mathbb{T}_{\mathbf{v}}^{(2)})$ constant on a class. The complete-invariant theorem makes every profile coordinate constant as well. Equality of $\mathcal{J}_{\mathbf{v}}^{\text{sim}}$ makes the physical overlap diagram constant. By theorem 12.8, every active constraint is a function of $\mathcal{P}_{\mathbf{v}}^+$ and is therefore constant on the same class. Admissibility congruence prevents a class from containing both an admissible and inadmissible simultaneous representative. $\square$

Corollary 13.9 (No standing transfer across quotient). *No witness-split or inadmissible presentation acquires membership in $\mathfrak{I}_{\mathbf{v}}^{\text{adm}}$ through equivalence to an admissible presentation. Such cross-status equivalence is excluded by pullback and admissibility congruence.*

13.4 Two-sided quotient control

Theorem 13.10 (No overquotient theorem). *If two simultaneous same-target presentations are distinguished by any target-admitted evaluator,*

$$\exists E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_{\mathbf{v}}^{(2)}) : E(I) \neq E(J),$$

with definedness included, then $I \not\simeq_{\text{int}} J$.

Proof. The evaluator difference is exactly failure of target-semantic equivalence by Definition 3.14. Lawful interface equivalence is the restriction of target-semantic equivalence to the simultaneous domain by definition 13.5. Hence no target-admitted physical difference can be erased by restricted record agreement or by a representative transformation. $\square$

Theorem 13.11 (No underquotient theorem). *If two simultaneous same-target presentations are indistinguishable by every target-admitted evaluator, then they define the same canonical interface class whether or not a constructive redescription zig-zag has been enumerated:*

$$I \equiv_{\mathbb{T}_{\mathbf{v}}^{(2)}} J \implies I \simeq_{\text{int}} J.$$

Equivalently, complete-profile equality is sufficient.

Proof. The first implication is definition 13.5. Complete-profile sufficiency follows from theorem 13.3. The explicit generator family $\mathcal{G}_{\mathbf{v}}^{\text{gen}}$ is constructive evidence in familiar cases but is not the semantic completeness premise. Because simultaneous-incidence evaluators belong to $\text{Eval}(\mathbb{T}_{\mathbf{v}}^{(2)})$, the semantic quotient cannot identify a shared-ledger interface with a witness-split tuple or two simultaneous interfaces with different overlap structure. $\square$

Theorem 13.12 (Exact lawful quotient). *For $I, J \in \mathcal{M}_{\text{pull}}$ in one fixed target fiber,*

$$I \simeq_{\text{int}} J \iff I \equiv_{\mathbb{T}_{\mathbf{v}}^{(2)}} J \iff I \equiv_{\mathcal{P}^+} J \iff \mathcal{P}_{\mathbf{v}}^+(I) = \mathcal{P}_{\mathbf{v}}^+(J). \quad (43)$$

Thus the canonical physical interface is the target-semantic quotient of the shared-ledger pullback, represented exactly by the finite complete profile.

Proof. The first equivalence is definition 13.5. The second and third are theorem 13.3 and definition 4.3. Constructive-generator soundness is theorem 4.2. Quotient ordering and pullback/admissibility congruence ensure the quotient is taken on the correct simultaneous physical domain and preserves the survivor subobject. $\square$

Theorem 13.13 (Profile-quotient identification). *On each fixed target fiber the maps*

$$\begin{aligned} F_{\text{sim}} : \mathfrak{S}_v^{\text{sim}} &\longrightarrow \text{Im}(\mathcal{P}_v^+ |_{\mathcal{M}_{\text{pull}}}), & F_{\text{sim}}([I]) &= \mathcal{P}_v^+(I), \\ F_{\text{adm}} : \mathfrak{S}_v^{\text{adm}} &\longrightarrow \text{Im}(\mathcal{P}_v^+ |_{\mathcal{M}_{\text{adm}}}), & F_{\text{adm}}([I]) &= \mathcal{P}_v^+(I) \end{aligned}$$

are bijections. Equivalently,

$$\mathfrak{S}_v^{\text{sim}} \cong \mathcal{M}_{\text{pull}} / \equiv_{\mathcal{P}^+} \cong \text{Im}(\mathcal{P}_v^+ |_{\mathcal{M}_{\text{pull}}}),$$

and

$$\mathfrak{S}_v^{\text{adm}} \cong \mathcal{M}_{\text{adm}} / \equiv_{\mathcal{P}^+} \cong \text{Im}(\mathcal{P}_v^+ |_{\mathcal{M}_{\text{adm}}}).$$

Proof. Well-definedness follows from theorem 13.12. Surjectivity holds by definition of each restricted image. If two classes have the same profile, their representatives are target-semantically equivalent by theorem 13.3; hence the classes coincide. Admissibility congruence protects the second map from nonadmissible representatives. $\square$

Corollary 13.14 (Componentwise agreement is insufficient). *Equality of $\mathcal{P}_v^0$ alone does not imply interface equality. The physical joint-incidence signature must also agree. In particular, a Frankenstein tuple and a shared-ledger interface can be componentwise identical while remaining target-semantically distinct through $\mathcal{G}_{\text{sim}}$.*

Remark 13.15 (Where the nontrivial completeness burden resides). *The exact quotient is not self-certifying. Semantic completeness comes from the independently defined target-facing trace, semantic use/evaluator extensionality, target-trace descent, and the trace/profile recovery theorem; the evaluator-generation theorem then proves that the saturated evaluator algebra represents exactly that semantic class. The remaining load-bearing work is coordinate formation and quotient soundness, shared-ledger factorization, generalized-symmetry relevance, and constraint congruence. A purported same-target evaluator that distinguishes equal target traces would depend on data outside the fixed target and would therefore expose a target-scope error rather than an omitted algebraic formula.*

13.5 Factorization through the canonical packet

For each primitive component there is a canonical factorization through the interface packet:

$$P_{\text{rec}} = F_{\text{rec}}(\mathcal{B}_{\text{CC}}, [I_\Lambda]_{\text{ch}}, \mathfrak{M}_{\text{vis}}), \quad (44)$$

$$C_{\text{cont}} = F_{\text{cont}}([I_\Lambda]_{\text{ch}}, \mathcal{L}_{\text{tr}}, \mathcal{E}_{\text{env}}, \mathcal{R}_{\text{EFT}}), \quad (45)$$

$$S_{\text{loc}} = F_{\text{loc}}(\mathcal{L}_{\text{tr}}, \mathcal{E}_{\text{env}}), \quad (46)$$

$$G_{\text{attach}} = F_{\text{gauge}}(\mathcal{B}_{\text{CC}}, \mathcal{G}_{\text{attach}}), \quad (47)$$

$$A_{\text{q}} = F_{\text{anom}}(\mathcal{G}_{\text{attach}}, \mathcal{A}_{\text{q}}, \mathfrak{S}_v^{\text{ext}}), \quad (48)$$

$$V_{\text{def}} = F_{\text{vac}}(\mathcal{V}_{\text{def}}, \mathcal{E}_{\text{env}}, \mathcal{G}_{\text{attach}}), \quad (49)$$

$$R_{\text{EFT}} = F_{\text{scale}}(\mathcal{R}_{\text{EFT}}, \mathcal{V}_{\text{def}}, [I_\Lambda]_{\text{ch}}), \quad (50)$$

$$X_{\text{ext}} = F_{\text{ext}}(\mathfrak{S}_v^{\text{ext}}) \quad \text{in activated fibers}, \quad (51)$$

$$\mathcal{J}_v^{\text{sim}} = F_{\text{sim}}(\text{Wit}_v, \text{Overlap}, \text{Cone}_{\text{req}}). \quad (52)$$

Lemma 13.16 (Complete-profile factorization). *Every component of $\mathcal{P}_v^+ = (\mathcal{P}_v^0, \mathcal{J}_v^{\text{sim}})$ factors through the canonical simultaneous interface packet by equations (44) to (52).*

Proof. The record profile is determined by boundary, channel, response, and lawful analysis data. Continuation and support are determined by the channel, transport, environment, and scale packets. Gauge attachment and anomaly profiles are the corresponding common-witness ledgers. Vacuum and deformation data determine the physical background response. Scale and matching are explicit in the EFT packet.

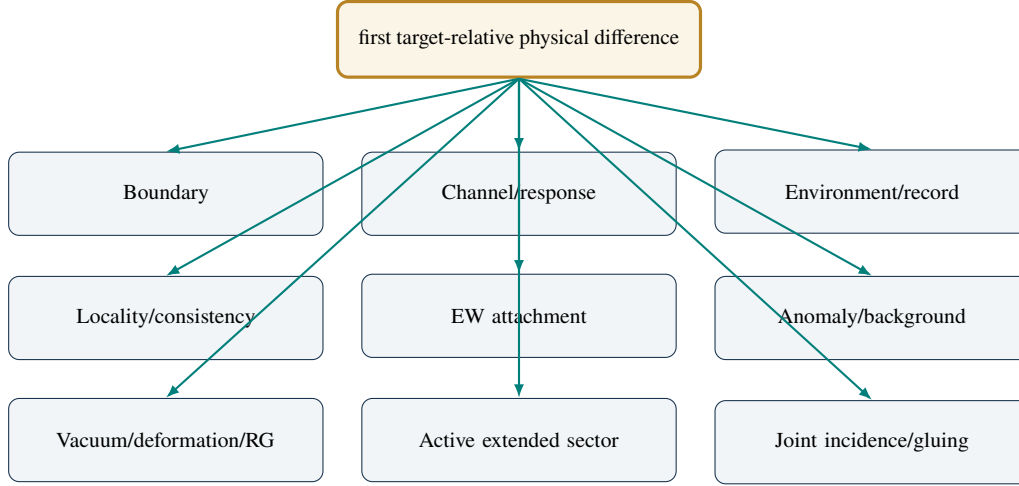


Figure 8: Every target-admitted first difference enters one of nine loci, including the physical incidence by which otherwise identical coordinates are jointly realized. This semantic grammar, not a finite catalogue, drives the completeness proof.

Extension data are present exactly in activated fibers. The joint profile is the isomorphism class of the physical witness-overlap diagram, including mandatory factorization maps and cone existence. Thus the factorization concerns one simultaneous physical interface rather than independently chosen projections. $\square$

Theorem 13.17 (Target-evaluator factorization). *For every $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) = \text{Eval}(\mathbb{T}_v^{(2)})$ there is a well-defined evaluator*

$$\widehat{E} : \mathfrak{I}_v^{\text{sim}} \longrightarrow V_E / \cong_E$$

such that, for every $I \in \mathcal{M}_{\text{pull}}$,

$$E(I) = \widehat{E}([I]_{\approx_{\text{int}}})$$

in the target-lawful output quotient.

Proof. Let $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$. By target-trace descent and theorem 3.11, E is a target-lawful function of the nine refined trace factors. By lemma 13.2, $\mathcal{P}_v^+$ is a complete invariant of that target trace, and by lemma 13.16 the complete profile factors through the canonical simultaneous packet. Hence E is constant on $\approx_{\text{int}}$ -classes. Define $\widehat{E}([I]) := E(I)$. This is well defined by target-semantic equivalence and gives the stated factorization. The equality $\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) = \text{Eval}(\mathbb{T}_v^{(2)})$ from theorem 3.13 gives the same result for the saturated evaluator algebra. No evaluator-term induction is required. $\square$

13.6 Evaluator-first difference routing

A first physical difference is defined semantically: some target-admitted evaluator returns different outputs. The dependency order used for exposition is

$$\begin{aligned} &\text{target/boundary} \prec \text{channel/response} \prec \text{transport/environment} \prec \text{gauge/anomaly} \\ &\prec \text{vacuum/deformation/EFT} \prec \text{activated extension data,} \end{aligned}$$

with simultaneous incidence checked across every stage.

Theorem 13.18 (First-difference routing completeness). *Let I, J be same-target simultaneous interfaces. If*

$$\exists E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) : E(I) \neq E(J),$$

then at least one interface locus changes:

1. target or charged-current boundary identity;
2. channel/instrument or transition-kernel structure;
3. visible response-mode structure and nested-role incidence;
4. locality, support, causal continuation, or environment response;
5. gauge/representation/charge attachment;
6. anomaly or quantum-consistency status;
7. vacuum or deformation behavior;
8. EFT, scale, threshold, or matching behavior;
9. activated global-form, line, higher-group, defect, boundary, or topological data;
10. physical joint incidence, shared-ledger cone, or a required overlap map;
11. lawful output typing or common definedness.

A purported standing-bearing difference outside the list either changes the target-facing trace, is internal to a candidate and inert at the interface, or exposes a failure in the declared target scope.

Proof. Let $E \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$ distinguish I and J . If $\tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J)$, target extensionality of E would force equal outputs, a contradiction. Hence the target traces differ. By lemma 13.2, the complete profiles differ. Therefore either one component of $\mathcal{P}_v^0$ differs or the physical joint-incidence profile $\mathcal{J}_v^{\text{sim}}$ differs. The component definitions and the factorizations equations (44) to (51) route the former to boundary/record, channel/response, continuation/support/environment, gauge, anomaly, vacuum/deformation, EFT/matching, or an activated extension locus. A $\mathcal{J}_v^{\text{sim}}$ difference routes to same-witness identity, overlap, commutativity, or common-cone structure by definition 3.17. Definedness differences are retained by totalization with $\perp_{\text{eval}}$. Thus every semantic evaluator difference changes an interface locus. $\square$

Remark 13.19 (Theoremic rather than enumerative closure). *The routing theorem does not rest on familiarity of a list, a finite catalogue of tests, or evaluator-term formation height. It starts from a semantic evaluator difference, forces a difference in the independently defined target-facing trace by extensionality, and then uses trace/profile recovery plus the already-proved packet factorization. A proposed evaluator that distinguishes equal target traces would depend on data outside the fixed target and therefore reopens the target scope rather than revealing an omitted same-target formula.*

13.7 Physical interface completeness

Theorem 13.20 (Physical interface completeness). *For all $X, Y \in \mathfrak{C}_v^{\text{sim}}$ in the same target fiber,*

$$\Pi_v(X) = \Pi_v(Y) \iff \widetilde{\mathfrak{F}}_v(X) \equiv_{\mathbb{T}_v^{(2)}} \widetilde{\mathfrak{F}}_v(Y) \iff \mathcal{P}_v^+(\widetilde{\mathfrak{F}}_v(X)) = \mathcal{P}_v^+(\widetilde{\mathfrak{F}}_v(Y)). \quad (53)$$

Proof. Equality of canonical classes is target-semantic equivalence by definition 13.5 and theorem 13.12. The equivalence with complete-profile equality is theorem 13.3. The restriction to $\mathfrak{C}_v^{\text{sim}}$ and pullback congruence ensure the statement concerns simultaneous physical interfaces rather than raw Frankenstein tuples. $\square$

Corollary 13.21 (Complete-profile separation and saturation). *Within one target fiber and the simultaneous candidate class,*

$$\mathcal{P}_v^+(\widetilde{\mathfrak{F}}_v(X)) \neq \mathcal{P}_v^+(\widetilde{\mathfrak{F}}_v(Y)) \iff \Pi_v(X) \neq \Pi_v(Y).$$

13.8 What completeness does not identify

The implication

$$\widetilde{\mathfrak{S}}_{\nu}(X) \equiv_{\mathbb{T}_{\nu}^{(2)}} \widetilde{\mathfrak{S}}_{\nu}(Y) \implies X \cong Y$$

is not claimed. Candidate theories can differ in UV sectors, inactive operator categories, hidden states, or parameters not evaluated by the fixed target. Such distinctions may remain physically meaningful at the microscopic-theory level even when they share one interface image.

Nonclaim 13.22 (No theory identity). *Physical interface completeness is a complete-invariant theorem for the target-evaluator algebra, not a reconstruction theorem for the full microscopic theory.*

13.9 Target-semantic interface exhaustion

Theorem 13.23 (Target-semantic interface exhaustion). *Assume the imported target and upstream role theorems, coordinate-admission results, generalized-symmetry relevance resolution, shared-ledger factorization, and pullback/constraint congruence established in this manuscript. Then no additional same-target target-admitted interface evaluator or joint-incidence datum survives outside the canonical packet.*

Proof. Suppose an additional datum c has target-visible interface force. Its variation changes a target-admitted physical judgment and therefore defines or contributes to a semantic evaluator $E_c \in \text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$. By target-trace descent, E_c depends only on $\tau_{\mathbb{T}_{\nu}^{(2)}}$; by lemma 13.2 and theorem 13.17, its value is therefore determined by the canonical packet. Hence c is a derived view or refinement of an existing target-facing locus, not an independent interface coordinate. If varying c changes no semantic target evaluator, it is interface-inert or candidate-internal. If the proposed distinction requires a primitive target-facing input, witness sort, operator category, or measurement relation not determined by $\mathcal{T}(\mathbb{T}_{\nu}^{(2)})$, it changes the target fiber. An unenumerated formula of existing trace data does not create a new case: theorems 3.8 and 3.13 already include it by saturated target-lawful postprocessing. These cases exhaust its status without appeal to a finite catalogue. $\square$

Scope is load-bearing

The exhaustion theorem is relative to $\mathbb{T}_{\nu}^{(2)}$. It does not classify arbitrary BSM theories, quantum gravity, every UV operator, or observables absent from the target. A claimed same-target evaluator that is a lawful functional of existing target-trace data is already in $\text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)}) = \text{Eval}(\mathbb{T}_{\nu}^{(2)})$. A distinction requiring data not determined by $\mathcal{T}(\mathbb{T}_{\nu}^{(2)})$ reopens the target specification and defines a new target fiber. The theorem therefore remains falsifiable through target-scope error, trace omission, or a failure of one of the stated factorization theorems.

14 Interface adequacy, no-carrier theorem, and realization boundary

14.1 Canonical admissible interface class

The canonical admissible interface space has already been constructed as

$$\mathfrak{S}_{\nu}^{\text{adm}} = \mathcal{M}_{\text{adm}} / \simeq_{\text{int}} \subseteq \mathfrak{S}_{\nu}^{\text{sim}} = \mathcal{M}_{\text{pull}} / \simeq_{\text{int}} .$$

Its classes satisfy:

1. the target and same-target predicates;
2. boundary reconstruction and channel adequacy;
3. $I_{\text{pq}} = 1$, $I_{\Delta} = 1$, and the nested incidence relation;
4. target visible-mode coverage with complete protocol/environment incidence;

5. transport, environment, gauge, anomaly, vacuum, deformation, and EFT constraints;
6. every relevance-gated extension predicate activated by the target fiber;
7. one shared physical witness diagram, all mandatory overlap clauses, and a required common cone;
8. lawful quotient, no-repair, and target-fiber conditions.

By theorem 12.9, these conditions are properties of canonical classes, not of a preferred representative. The definition does not assert that every abstract class has a microscopic realization.

14.2 Necessity

Theorem 14.1 (Interface necessity). *If $X \in \mathfrak{C}_v$ is interface-adequate for T_v^{osc} , then*

$$X \in \mathfrak{C}_v^{\text{adm}} \subseteq \mathfrak{C}_v^{\text{sim}} \quad \text{and} \quad \Pi_v(X) \in \mathfrak{S}_v^{\text{adm}}.$$

Proof. Interface adequacy preserves the target boundary and records, occupies the nested role complex, satisfies every mandatory support and consistency condition, and requires one shared physical witness diagram by definition 4.9. Shared-ledger necessity gives $\tilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{adm}}$, hence $X \in \mathfrak{C}_v^{\text{adm}} \subseteq \mathfrak{C}_v^{\text{sim}}$. The canonical projection is therefore defined. Quotienting the admissible simultaneous presentation places its class in $\mathfrak{S}_v^{\text{adm}}$. $\square$

14.3 Sufficiency at interface level

Theorem 14.2 (Interface sufficiency). *Assume the imported target and physical-role theorems, the target-evaluator generation result (theorem 3.13), resolved relevance gates, complete shared-ledger and constraint factorization, and the congruence statements in theorems 12.7 and 12.9. If an already-specified completion $X \in \mathfrak{C}_v^{\text{sim}}$ satisfies*

$$\Pi_v(X) \in \mathfrak{S}_v^{\text{adm}},$$

then $\text{InterfaceAdeq}(X, T_v^{\text{osc}})$ holds.

Proof. Membership in $\mathfrak{S}_v^{\text{adm}}$ means that the class has a representative in $\mathcal{M}_{\text{adm}}$. Since $X \in \mathfrak{C}_v^{\text{sim}}$, $\tilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{pull}}$ and represents the same class. Admissibility congruence forces every simultaneous representative of that class, including $\tilde{\mathfrak{S}}_v(X)$, to lie in $\mathcal{M}_{\text{adm}}$. The target, boundary, channel, visible-response, nesting, transport, attachment, active extension, and shared-incidence predicates therefore all hold for X . These are exactly the clauses of definition 4.9. No abstract interface class has been converted into a completion; the completion X was specified independently. $\square$

Theorem 14.3 (Interface adequacy characterization). *Under the hypotheses of theorem 14.2, every already-specified $X \in \mathfrak{C}_v$ satisfies*

$$\text{InterfaceAdeq}(X, T_v^{\text{osc}}) \iff \tilde{\mathfrak{S}}_v(X) \in \mathcal{M}_{\text{adm}}. \quad (54)$$

For $X \in \mathfrak{C}_v^{\text{sim}}$, this is equivalent to

$$\text{InterfaceAdeq}(X, T_v^{\text{osc}}) \iff \Pi_v(X) \in \mathfrak{S}_v^{\text{adm}}. \quad (55)$$

Proof. The first equivalence is shared-ledger necessity together with the definition of $\mathcal{M}_{\text{adm}}$, the target-evaluator generation theorem, and complete constraint factorization. The second follows from theorems 14.1 and 14.2. Pullback and admissibility congruence ensure that quotient membership cannot transfer standing across representatives. $\square$

14.4 No-carrier-identification theorem

Theorem 14.4 (No carrier identity from interface closure). *Neither theorem 13.20 nor theorem 14.3 licenses any of the implications*

$$r_{\Delta} \Rightarrow m_{\nu}, \quad r_{\bowtie} \Rightarrow U_{\text{PMNS}}, \quad \mathfrak{I}_{\nu}^{\text{sim}} \Rightarrow \nu_R, \quad \mathfrak{I}_{\nu}^{\text{sim}} \Rightarrow O_5, \quad \mathfrak{I}_{\nu}^{\text{sim}} \Rightarrow \text{new local matter},$$

or analogous implications to a sterile sector, nonunitary mechanism, topological sector, defect, non-invertible symmetry, or unique microscopic completion.

Proof. The canonical interface forgets microscopic variables that change neither the complete target profile nor the active constraints. The same completely positive channel, for example, admits different Kraus and Stinespring presentations, and nonminimal dilations can contain additional decoupled or target-invisible sectors [12, 29]. A standard unitary mass-mixing model and a distinct enlarged representation can therefore share the same event-level values and the same physical witness-incidence diagram without being the same candidate record. The interface contains no field-species, representation, operator-dimension, mass, or carrier-type coordinate from which the listed conclusions could be read. Establishing that all alternative preimages fail requires an independently complete candidate universe and coupled survivor analysis, which the present interface theorem does not provide. $\square$

Remark 14.5 (Logical examples are not candidate completeness). *The proof uses distinct formal preimages to block definitional identity. It does not assert that every formal preimage is a complete, local, anomaly-free physical theory. Determining which microscopic candidate classes are admissible and survive requires a separate candidate-completeness and simultaneous-survivor analysis.*

14.5 Interface image can be many-to-one

Proposition 14.6 (Many-to-one interface projection). *The projection*

$$\Pi_{\nu} : \mathfrak{C}_{\nu}^{\text{sim}} \longrightarrow \mathfrak{I}_{\nu}^{\text{sim}}$$

need not be injective on candidate records. Candidate-distinct simultaneous completions can have the same canonical interface image when their differences lie outside the complete target profile.

Proof. Take a simultaneous completion and adjoin a sector that is completely decoupled from the target boundary, channel, support, anomaly, vacuum, scale, extension, and physical witness-incidence data. The candidate record changes, but every evaluator in $\text{Eval}(\mathbb{T}_{\nu}^{(2)})$ has the same outcome because the added sector is target-invisible. Hence the presentations are $\equiv_{\mathbb{T}_{\nu}^{(2)}}$ -equivalent, and theorem 13.20 gives equal canonical interface images. If the added sector becomes coupled, changes a common witness, or becomes target-visible, the complete profiles can separate. $\square$

14.6 No realization promotion

Theorem 14.7 (Interface adequacy is not licensed realization). *For a specified candidate $X \in \mathfrak{C}_{\nu}^{\text{adm}}$,*

$$\Pi_{\nu}(X) \in \mathfrak{I}_{\nu}^{\text{adm}} \not\Rightarrow \text{RealizationLicensed}(C, X).$$

Proof. Role-Occupancy Closure requires, relative to a fixed realization specification and target, candidate completeness, presentation-quotient completeness, role-set completeness, eligibility completeness, joint-compatibility completeness, live-residue completeness, and failure completeness [18]. The present theorem supplies a target-relative interface and compatibility predicates for an already-specified candidate. It does not generate every candidate, prove the candidate quotient complete, determine the live residue, or eliminate all rival assignment orbits. The antecedent therefore lacks the hypotheses required for licensed realization. $\square$

Corollary 14.8 (No unique particle prediction). *The interface theorem does not by itself establish a new particle, field representation, mass mechanism, or experimental signal, even if only one familiar realization has been exhibited.*

14.7 Eligibility and compatibility predicates

The interface construction nevertheless provides nontrivial information for any subsequent candidate analysis. For any generated candidate X , define

$$E_{\nu}^{(2)}(X) := [\tilde{\mathfrak{I}}_{\nu}(X) \in \mathcal{M}_{\text{adm}}],$$

$$J_{\nu}^{(2)}(X_1, \dots, X_n) := \text{joint shared-ledger compatibility of the proposed occupancy package.}$$

The first is a total interface-eligibility predicate on generated candidate records; it does not require Π_{ν} to be defined for witness-split candidates. For $X \in \mathfrak{C}_{\nu}^{\text{sim}}$, it is equivalent to $[\Pi_{\nu}(X) \in \mathfrak{I}_{\nu}^{\text{adm}}]$. The second records whether several carrier components can jointly realize one interface without witness conflict. Neither supplies candidate completeness or realization by itself.

Theorem 14.9 (Candidate-comparison sufficiency theorem). *The canonical interface data are sufficient to pose an independent source-first candidate-generation problem without altering the target. The required packet contains:*

1. *target specification and target-fiber graph;*
2. *canonical core interface schema and extension-fiber schema;*
3. *independently derived target-evaluator grammar, target-relative indistinguishability, component profile, physical joint-incidence grammar, and the proof that $\mathcal{P}_{\nu}^{+}$ is their complete invariant;*
4. *constructive redescription-generator library;*
5. *shared witness ledger, physical overlap maps, and pullback test;*
6. *constraint family, constraint-factorization map, and total interface-eligibility predicate;*
7. *generalized-symmetry relevance decisions;*
8. *first-difference and reopening rules;*
9. *explicit prohibition on carrier ranking or realization promotion.*

Proof. An independent candidate analysis requires a target-fixed criterion for whether a generated candidate can occupy the forced role complex. The interface schema, pullback test, complete profile, quotient, constraints, and witness ledger provide that criterion. Witness-split candidates fail before projection, while simultaneous candidates are compared only after the shared-ledger condition is met. The target remains fixed because no candidate variable appears in its grammar. The reopening rules preserve the possibility that a later candidate analysis exposes a missing interface value or incidence coordinate. $\square$

14.8 Range, reopening, and stability of the comparison object

Definition 14.10 (Occupancy-positive same-target interface signature). *A canonical packet σ is an occupancy-positive same-target interface signature when it:*

1. *has the target fingerprint and active target-fiber typing of $\mathbb{T}_{\nu}^{\text{if}}$;*
2. *contains the complete core role interface, including $r_{\Delta} \succ r_{\text{psd}}$;*
3. *satisfies every mandatory attachment and shared-ledger compatibility predicate;*
4. *carries every coordinate activated by the fixed relevance policy;*

5. *includes a lawful physical joint-incidence class with a required common cone; and*
6. *is supported by a target-admitted operational witness packet rather than by a label, ranking, or microscopic expectation.*

The definition does not assert that a microscopic completion realizing σ has been constructed.

Theorem 14.11 (Interface range and reopening). *Let σ be an occupancy-positive same-target interface signature. If*

$$\sigma \notin \text{Ran}(\Pi_\nu),$$

then σ is a genuine same-target counterexample to the declared interface range. At least one of the target-evaluator generation theorem, complete-invariant theorem, coordinate schema, activation policy, coordinate-specific quotient, shared-ledger conditions, constraint factorization, or first-difference routing theorem must reopen. The signature may not be discarded by an unregistered quotient, empirical ranking, or candidate-specific repair.

Proof. By definition, σ is target-admissible, supports the full nested role interface, and satisfies every declared attachment, activation, and simultaneous-incidence predicate. It therefore cannot be disposed of as target-changing, ill typed, or merely candidate-internal. If it is not in the image of Π_ν , then the target-evaluator grammar has omitted a standing-bearing comparison, the complete-invariant theorem fails, a coordinate-specific quotient has been mistyped, the shared-ledger or constraint-congruence conditions are incomplete, or the first-difference theorem fails to route the new witness. Ranking or exclusion by preference would introduce a selector not present in the target, while modifying the packet after failure would violate the no-repair discipline. $\square$

Theorem 14.12 (Comparison-object stability). *Once the target fingerprint, target-evaluator grammar, complete-invariant theorem, canonical interface schema, exact simultaneous-domain quotient, constructive generator library, activation decisions, shared witness ledger, constraint-factorization theorem, and reopening rule have been established, they are fixed as the comparison object for any subsequent candidate analysis. Any later alteration is either an explicit revision triggered by theorem 14.11 or a declared target update; it is not an in-candidate repair.*

Proof. Those objects determine what counts as the same target, which physical differences survive the quotient, what constitutes one simultaneous interface, and what every candidate must reproduce. Allowing them to vary after candidate inspection would permit target backfitting and would destroy the independence of the candidate universe. The only lawful changes are a theorem revision supported by a genuine counterexample or a new target fiber with its own comparison analysis. $\square$

Theorem 14.13 (No backflow from candidate generation). *Candidate generation, candidate abundance, familiarity, fit quality, or eventual survivor status cannot alter the fixed target, target-evaluator grammar, interface coordinates, physical incidence grammar, equivalence, activation policy, or compatibility predicates. A candidate that requires such an alteration is a target-update candidate or a trigger for explicit reopening, not a realization of the original interface.*

Proof. Candidate generation is downstream of the fixed interface comparison object. If a candidate could modify the comparison object used to test it, the candidate would authorize its own eligibility and the separation between target formation and candidate evaluation would collapse. The fixed-domain, AMetric, and no-repair constraints exclude that route. The only remaining dispositions are target update or theorem reopening under the preceding results. $\square$

14.9 Rival interface disposition

Every proposed rival to the canonical interface is classified as one of:

1. lawfully equivalent;
2. skin-only;
3. an optional extension-fiber coordinate;
4. a target update;
5. a microscopic candidate distinction invisible at the declared interface scope;
6. a different physical joint-incidence class;
7. an open burden exposing an incomplete theorem or quotient;
8. a genuine same-target counterexample to interface completeness.

This list is a disposition protocol rather than a device for rejecting novel physics. The final categories are scientifically valuable and force revision where appropriate.

Principal endpoint

Given the fixed target, the independently derived target-evaluator generation theorem, the proof that $\mathcal{P}_\nu^+$ is a complete invariant for $\text{Eval}(\mathbb{T}_\nu^{(2)})$, coordinate-admission proofs, resolved generalized-symmetry relevance, shared-ledger closure, constraint congruence, and lawful first-difference routing, the preceding results yield a carrier-neutral canonical simultaneous interface satisfying

$$\Pi_\nu(X) = \Pi_\nu(Y) \iff \tilde{\mathfrak{I}}_\nu(X) \equiv_{\mathbb{T}_\nu^{(2)}} \tilde{\mathfrak{I}}_\nu(Y) \iff \mathcal{P}_\nu^+(\tilde{\mathfrak{I}}_\nu(X)) = \mathcal{P}_\nu^+(\tilde{\mathfrak{I}}_\nu(Y))$$

for $X, Y \in \mathfrak{C}_\nu^{\text{sim}}$, and

$$\text{InterfaceAdeq}(X, T_\nu^{\text{osc}}) \iff \tilde{\mathfrak{I}}_\nu(X) \in \mathcal{M}_{\text{adm}}$$

for every already-specified completion in the comparison class. For $X \in \mathfrak{C}_\nu^{\text{sim}}$, the latter is equivalent to $\Pi_\nu(X) \in \mathfrak{I}_\nu^{\text{adm}}$. The result does not identify an occupant or license realization.

15 Representative realizations and limiting cases

15.1 Purpose of the comparison examples

The following examples test whether one interface schema can type conventional, nonunitary, enlarged, effective, and generalized-symmetry cases without redefining its fields. An example is not evidence that its candidate class is complete, uniquely admissible, or physically realized. Their function is to expose overquotienting, underquotienting, target drift, and hidden coupling among interface coordinates.

15.2 Standard unitary representative

Consider a finite-dimensional unitary representative with interaction-defined boundary states $|\nu_\alpha\rangle$, a unitary matrix U , and propagation phases $\phi_i(\lambda)$:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |i\rangle, \tag{56}$$

$$U_\lambda^{\text{prop}} = \sum_i e^{-i\phi_i(\lambda)} |i\rangle\langle i|, \tag{57}$$

$$K_{\beta\alpha}(\lambda) = \left| \sum_i U_{\beta i} e^{-i\phi_i(\lambda)} U_{\alpha i}^* \right|^2. \tag{58}$$

In the relativistic mass representative, $\phi_i \simeq m_i^2 L / (2E)$. This formula is a standard realization of the interface, not an interface definition [2, 3].

If U is nontrivial relative to the charged-current boundary and the phase differences are nonconstant, the model occupies $r_{\triangleright\triangleleft}$. If the event-response projection contains the two target mode classes, it occupies $r_{\triangle}$. Weak-basis and rephasing changes are removed by the lawful groupoid. The mass values, ordering, state-space dimension, and matrix coordinates remain candidate data.

Proposition 15.1 (Standard-model comparator mapping). *The conventional unitary oscillation representation maps to the core interface by:*

$$(U, \{\phi_i\}) \mapsto (\mathcal{B}_{\text{CC}}, [\mathcal{E}_\lambda]_{\text{ch}}, \mathcal{K}_{\text{CC}}, \mathfrak{M}_{\text{vis}}, \mathcal{N}_{\triangle\triangleright\triangleleft}, \dots),$$

with all basis and phase conventions quotiented. The map is many-to-one and does not invert to recover U or $\{m_i\}$ from the interface alone.

15.3 One-mode nonaligned countermodel

The two-state model of theorem 7.9 has

$$U_t = e^{-it\omega\sigma_x}, \quad K_{10}(t) = \sin^2(\omega t).$$

It passes the boundary/channel nonalignment test and fails the target visible-mode coverage. This is the principal anti-collapse benchmark: $r_{\triangleright\triangleleft}$ and $r_{\triangle}$ are not the same coordinate.

15.4 Open-system channel

Let

$$\mathcal{E}_t(\rho) = \sum_a M_a(t) \rho M_a(t)^\dagger, \quad \sum_a M_a(t)^\dagger M_a(t) = \mathbf{1}.$$

The interface reads only the CP map, target-visible outcomes, support/environment packet, and any admitted dilation witness. It does not promote the chosen Kraus list. Decoherence parameters are candidate tensor data if they alter the response profile.

Proposition 15.2 (Kraus covariance benchmark). *Two Kraus families representing the same CP map and preserving every active witness define the same interface. A change in the CP map or in a dilation-sensitive witness defines a new interface candidate.*

15.5 Decay, absorption, and loss

A trace-decreasing active-sector map is incomplete unless paired with loss outcomes. The event-complete instrument distinguishes:

1. active flavor transfer;
2. disappearance into a declared sterile or decay sector;
3. absorption or detector inefficiency carried by the boundary/environment ledger;
4. conditional normalization used only when the target analysis is explicitly conditional.

This benchmark blocks the false inference that any reduction of active normalization is a propagation mode or that detector loss can be used as hidden flavor transfer.

15.6 Enlarged state space

Let V_t be a unitary on $\mathcal{H}_{\text{active}} \otimes \mathcal{H}_{\text{env}}$ and

$$\mathcal{E}_t(\rho) = \text{Tr}_{\text{env}} \left[V_t(\rho \otimes \eta) V_t^\dagger \right].$$

If the environment is target-invisible, the dilation and reduced description share one interface image. If environment states, charges, decay products, line sectors, or causal support are admitted, the dilation activates additional coordinates. State-space cardinality is never inferred from the reduced interface.

15.7 Nonuniform matter profile

A standard matter realization uses a path-dependent Hamiltonian and path ordering. The interface stores the physical profile and the induced channel, not a particular discretization. Two coordinate descriptions of the same density profile are lawful; two different physical profiles are environment variations. Matter-induced changes cannot be reclassified as source/detector effects without a target-fiber map.

15.8 Source/detector interaction variants

Nonstandard source or detector interactions modify $\mathcal{B}_{CC}$. They are controlled target variants, not core propagation-interface coordinates. A phenomenological reparameterization that pushes a boundary interaction into an “effective mixing matrix” does not preserve target identity unless the complete boundary profile is shown equivalent.

15.9 Global-form and defect fibers

Two candidates can share the same point-local core interface image while differing in line or defect sectors. In the core target they are interface-equivalent and candidate-distinct. In an activated extension fiber they can become interface-distinct. A non-invertible defect supported only on a defect-localized TQFT is therefore not a bulk-matter prediction and not a mandatory core coordinate.

15.10 Pairwise but not global benchmark

The witness-split obstruction of section 12 is included in the benchmark suite. A candidate that passes channel, anomaly, and defect checks on three incompatible ledgers fails the shared pullback even if every pair has a reduced common description.

15.11 Benchmark matrix

Table 9: Representative hard-case benchmark matrix.

ID	Case	Required classification	Failure exposed
B01	Weak-basis/rephasing variants	Same lawful interface.	Underquotienting or PMNS backflow.
B02	Equivalent Kraus lists	Same channel/interface.	Representation multiplicity.
B03	Equivalent Stinespring dilations, no dilation witness	Same interface.	Dilation overcounting.
B04	Dilation with visible leakage/charge/support witness	Distinct extension interface or target fiber.	Overquotienting.
B05	One-mode nonaligned model	Occupies r_{DQ} , not r_{Δ} .	Role collapse.
B06	Standard unitary comparator	Benchmark realization only.	Candidate backflow.
B07	Decohering CPTP channel	Channel-native typing.	Unitary overrestriction.
B08	Decay/absorption instrument	Explicit loss and normalization ledger.	Hidden loss or detector retyping.
B09	Nonuniform matter profile	Environment packet and covariance.	Hidden medium.
B10	Source/detector new interaction	Target variant unless full boundary equivalence is proved.	Silent target shift.
B11	Sterile/enlarged state space	Candidate-level stress form mapping to interface.	State-space cardinality inference.

ID	Case	Required classification	Failure exposed
B12	Two global-form fibers, same local records	Same core interface; possibly distinct extension fibers.	Global-form omission or overpromotion.
B13	One-form sector absent from target category	Optional/irrelevant, not mandatory.	Generalized-symmetry overreach.
B14	Defect-localized non-invertible TQFT	Candidate extension; no bulk-matter inference.	New-particle backflow.
B15	Pairwise-compatible, globally inconsistent ledgers	Excluded by shared pullback.	Checklist fallacy.
B16	Same records, different active anomaly status	Distinct interface in anomaly-sensitive fiber.	Record-only undercompleteness.
B17	Same interface, different UV-invisible sectors	Interface-equivalent, candidate-distinct.	Interface-to-theory identity.
B18	Hidden microscopic richness, one visible mode	Fails r_Δ .	Microscopic-to-visible promotion.
B19	Adequacy at one unlicensed tuned point	Fails deformation continuity.	Parameter selector.
B20	EFT/UV pair with full matching	Same interface, distinct candidate records.	Matching confusion.

Theorem 15.3 (Benchmark adequacy). *The canonical interface schema classifies every benchmark in table 9 without adding a case-specific coordinate or altering the meaning of an existing coordinate.*

Proof. B01–B04 are resolved by the lawful groupoid and dilation relevance rule. B05–B08 are resolved by the channel and nested-response theorems. B09–B10 are resolved by environment/boundary separation. B11 and B17 distinguish candidate data from interface data. B12–B14 are resolved by generalized-symmetry relevance. B15 is the pairwise-insufficiency theorem. B16 uses the anomaly-sensitive profile. B18 is the visible/microscopic multiplicity split. B19 is deformation continuity. B20 is EFT-fiber discipline. No benchmark requires a new field. $\square$

Interpretive criterion. A benchmark is informative only when it can be classified without silently changing the target, promoting a candidate variable into a target coordinate, or adding an ad hoc exception. Failure of this criterion would reopen the interface schema rather than be absorbed into a special case.

16 Robustness and falsifiability

16.1 Scope of admissible challenges

The theorem is relative to a fixed target and to the semantic evaluator class derived from that target. A substantive counterexample must therefore preserve the target while breaking at least one load-bearing implication. Examples include an omitted primitive target-facing datum, failure of target-trace descent, failure of profile recovery, a lawful equivalence that erases a target-visible distinction, a retained coordinate that cannot be jointly realized with the rest of the packet, or a relevance assignment that fails in an active target fiber. By contrast, a microscopic distinction invisible to every target-admitted evaluator is not a counterexample to a target-relative interface theorem; it is a distinction outside the declared interface scope.

Several recurring alternatives are useful stress tests because they probe different parts of the theorem rather than different parameter choices within one model.

Potential challenge	Required test	Controlling result
Nonunitary propagation or decoherence	Preserve every charged-current sector while reproducing the target cross-boundary records.	Event-complete channel and loss-tolerant nonmixing theorems.
Different Kraus or dilation descriptions	Produce a target-visible evaluator or witness-incidence difference.	Constructive-generator soundness and complete-profile equivalence.
Source/detector new physics	Preserve the operational charged-current boundary under the full target profile.	Boundary reconstruction and target-fiber typing.
Matter/environment reparameterization	Preserve the physical medium profile, channel family, and record transport.	Environment covariance and shared-realization conditions.
Alternative gauge or charge presentation	Preserve the quotient-stable boundary and anomaly/attachment witnesses.	Gauge attachment and anomaly compatibility.
Generalized-symmetry structure	Exhibit a target-admitted line, higher-form, higher-group, defect, or anomaly witness in the active fiber.	Generalized-symmetry activation theorem.
Componentwise but incompatible data	Exhibit one common physical witness cone for all retained coordinates.	Shared-realization pullback and pairwise-insufficiency theorem.
Microscopic structure absent from the profile	Show that it changes a semantic evaluator of the fixed target.	Target-trace descent, evaluator saturation, and profile recovery.

Table 10: Representative robustness tests and the theorem layer each probes.

16.2 Explicit falsifiers

The strongest interface endpoint fails if any of the following is established:

1. a faithful same-target completion realizes the established active flavor-transformation records while remaining charged-current nonmixing;
2. the inherited visible-mode input is lawfully collapsed to one class on the fixed charged-current boundary;
3. a lawful standing-bearing comparison uses only the fixed target but is not extensional with respect to the declared target-facing trace, showing that $\mathcal{T}(\mathbb{T}_V^{(2)})$ omitted a same-target primitive datum;
4. an interface coordinate cannot be defined for a lawful completion in the declared class and cannot be placed in a controlled relevance fiber;
5. a transformation currently classified as a constructive lawful generator changes $\mathcal{P}_V^+$, invalidating generator soundness or an affected coordinate quotient;
6. two same-target presentations with equal target-facing traces are distinguished by a semantic target evaluator, or equal complete profiles fail to recover equal target traces;
7. pullback or admissible-survivor membership differs inside one $\equiv_{\mathcal{P}^+}$ -class, contradicting congruence;
8. a generalized-symmetry coordinate is proven universally target-relevant after being classified optional, or conversely;
9. the interface-adequacy biconditional is applied to an abstract tuple with no specified completion, or without the stated target, relevance, factorization, and congruence hypotheses.

Theorem 16.1 (Falsification completeness). *Every substantive same-target defeat of the interface endpoint alters at least one of: the established role input, target-facing trace scope, semantic target-use descent, target-evaluator generation theorem, trace/profile recovery or complete-invariant theorem, coordinate admission or quotient soundness, shared-ledger factorization, pullback/constraint congruence, relevance resolution, first-difference routing, or adequacy hypotheses.*

Proof. The endpoint is the composition of exactly those theorem layers. A defeat must break one implication in the chain or provide a counterexample to the composed conclusion. Breaking a layer yields the corresponding category. A purported defeat that changes none of them leaves the premises and conclusion unchanged and is not a substantive counterexample. $\square$

16.3 Scope limitations

Three limitations are intrinsic to the result and should be kept distinct from failures of the theorem.

First, r_Δ is conditional on the phenomenological response input established in the preceding target-fixed analysis. The present paper proves its interface consequences, not a new reconstruction from raw experimental data. Second, interface completeness is target-relative. A genuinely new primitive observable or physical witness relation absent from the target-facing trace either exposes a target-scope omission or defines a richer target fiber. A new lawful function of existing target-trace data, however, is already included by saturated postprocessing. Third, the generalized-symmetry result is a relevance statement for the point-local target; it is not a claim that generalized symmetries are absent from every microscopic completion.

17 Discussion and conclusion

17.1 Summary of the result

The analysis leaves the microscopic occupant open while removing ambiguity about the target-relative physical interface that any occupant must reproduce. Its central achievements are:

1. the established role complex is imported without mass, mixing, field, or mechanism backflow;
2. the charged-current boundary is reconstructed from operational source/detection records before propagation is specified;
3. event-complete completely positive instruments provide a common channel interface for unitary, reduced nonunitary, decay/loss, and enlarged-state-space presentations;
4. loss-tolerant charged-current nonalignment and the complete visible-response packet preserve the strict nesting $r_\Delta \succ r_{\text{pct}}$;
5. observable transport, environment, gauge attachment, anomaly, vacuum, deformation, and EFT data are treated as compatibility conditions of one physical interface;
6. global-form and generalized-symmetry data are not universal discriminators of the point-local core target, but become active in target fibers that admit the corresponding probes;
7. the raw coordinate product is reduced first to a shared-realization pullback and then to a fail-closed admissible subobject, so pairwise compatibility cannot substitute for global physical co-instantiation;
8. the target-facing trace is defined directly from the fixed target and upstream physical roles before the use grammar, evaluator algebra, interface coordinates, or microscopic candidates are introduced;
9. semantic uses and evaluators descend through that trace, while the complete profile $\mathcal{P}_\nu^+ = (\mathcal{P}_\nu^0, \mathcal{J}_\nu^{\text{sim}})$ is proved to recover it and therefore to be a complete target-relative invariant;
10. pullback membership and every active constraint factor through the complete profile, making the simultaneous and admissible quotients well defined;
11. equality of canonical interfaces is equivalent to indistinguishability under every target-admitted evaluator; and
12. interface adequacy neither identifies a microscopic carrier nor licenses physical realization.

17.2 Compressed main theorem

Theorem 17.1 (Complete physical interface of the neutrino propagation role complex). *Fix the oscillation target T_v^{osc} , the comparison class $\mathfrak{C}_v$, the independently defined target-facing trace $\tau_{T_v^{(2)}}$, the semantic use/evaluator classes with their proved saturated realizations $\text{Use}(T_v^{(2)})$ and $\text{Eval}(T_v^{(2)})$, the complete invariant $\mathcal{P}_v^+ = (\mathcal{P}_v^0, \mathcal{J}_v^{\text{sim}})$, and the relevance policy stated above. Let $\mathfrak{C}_v^{\text{sim}}$ and $\mathfrak{C}_v^{\text{adm}}$ be the simultaneous and admissible subclasses defined by the shared-realization pullback and active constraint family.*

For every $X \in \mathfrak{C}_v^{\text{sim}}$, the canonical projection

$$\Pi_v : \mathfrak{C}_v^{\text{sim}} \longrightarrow \mathfrak{I}_v^{\text{sim}}$$

records a carrier-neutral simultaneous packet consisting of the charged-current boundary, event-complete channel class, transition kernels, visible-mode packet, nested-role incidence, transport/environment, gauge/anomaly, vacuum/deformation, EFT/matching, and relevance-gated extension data, together with their physical joint-incidence class. For $X, Y \in \mathfrak{C}_v^{\text{sim}}$ in one target fiber,

$$\Pi_v(X) = \Pi_v(Y) \iff \tilde{\mathfrak{I}}_v(X) \equiv_{T_v^{(2)}} \tilde{\mathfrak{I}}_v(Y) \iff \mathcal{P}_v^+(\tilde{\mathfrak{I}}_v(X)) = \mathcal{P}_v^+(\tilde{\mathfrak{I}}_v(Y)).$$

Under the target-use exhaustion, target-evaluator generation, complete-invariant, coordinate-admission, relevance, shared-realization, constraint-factorization, and congruence theorems,

$$\text{InterfaceAdeq}(X, T_v^{\text{osc}}) \iff \tilde{\mathfrak{I}}_v(X) \in \mathcal{M}_{\text{adm}}$$

for every already-specified $X \in \mathfrak{C}_v$; if $X \in \mathfrak{C}_v^{\text{sim}}$, this is equivalent to $\Pi_v(X) \in \mathfrak{I}_v^{\text{adm}}$.

The point-local core target does not force an independent global-form, higher-form, higher-group, or non-invertible coordinate; those structures are activated only in explicit extension fibers. No conclusion about a mass term, PMNS matrix, field content, representation, particle, topological sector, defect, or physical realization follows from the interface theorem alone.

Proof. The target, target-facing trace, semantic use/evaluator definitions, target-trace descent, saturated use/evaluator generation, component and incidence profiles, complete-invariant setup, and coordinate admission are established in section 3. Boundary and channel necessity are sections 5 and 6. Visible response and strict nesting are section 7. Transport, gauge, consistency, vacuum, deformation, and scale closure are sections 8 to 10. Generalized-symmetry relevance is theorem 11.9. Simultaneous shared-ledger closure and congruence are theorems 12.3, 12.6 to 12.9 and 12.14. The independent semantic spine is definitions 3.3 and 3.4, lemmas 3.5 and 13.2, and theorems 3.8, 3.11, 3.13 and 13.3. The exact quotient, profile identification, evaluator-first routing, and physical interface completeness are theorems 13.12, 13.13, 13.18 and 13.20. The adequacy biconditional is theorem 14.3. The no-carrier and no-realization boundaries are theorems 14.4 and 14.7. $\square$

17.3 Physical interpretation

The preceding role theorem showed that the realized oscillation target contains physical work absent from the minimal massless-neutrino Standard Model. The present result identifies the canonical target-visible simultaneous interface of that work. This is stronger than a list of phenomenological constraints and weaker than a microscopic particle prediction. Any lawful completion must reproduce the same boundary, channel, response, continuation, consistency, scale, and physical-incidence packet. No target-visible component or joint-incidence difference is erased, no semantically identical simultaneous presentation survives as false multiplicity, and no witness-split non-survivor shares an admissible class with a coherent interface.

The result also locates generalized-symmetry structure at the correct logical level. Non-invertible, higher-form, higher-group, and global-form data can be physically significant in particular completions, but the point-local neutrino-propagation target does not force them as universal interface coordinates. They

become interface data only when a target-admitted operator, anomaly, support, or defect witness activates the corresponding extension fiber.

17.4 Outlook

The canonical interface is intentionally not inverted into a microscopic model. The appropriate next question is independent: which candidate carrier classes can realize this interface while satisfying the complete shared-realization and consistency conditions? That problem requires a source-first candidate universe, simultaneous constraint evaluation, and role-occupancy analysis. Its possible outcomes include underdetermination, inconsistency, realization by already known structure, realization by a non-matter sector, or the necessity of additional matter. None is presupposed here.

Conclusion

The nested neutrino-propagation role complex has a complete target-relative simultaneous physical interface at the declared scope. Its completeness basis is independent of the interface coordinate list: the target-facing trace is fixed first, semantic target uses and evaluators descend through that trace, and the complete profile is proved to recover it. The resulting interface is carrier-neutral, exact-quotient, shared-realization-congruent, and relevance-resolved. Equality of canonical packets is equivalent to target-semantic indistinguishability, while interface adequacy remains logically distinct from microscopic carrier identity and physical realization.

A Formal target and comparison specification

A.1 Target coordinates

The target used throughout the paper is the fixed oscillation target inherited from the preceding role theorem, augmented only by the information required to compare physical interfaces. A standing-bearing change to any coordinate in table 11 defines a different target fiber unless an explicit equivalence theorem preserves the entire target-relative semantics.

Table 11: Target-coordinate dictionary.

Coordinate	Fixed content	Target-change trigger
Parent fingerprint	Established $\mathfrak{f}(T_{\nu}^{\text{osc}})$.	Change to the role definitions, empirical packet, or nesting relation.
Background	Four-dimensional flat-background operational target.	Curved or dynamical background without an equivalence theorem.
Boundary	Interaction-defined charged-current preparation/effect classes.	New source/detector current or operational retagging.
Record packet	SNO, Super-Kamiokande, KamLAND, and Daya Bay target records used by the preceding analysis.	Added or removed empirical anchor, or changed response analysis.
Protocol domain	Baseline, energy, path, environment, and calibration domains used by target records.	Changed protocol semantics or medium target.
Visible-mode packet	Finite lawful mode classes with multiplicity at least two.	Lawful common-boundary collapse or a new source analysis.
Operator category	Point-local source/detector and event-response core.	Admission of line, surface, defect, or junction observables.
Gauge boundary	Fixed electroweak charged-current attachment and residual electromagnetic compatibility.	Changed representation/charge boundary.

Coordinate	Fixed content	Target-change trigger
Consistency policy	No uncanceled gauge/global inconsistency; explicit compensation required.	New anomaly, background, or global-form fiber.
Completion coordinate	Open.	A microscopic proposal changes the target only when it changes an anchor.
Lawful quotient floor	Established boundary, channel, basis, phase, and calibration equivalences.	New target-preserving representative transformation.
Generalized-symmetry policy	Relevance-gated extension fibers.	A universal target witness establishes mandatory relevance.

A.2 Comparison-class criteria

A completion belongs to $\mathfrak{C}_\nu$ only when the following conditions hold, or when any departure is explicitly assigned to a controlled target fiber:

1. it computes every target source-to-detection record;
2. its source and detector classes are those of $\mathcal{B}_{CC}$;
3. loss, decoherence, or leakage is represented by an event-complete instrument or by a declared conditional target;
4. protocol, environment, and calibration maps are explicit;
5. it admits a stable map to the target visible-mode packet;
6. transport/support and causal records are well defined;
7. attachment to the electroweak boundary is explicit;
8. anomaly and vacuum conditions are evaluated on the same physical candidate data;
9. EFT/UV scope and matching maps are explicit;
10. every extended operator is tied to an activated operator-category or global-form fiber;
11. lawful representative transformations are distinguished from physical tensor changes; and
12. fit quality, typicality, naturalness, or familiarity is not used as interface authority.

A.3 Target trace, semantic uses, and evaluators

The semantic completeness basis is fixed before the comparison-profile fields. The target-facing trace

$$\tau_{\mathbb{T}_\nu^{(2)}} : \mathcal{R}_\nu^{\text{raw}} \rightarrow \mathcal{T}(\mathbb{T}_\nu^{(2)})^\perp$$

is the full target-relative restriction to record, continuation/support, known-substrate attachment, activated extended-probe, and simultaneous-realization data. A semantic physical use is any lawful, same-target, standing-bearing map that is extensional with respect to this trace. Target-trace descent proves that every such use is a lawful postprocessing of the five trace restrictions, yielding $\text{PhysUse}_{\text{sem}}(\mathbb{T}_\nu^{(2)}) = \text{Use}(\mathbb{T}_\nu^{(2)})$.

The trace refines into the nine evaluator-bearing families

$$\mathbf{G}_{\text{rec}}, \mathbf{G}_{\text{cont}}, \mathbf{G}_{\text{loc}}, \mathbf{G}_{\text{gauge}}, \mathbf{G}_{\text{anom}}, \mathbf{G}_{\text{vac}}, \mathbf{G}_{\text{EFT}}, \mathbf{G}_{\text{ext}}, \mathbf{G}_{\text{sim}}.$$

Their trace projections are themselves primitive evaluators. The saturated evaluator algebra admits every lawful physical postprocessing of finitely many primitive outputs whether or not that functional was named in advance. Thus $\text{PhysEval}_{\text{sem}}(\mathbb{T}_\nu^{(2)}) = \text{Eval}(\mathbb{T}_\nu^{(2)})$. Candidate labels, microscopic carrier coordinates, fit

rankings, and administrative identifiers remain outside the semantic target trace and therefore cannot become evaluator authority.

A proposed new same-target physical comparison is treated by the following logical test. First determine whether it is lawful and extensional with respect to the complete target-facing trace. If it is, its unique trace descent determines the refined trace factors on which it depends, and hence its representation in $\text{Eval}(\mathbb{T}_v^{(2)})$ and its factorization through $\mathcal{P}_v^+$. If instead it distinguishes two presentations with equal target-facing traces, then it uses an additional primitive target-facing datum; that is a target-scope omission or target update rather than a merely unenumerated formula on the existing target.

A.4 Comparison-profile field dictionary

Field	Object type	Equality condition	Principal interface factors
P_{rec}	Family of event-response functionals and source analyses.	Equality on all target protocols and record maps.	Boundary, channel, visible modes.
C_{cont}	Continuation/environment functor over protocol domain.	Same lawful response under every admitted continuation.	Channel, transport, environment, EFT.
S_{loc}	Support and causal-relation packet.	Same source/detector support and admissible causal links.	Transport and environment.
G_{attach}	Boundary representation/charge quotient.	Same operational charged-current attachment.	Boundary and gauge data.
A_q	Quantum-consistency/anomaly packet.	Same active obstruction and compensation status.	Gauge, anomaly, extension packet.
V_{def}	Vacuum/background and deformation continuation.	Same target vacuum and lawful perturbation behavior.	Vacuum, environment, gauge.
R_{EFT}	Scale/order/matching packet.	Full profile preserved under matching, not finite-fit equality.	EFT, channel, vacuum.
X_{ext}	Activated extended-operator packet.	Same global form, operator action, anomaly, support, fusion/junction data.	Extension packet.

Table 12: Independent comparison-profile fields.

A.5 Target-fiber graph

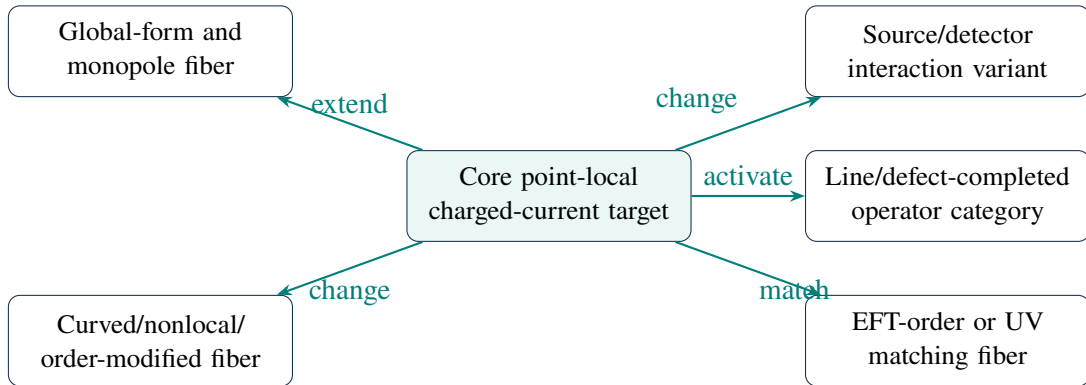


Figure 9: Controlled target fibers. Edges are typed maps and do not transfer standing automatically.

A.6 Coordinate-admission criteria

A proposed interface coordinate is retained only when its target fiber, mathematical type, physical evaluator, lawful quotient, shared-realization incidence, carrier-neutral definition, and variation witness are all explicit. In particular, the coordinate must satisfy the five conditions of [definition 3.21](#); absence of an explicit condition is not interpreted as a default pass.

A.7 Completeness and target update

The target and semantic evaluator structure are fixed before the component and joint-incidence profiles are formed. A new experiment, new record semantics, or newly admitted operator category can therefore enlarge or change the target. A new microscopic mechanism by itself does not. The completeness theorem must be reconsidered whenever the target changes, whenever target-trace descent or profile recovery is defeated, or whenever a genuine same-target interface counterexample is found.

B Supplementary proof details

B.1 Kraus-representation covariance

Let

$$\mathcal{E}(\rho) = \sum_{a=1}^m K_a \rho K_a^\dagger = \sum_{b=1}^n L_b \rho L_b^\dagger$$

be two Kraus representations of the same completely positive map. After padding the shorter list with zero operators, the representations are related by an isometry v on the Kraus index space:

$$L_b = \sum_a v_{ba} K_a.$$

For every boundary effect F and source state ρ ,

$$\begin{aligned} \text{Tr} \left[F \sum_b L_b \rho L_b^\dagger \right] &= \sum_{a,c} \left(\sum_b v_{ba} \bar{v}_{bc} \right) \text{Tr}(F K_a \rho K_c^\dagger) \\ &= \sum_a \text{Tr}(F K_a \rho K_a^\dagger), \end{aligned}$$

using $v^\dagger v = \mathbf{1}$ on the supported Kraus subspace. Thus transition kernels and all response records are unchanged. If an interface record stores a Kraus index as physical without an admitted environment witness, it is underquotiented.

B.2 Stinespring presentation and dilation sensitivity

A normal CP map admits a Stinespring representation

$$\mathcal{E}(\rho) = \text{Tr}_{\mathcal{K}}[V \rho V^\dagger]$$

for an isometry V into an enlarged output space. Minimal Stinespring representations are unique up to a unitary on the environment [29]. Nonminimal dilations can differ by additional environment factors.

For the present interface, two dilations are lawfully equivalent when:

1. their reduced CP maps agree on the record-relevant operator system;
2. the environment state and isometry are related by an admitted intertwiner;
3. no active support, charge, anomaly, decay-product, line, defect, or causal witness distinguishes the environment.

If clause 3 fails, the environment becomes an active extension coordinate. The reduced channel remains the same core coordinate, but the full interfaces can differ.

B.3 General face-invariance formulation

The projective proof in lemma 6.5 can be written for convex faces. Let $\mathfrak{F}_\alpha$ be a face of the state space associated with preparation class α , and let F_β be an effect vanishing on $\mathfrak{F}_\alpha$ for $\beta \neq \alpha$. If

$$\mathcal{E}(\mathfrak{F}_\alpha) \subseteq \mathfrak{F}_\alpha,$$

then

$$F_\beta(\mathcal{E}(\rho_\alpha)) = 0.$$

Thus target-visible cross-boundary transfer implies failure of charged-current sector nonmixing without requiring sharp projectors. In an operator-algebraic presentation, the visible sectors may be represented by hereditary subalgebras or support projections under suitable regularity; an explicit null/loss sector is permitted and does not by itself count as nonalignment.

B.4 Leakage-bound derivation

Suppose the source preparation ρ_α has calibrated off-face weight at most ϵ_{src} , the detector effect F_β has calibrated cross-tag response at most ϵ_{det} , and a boundary-preserving channel is contractive in trace norm. Then the cross-tag probability is bounded schematically by

$$K_{\beta\alpha} \leq \epsilon_{\text{src}} + \epsilon_{\text{det}} + O(\epsilon_{\text{src}}\epsilon_{\text{det}}).$$

The precise bound is experiment- and representation-dependent and belongs to the boundary specification. The present analysis uses only the general implication: a certified excess over the full calibrated boundary floor is a channel nonalignment witness.

B.5 Visible-mode injection

Let $m_1, m_2 \in \mathfrak{M}_{\text{vis}}(C_\nu)$ be distinct target classes. A target-realizing completion supplies candidate records r_1, r_2 in the same source-analysis fibers. If their candidate classes were equal, there would be a lawful source-authorized transport identifying r_1 and r_2 . Composing with the target-to-candidate record equivalences would identify m_1 and m_2 , contrary to target inequivalence. Hence the target-to-candidate class map is injective.

The proof does not claim that every candidate mode is target-required. It requires only that no target class is lost or merged.

B.6 Nonalignment from nonconstant cross-boundary response

Let $m \in \mathfrak{M}_{\text{vis}}(X)$ be a target mode class whose source analysis contains a nonzero cross-boundary kernel or a survival modulation that, together with the fixed target packet, requires flavor transfer. If every boundary face were invariant, all cross-boundary kernels would vanish by lemma 6.5, and the source analysis could not reproduce the target class. Thus $I_\Delta \Rightarrow I_{\text{ns}}$ in the fixed target.

The statement is target-relative. A nonconstant attenuation mode with no flavor transfer in a different target need not imply nonalignment. Such a record would not be one of the imported target mode classes.

B.7 Pairwise-pullback countermodel

Let the full witness universe be

$$W = E \times Q \times \Gamma,$$

where $E = \{e_0, e_1\}$ is environment data, $Q = \{q_0, q_1\}$ is representation/charge data, and $\Gamma = \{g_0, g_1\}$ is global-form/operator-category data. Define coordinate subobjects

$$\begin{aligned} C &= \{(e_0, q_0, g_0), (e_0, q_0, g_1)\}, \\ A &= \{(e_0, q_1, g_0), (e_1, q_1, g_0)\}, \\ D &= \{(e_1, q_0, g_1), (e_1, q_1, g_1)\}. \end{aligned}$$

Every pair has compatible projections after forgetting one coordinate:

$$\pi_{E,Q}(C) \cap \pi_{E,Q}(A) \neq \emptyset, \quad \pi_{Q,\Gamma}(A) \cap \pi_{Q,\Gamma}(D) \neq \emptyset, \quad \pi_{E,\Gamma}(C) \cap \pi_{E,\Gamma}(D) \neq \emptyset.$$

But

$$C \cap A \cap D = \emptyset.$$

Thus pairwise nonempty compatibility does not imply a global shared witness.

B.8 Target trace, semantic use descent, evaluator saturation, and profile recovery

Fix one target fiber. Before the interface coordinate packet is formed, the fixed target specification and imported physical-role architecture define the complete target-facing trace

$$\tau_{\mathbb{T}_V^{(2)}} : \mathcal{R}_V^{\text{raw}} \longrightarrow \mathcal{T}(\mathbb{T}_V^{(2)})^\perp, \quad \mathcal{T}(\mathbb{T}_V^{(2)}) = \mathcal{T}_{\text{rec}} \times \mathcal{T}_{\text{cont}} \times \mathcal{T}_{\text{att}} \times \mathcal{T}_{\text{ext}} \times \mathcal{T}_{\text{sim}}.$$

This is the full restriction of a raw presentation to the target-declared record, continuation/support, known-substrate attachment, activated-operator, and simultaneous-incidence data. It is not the later interface profile.

A semantic target-admitted use is defined independently of any grammar by same-target typing, lawful-redescription invariance, standing-bearing output, no selector authority, and target extensionality

$$\tau_{\mathbb{T}_V^{(2)}}(I) = \tau_{\mathbb{T}_V^{(2)}}(J) \implies U(I) = U(J).$$

Therefore every such use descends uniquely along $\tau_{\mathbb{T}_V^{(2)}}$:

$$U = \bar{U} \circ \tau_{\mathbb{T}_V^{(2)}}.$$

The five coordinate restrictions of $\tau_{\mathbb{T}_V^{(2)}}$ are the primitive use bearers. The saturated target-use grammar $\text{Use}(\mathbb{T}_V^{(2)})$ consists of all lawful standing-bearing postprocessings of their finite tuple. Hence

$$\text{PhysUse}_{\text{sem}}(\mathbb{T}_V^{(2)}) = \text{Use}(\mathbb{T}_V^{(2)}).$$

This normal form closes the possibility of an unenumerated lawful functional directly: a newly written lawful functional of existing target data is simply another $\bar{U}$; it does not require a new primitive formation rule. Only a use requiring target-facing data not determined by $\mathcal{T}(\mathbb{T}_V^{(2)})$ reopens the target scope.

Target-admitted evaluators are likewise defined semantically, with no use-grammar derivation in their definition. The five trace factors refine canonically into nine target output sorts,

$$\mathcal{T}_{\text{rec}}, \mathcal{T}_{\text{cont}}, \mathcal{T}_{\text{loc}}, \mathcal{T}_{\text{gauge}}, \mathcal{T}_{\text{anom}}, \mathcal{T}_{\text{vac}}, \mathcal{T}_{\text{EFT}}, \mathcal{T}_{\text{ext}}, \mathcal{T}_{\text{sim}},$$

whose projections belong to

$$\text{Gen}(\mathbb{T}_V^{(2)}) = \mathbf{G}_{\text{rec}} \sqcup \mathbf{G}_{\text{cont}} \sqcup \mathbf{G}_{\text{loc}} \sqcup \mathbf{G}_{\text{gauge}} \sqcup \mathbf{G}_{\text{anom}} \sqcup \mathbf{G}_{\text{vac}} \sqcup \mathbf{G}_{\text{EFT}} \sqcup \mathbf{G}_{\text{ext}} \sqcup \mathbf{G}_{\text{sim}}.$$

Every semantic evaluator therefore has a finite-root normal form

$$E = h_E \circ (\tau_{\text{rec}}, \tau_{\text{cont}}, \tau_{\text{loc}}, \tau_{\text{gauge}}, \tau_{\text{anom}}, \tau_{\text{vac}}, \tau_{\text{EFT}}, \tau_{\text{ext}}, \tau_{\text{sim}}),$$

with h_E any lawful physical postprocessing on the existing output quotients. This yields

$$\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)}) = \text{Eval}(\mathbb{T}_v^{(2)})$$

without assuming finite derivability in the definition of $\text{PhysEval}_{\text{sem}}(\mathbb{T}_v^{(2)})$.

The component profile $\mathcal{P}_v^0$ is the joint lawful output signature of the non-incidence primitive families, while $\mathcal{J}_v^{\text{sim}}$ is the physical output signature of $\mathbf{G}_{\text{sim}}$. Because the refined trace projections themselves are members of those families, the primitive signatures recover the target trace. Consequently

$$\mathcal{P}_v^+(I) = \mathcal{P}_v^+(J) \iff \tau_{\mathbb{T}_v^{(2)}}(I) = \tau_{\mathbb{T}_v^{(2)}}(J) \iff I \equiv_{\mathbb{T}_v^{(2)}} J.$$

The first equivalence is trace recovery; the second is semantic target extensionality together with the inclusion of the trace projections among the semantic evaluators. The complete-invariant result therefore does not depend on quotient design, finite enumeration, or an assumption that every physical use has a pre-enumerated grammar derivation.

B.9 Constructive-generator soundness and complete-profile quotient after pullback

Let

$$I_0 \xrightarrow{g_1} I_1 \xleftarrow{g_2} I_2 \xrightarrow{g_3} \dots \xrightarrow{g_n} I_n$$

be a constructive zig-zag in $\mathcal{G}_v^{\text{gen}}$. For every generator g_i ,

$$\mathcal{P}_v^+(I_{i-1}) = \mathcal{P}_v^+(I_i).$$

Equality is symmetric, so arrow orientation does not matter. Transitivity gives

$$\mathcal{P}_v^+(I_0) = \mathcal{P}_v^+(I_n).$$

This proves soundness of the explicit redescription library, including preservation of the physical witness-incidence diagram. The canonical physical quotient is then formed on $\mathcal{M}_{\text{pull}}$. For $I, J \in \mathcal{M}_{\text{pull}}$,

$$I \simeq_{\text{int}} J \iff \mathcal{P}_v^+(I) = \mathcal{P}_v^+(J),$$

which is theorem 13.12. A raw witness-split tuple lies outside the quotient domain even if its component values resemble those of a simultaneous interface.

B.10 Coordinate factorization and exact complete-profile quotient

Define

$$F_{\text{sim}} : \mathfrak{J}_v^{\text{sim}} \longrightarrow \text{Im}(\mathcal{P}_v^+ | \mathcal{M}_{\text{pull}})$$

by

$$\begin{aligned} F_{\text{sim}}([I]) = (&F_{\text{rec}}([I]), F_{\text{cont}}([I]), F_{\text{loc}}([I]), \\ &F_{\text{gauge}}([I]), F_{\text{anom}}([I]), F_{\text{vac}}([I]), \\ &F_{\text{scale}}([I]), F_{\text{ext}}([I]), F_{\text{sim}}^{\text{inc}}([I])). \end{aligned}$$

The final entry is the physical joint-incidence class. Quotient well-definedness makes the map independent of the representative. For every $X \in \mathfrak{C}_v^{\text{sim}}$,

$$F_{\text{sim}}(\Pi_v(X)) = \mathcal{P}_v^+(\tilde{\mathfrak{J}}_v(X)).$$

By theorem 13.13, the map is bijective. Its restriction to $\mathfrak{S}_v^{\text{adm}}$ is a bijection onto $\text{Im}(\mathcal{P}_v^+|_{\mathcal{M}_{\text{adm}}})$.

B.11 Pullback and survivor congruence detail

Let $D_I = \mathcal{D}_{\text{sim}}(I)$. A shared ledger is a typed cone over D_I whose overlap triangles commute. If $I \equiv_{\mathcal{P}^+} J$, then $D_I \cong_{\text{phys}} D_J$. Transporting a required cone across the diagram isomorphism gives a required cone over D_J , and the inverse isomorphism gives the converse. Hence

$$I \equiv_{\mathcal{P}^+} J \implies (I \in \mathcal{M}_{\text{pull}} \iff J \in \mathcal{M}_{\text{pull}}).$$

Every active constraint is a predicate of the component values and their physical incidence. Thus for each $K \in \mathcal{K}_v^{(2)}$,

$$K(I) = \widehat{K}(\mathcal{P}_v^+(I)).$$

Consequently, for simultaneous presentations,

$$I \equiv_{\mathcal{P}^+} J \implies (I \in \mathcal{M}_{\text{adm}} \iff J \in \mathcal{M}_{\text{adm}}).$$

These are the representative-independence results needed for

$$\mathfrak{S}_v^{\text{sim}} = \mathcal{M}_{\text{pull}} / \equiv_{\mathcal{P}^+}, \quad \mathfrak{S}_v^{\text{adm}} = \mathcal{M}_{\text{adm}} / \equiv_{\mathcal{P}^+}.$$

B.12 Adequacy biconditional dependency

The reverse implication in theorem 14.2 uses the following exact assumptions:

1. the target-evaluator generation theorem and complete-invariant theorem hold for the fixed target fiber;
2. every target-admitted consistency condition appears in the constraint family;
3. relevance gates have no unresolved mandatory coordinate;
4. shared-ledger factorization and the physical overlap diagram are complete;
5. the exact complete-profile-saturated lawful quotient of theorem 13.12 is used on $\mathcal{M}_{\text{pull}}$;
6. pullback and active-constraint membership descend to complete-profile classes;
7. the completion X is already specified and all interface coordinates are evaluable.

Without the last clause, an abstract interface tuple need not have a physical preimage. Without the scientific completeness clauses, membership in a partial admissible set does not establish complete interface adequacy.

B.13 No-carrier noninjectivity example

Let $\mathcal{E}$ be any target-adequate CP channel with minimal dilation $(\mathcal{K}, V)$. Define a nonminimal dilation on $\mathcal{K} \otimes \mathcal{K}'$ by

$$V'\psi = V\psi \otimes |0\rangle.$$

If $\mathcal{K}'$ is decoupled and carries no active target witness, both dilations lie in $\mathfrak{C}_v^{\text{sim}}$ and have the same complete interface profile. Their candidate records differ by the additional sector, but their canonical interface images agree. If $\mathcal{K}'$ is coupled, charged, or changes a common witness relation, the component or joint-incidence profile changes and the interfaces can separate.

C Benchmark model details

C.1 Three-flavor unitary representative

In the conventional relativistic approximation,

$$H_{\text{vac}} \simeq \frac{1}{2E} U \text{diag}(m_1^2, m_2^2, m_3^2) U^\dagger, \quad \mathcal{E}_{L,E}(\rho) = e^{-iH_{\text{vac}}L} \rho e^{iH_{\text{vac}}L}.$$

For interaction boundary projectors $P_\alpha = |\nu_\alpha\rangle\langle\nu_\alpha|$, the transition probability is

$$K_{\beta\alpha}(L, E) = \text{Tr}[P_\beta \mathcal{E}_{L,E}(P_\alpha)].$$

The model maps to the interface as follows:

- $\mathcal{B}_{\text{CC}}$: charged-current flavor tags;
- $[\mathcal{E}]_{\text{ch}}$: unitary channel class;
- I_{bc} : noncommutation of H_{vac} with all flavor projectors;
- I_Δ : target response-mode coverage;
- $\mathcal{E}_{\text{env}}$: matter potential when included;
- $\mathcal{G}_{\text{attach}}$: standard electroweak boundary;
- candidate-only fields: m_i, U , ordering, phases, and microscopic state-space description.

The last fields do not belong to the carrier-neutral interface.

C.2 Matter-dependent unitary representative

A standard matter presentation has

$$H(x) = H_{\text{vac}} + V(x), \quad U(L) = \mathcal{P} \exp \left[-i \int_0^L H(x) dx \right].$$

The physical environment packet contains the density/profile data that determine $V(x)$, the path ordering, and the induced channel. Replacing one discretization of the same profile by another convergent representation is lawful. Replacing the physical profile is a tensor-level environment variation.

C.3 Pure-dephasing benchmark

Let $\{|i\rangle\}$ be a candidate propagation basis and

$$\mathcal{E}_t(\rho) = \sum_{i,j} D_{ij}(t) e^{-i(\phi_i - \phi_j)t} \rho_{ij} |i\rangle\langle j|,$$

with $D_{ii} = 1$ and a positive-semidefinite decoherence matrix. The channel coordinate is the CP map, not the basis in which the formula is written. Decoherence changes the response profile when it damps or reshapes target-visible interference. It is candidate tensor data and must pass deformation, locality, environment, and EFT constraints.

C.4 Decay/loss benchmark

For visible active outcomes a and a loss outcome ℓ , let

$$\mathcal{I}_t = \{\mathcal{E}_{t,a}, \mathcal{E}_{t,\ell}\}, \quad \sum_a \mathcal{E}_{t,a} + \mathcal{E}_{t,\ell} \text{ trace preserving.}$$

The interface stores active transition kernels and the loss record separately. A conditional active-only probability is allowed only if the target source analysis is conditional on active detection. Otherwise renormalizing away ℓ changes the record semantics.

C.5 Enlarged-state-space benchmark

Let

$$\mathcal{H} = \mathcal{H}_{\text{active}} \oplus \mathcal{H}_{\text{extra}}$$

and U_t be unitary on $\mathcal{H}$. The active boundary effects project only onto $\mathcal{H}_{\text{active}}$. The reduced active channel can be nonunitary. The interface theorem does not infer the dimension or ontology of $\mathcal{H}_{\text{extra}}$. If extra states produce visible loss, charges, anomalies, decay products, support, or generalized-symmetry data, those become active candidate/interface coordinates.

C.6 Source/detector interaction benchmark

Suppose a candidate modifies the source preparation by S and the detector effect by D :

$$\rho_\alpha \mapsto S(\rho_\alpha), \quad F_\beta \mapsto D^\dagger(F_\beta).$$

Writing the resulting probability as

$$\text{Tr}[F_\beta D \circ \mathcal{E} \circ S(\rho_\alpha)]$$

does not make S and D propagation data. Unless they preserve the operational boundary equivalence, the candidate lies in a source/detector target fiber.

C.7 Global-form benchmark

Let two completions have faithful gauge groups G_{Γ_1} and G_{Γ_2} but the same local charged-current action and point-local target records. Their core interfaces coincide. If the target is extended to genuine line or monopole operators, the global-form field activates and the interfaces can differ. This benchmark separates interface completeness from full theory identity.

C.8 Non-invertible defect benchmark

Let a candidate carry a codimension-one defect $\mathcal{D}$ with non-group fusion and an auxiliary defect TQFT. If no target source, detector, anomaly, support, or response witness couples to $\mathcal{D}$, the defect is candidate-internal at the core target scope. If an activated target fiber admits its selection rule or junction data, $\mathcal{D}$ enters $\mathfrak{F}_v^{\text{ext}}$. Neither branch implies a new bulk asymptotic particle.

C.9 One visible mode with hidden spectral richness

A candidate can have many microscopic eigenvalues but project to one visible response class because the boundary couples only to one effective phase combination or because all other modes are degenerate, decoupled, or unresolved under the target analysis. Such a candidate can occupy $r_{\bowtie}$ and fail r_Δ . This is why the interface records visible-mode coverage rather than microscopic spectral cardinality.

C.10 Full-matching benchmark

Let a UV theory X_{UV} match to an EFT X_{EFT} through M_μ . If

$$\mathcal{P}_v^+(X_{\text{UV}}) = \mathcal{P}_v^+(X_{\text{EFT}})$$

under the matching map throughout the target scale interval, they share one interface image. If only selected amplitudes or best-fit parameters agree, interface identity is unproved.

D Generalized-symmetry relevance details

D.1 Relevance criterion

For each proposed generalized-symmetry coordinate c , the relevant questions are:

1. Which target operator or background detects c ?

2. Is that object part of the point-local core operator category?
3. Does changing c alter a component of $\mathcal{P}_\nu^+$ or the physical joint-incidence profile $\mathcal{J}_\nu^{\text{sim}}$?
4. Is the effect universal across every lawful completion or only completion-specific?
5. Does the coordinate possess a physical quotient beyond a categorical label?
6. Can it be evaluated on the same global-form, anomaly, support, vacuum, and EFT witness system?

Table 13: Generalized-symmetry relevance classification.

Structure	Detecting witness	Target-relative role	Status
Faithful SM global form G_Γ	Genuine lines, monopoles, theta periodicity, global anomalies.	Target metadata at core; active in extension fibers.	Relevance-gated.
Electric one-form symmetry	Unscreened Wilson lines/endability.	Not detected by point-local oscillation records.	Optional/fiber.
Magnetic one-form symmetry	't Hooft lines/monopole policy.	Requires an extended operator and global-form target.	Optional/fiber.
Higher flavor group	Mixed zero/one-form background law and anomaly.	Active only if corresponding backgrounds or lines are admitted.	Optional/fiber.
Non-invertible chiral defect	Defect action, fusion, junction, selection rules.	No universal core witness in the fixed point-local target.	Optional or target-changing.
Defect-localized TQFT	Defect partition function/junction observables.	Completion support class; no bulk-matter inference.	Optional/fiber.
Theta periodicity/CP special point	Topological sector and global-form data.	Not a universal core propagation coordinate.	Target-fiber data.
Anomaly inflow	Boundary/bulk/defect anomaly data.	Mandatory only when a completion uses it to preserve consistency.	Activated constraint.
Higher-group anomaly	Mixed background consistency.	Mandatory in an active higher-group fiber.	Relevance-gated.
Non-invertible RG transport	Defect survival/deformation across scale.	Completion-specific EFT extension data.	Optional/fiber.

D.2 Core non-necessity

The point-local core target contains no witness in the detecting-witness column of table 13. Its source/detection effects and response records are local, and its mandatory anomaly conditions concern the consistency of the fixed charged-current boundary. Therefore the core-forgetful projection preserves the complete core comparison profile, including physical joint incidence when generalized coordinates are forgotten. This establishes non-necessity at core scope.

D.3 Activation examples

Line-coupled completion. A completion whose role occupant carries a genuine unscreened line charge activates the electric/magnetic one-form and global-form coordinates. Screening, endability, and global-form data then become part of the interface even if the point-local response remains unchanged.

Inflow-supported completion. A completion whose local boundary anomaly is canceled by bulk or defect inflow activates the inflow coordinate and, where applicable, defect/TQFT data. The compensating sector belongs to the same shared physical witness system and cannot be treated as an independent appendage.

Purely point-local matter completion. If a completion changes no extended operator or background witness, the generalized coordinates remain inactive. This does not imply that the full theory lacks generalized symmetries; it means only that they do not distinguish the target-relative core interface.

D.4 Conditions for revisiting the relevance classification

The generalized-symmetry conclusion must be revisited if a universal defect or line observable is derived from the established role complex; if an active anomaly condition cannot be formulated without a higher-form or higher-group background; if an independent candidate-completeness result shows that every interface-adequate completion activates the same extension packet; or if a new experiment directly probes an extended operator, topological sector, or defect selection rule. In each case the new information strengthens or changes the target-relative analysis rather than being assumed in advance.

E Glossary and symbol register

Table 14: Principal symbols and meanings.

Symbol/term	Meaning
T_{ν}^{osc}	Fixed realized neutrino-oscillation target established in the preceding role analysis.
$\mathbb{T}_{\nu}^{\text{if}}$	Interface target, including comparison class, attachment, activation, and open carrier coordinate.
$\mathfrak{C}_{\nu}^{(2)}$	Fixed class of lawful completions compared by the present interface theorem.
$\text{Real}_{\mathfrak{C}_{\nu}}(T_{\nu}^{\text{osc}})$	Lawful realizations of the fixed target in $\mathfrak{C}_{\nu}^{(2)}$.
$\mathfrak{B}_{\text{CC}}^{\text{op}}$	Operational charged-current preparation/detection boundary.
$\mathcal{E}_{\lambda,r}$	Record-complete CP instrument branch for protocol λ .
$K_{\beta\alpha}$	Transition/record probability functional.
$r_{\triangleright\triangleleft}$	Interaction–propagation nonalignment role: certified failure of loss-tolerant charged-current sector nonmixing.
r_{Δ}	Boundary-visible response-multiplicity role.
$\mathfrak{M}_{\text{vis}}$	Finite lawful boundary-visible mode-class set.
ρ_{vis}	Cardinality of $\mathfrak{M}_{\text{vis}}$; not microscopic rank.
$\mathcal{T}(\mathbb{T}_{\nu}^{(2)})$	Complete target-facing trace space: record, continuation/support, known-substrate attachment, activated extension, and simultaneous-incidence restrictions of the fixed target.
$\tau_{\mathbb{T}_{\nu}^{(2)}}$	Target-facing trace map from a raw evaluation presentation to $\mathcal{T}(\mathbb{T}_{\nu}^{(2)})^{\perp}$, defined before any grammar, interface coordinate, or candidate.
$\text{PhysUse}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$	Semantic same-target standing-bearing physical uses, defined by lawful redescription and extensionality with respect to $\tau_{\mathbb{T}_{\nu}^{(2)}}$, not by grammar membership.
$\text{Use}(\mathbb{T}_{\nu}^{(2)})$	Saturated target-use grammar; proved equal to $\text{PhysUse}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$ by target-trace descent.
$\text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$	Semantic target-admitted physical evaluators, defined independently of evaluator-grammar derivability.
$\text{Eval}(\mathbb{T}_{\nu}^{(2)})$	Saturated target-evaluator algebra; proved equal to $\text{PhysEval}_{\text{sem}}(\mathbb{T}_{\nu}^{(2)})$ by refinement of the target trace and lawful postprocessing.

Symbol/term	Meaning
$I \equiv_{\mathbb{T}_\nu^{(2)}} J$	Same-target physical indistinguishability: every semantic target evaluator has the same definedness and lawful output on I and J .
$\mathcal{P}_\nu^0$	Target-derived component comparison profile.
$\mathcal{J}_\nu^{\text{sim}}$	Physical joint-incidence profile: overlap-diagram isomorphism class with archival metadata removed.
$\mathcal{P}_\nu^+$	Complete comparison profile $(\mathcal{P}_\nu^0, \mathcal{J}_\nu^{\text{sim}})$; its primitive signatures recover $\tau_{\mathbb{T}_\nu^{(2)}}$, so it is a complete invariant for $\text{PhysEval}_{\text{sem}}(\mathbb{T}_\nu^{(2)}) = \text{Eval}(\mathbb{T}_\nu^{(2)})$.
Π_ν	Complete carrier-neutral physical-interface projection.
$\mathfrak{Z}_\nu^{\text{core}}$	Boundary/channel/response/environment/record core packet.
$\mathfrak{Z}_\nu^{\text{att}}$	Locality/gauge/anomaly/vacuum/deformation/operator/RG attachment packet.
$\mathfrak{Z}_\nu^{\text{ext}}$	Conditional global-form/generalized-symmetry packet.
$\mathcal{G}_\nu^{\text{law}}$	Groupoid of simultaneous lawful interface redescrptions.
$X \simeq_{\text{int}} Y$	Same-target complete-profile-lawful interface equivalence: equality of component values and physical joint incidence, not full-theory identity.
Wit_ν	Raw shared witness ledger through which all active coordinates factor; its physical quotient is $\mathcal{J}_\nu^{\text{sim}}$.
Activation	Target-relative theorem deciding whether a conditional coordinate is required, optional, irrelevant, or target-updating.
Interface completeness	Equality of complete packets iff target-evaluator indistinguishability, equivalently equality of $\mathcal{P}_\nu^+$.
Candidate completeness	Exhaustion of the allowed microscopic carrier universe; logically distinct from interface completeness.
Derived view	A named field functorially determined by primitive coordinates, retained for traceability but adding no multiplicity.
Compatibility condition	A condition needed for lawful attachment or subsequent typing, possibly constant on admitted interfaces.
Target fiber	A controlled target obtained by changing an explicit fingerprint coordinate.
Reopening	Reconsideration of the interface theorem when a genuine same-target signature lies outside the established interface range.

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